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(54) **MICROELECTRONIC DEVICES, AND
RELATED MEMORY DEVICES AND
ELECTRONIC SYSTEMS**

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See application file for complete search history.

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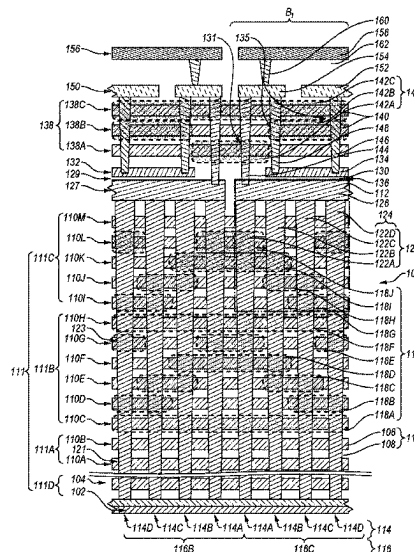
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(57) **ABSTRACT**

A microelectronic device comprises a stack structure, pillar structures, a conductive plug structure, a sense transistor, and selector transistors. The stack structure comprises a vertically alternating sequence of conductive material and insulative material, and is divided into blocks separated by dielectric slot structures. The blocks individually include sub-blocks horizontally extending in parallel with one another. The pillar structures vertically extend through one of the blocks of the stack structure. Each pillar structure of a group of the pillar structures is positioned within a different one of the sub-blocks of the one of the blocks than each other pillar structure of the group. The conductive plug structure is coupled to multiple of the pillar structures of the group of the pillar structures. The sense transistor is gated by the conductive plug structure. The selector transistors couple the sense transistor to a read source line structure and a digit line structure.

36 Claims, 12 Drawing Sheets



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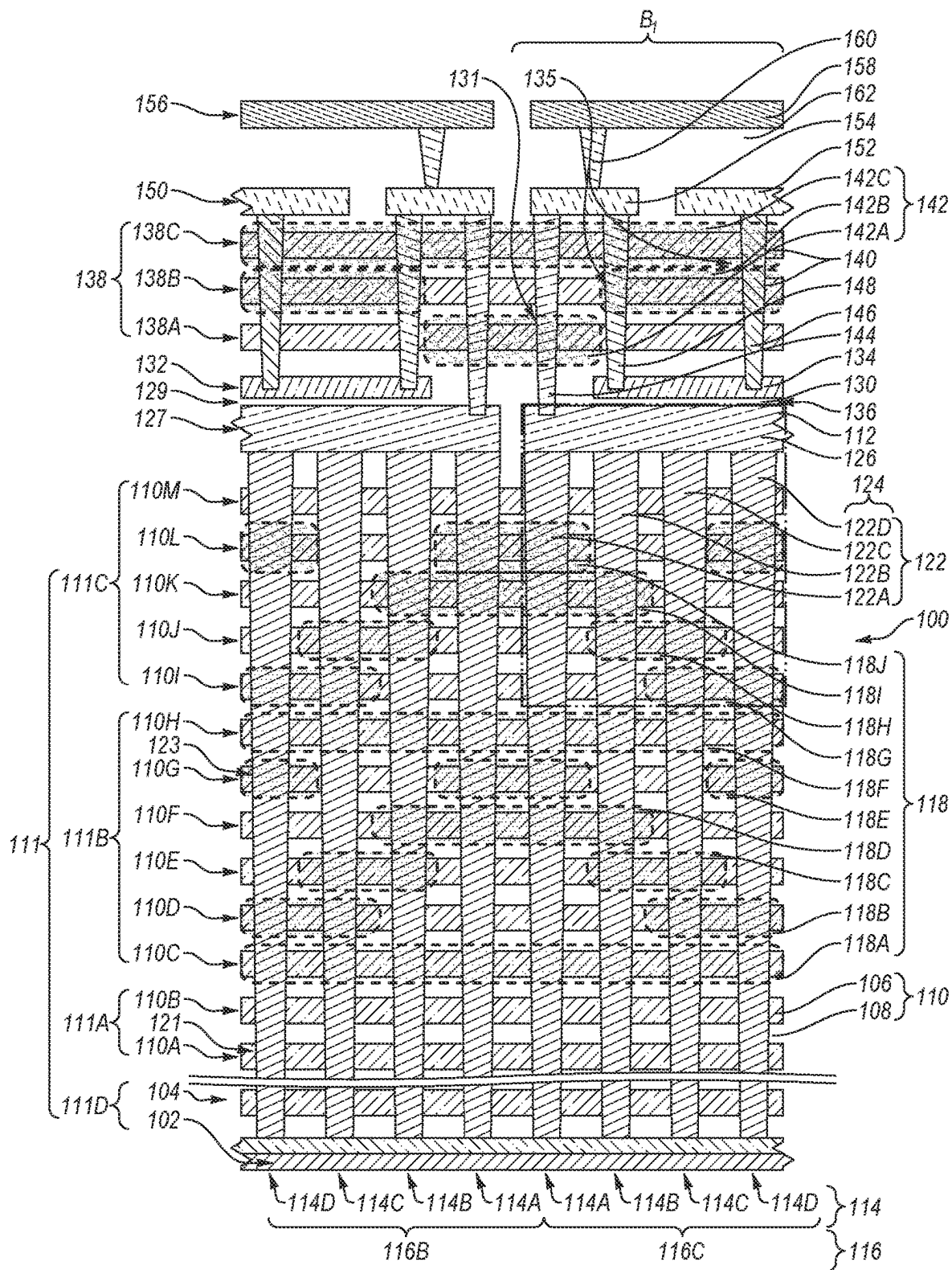


FIG. 1A

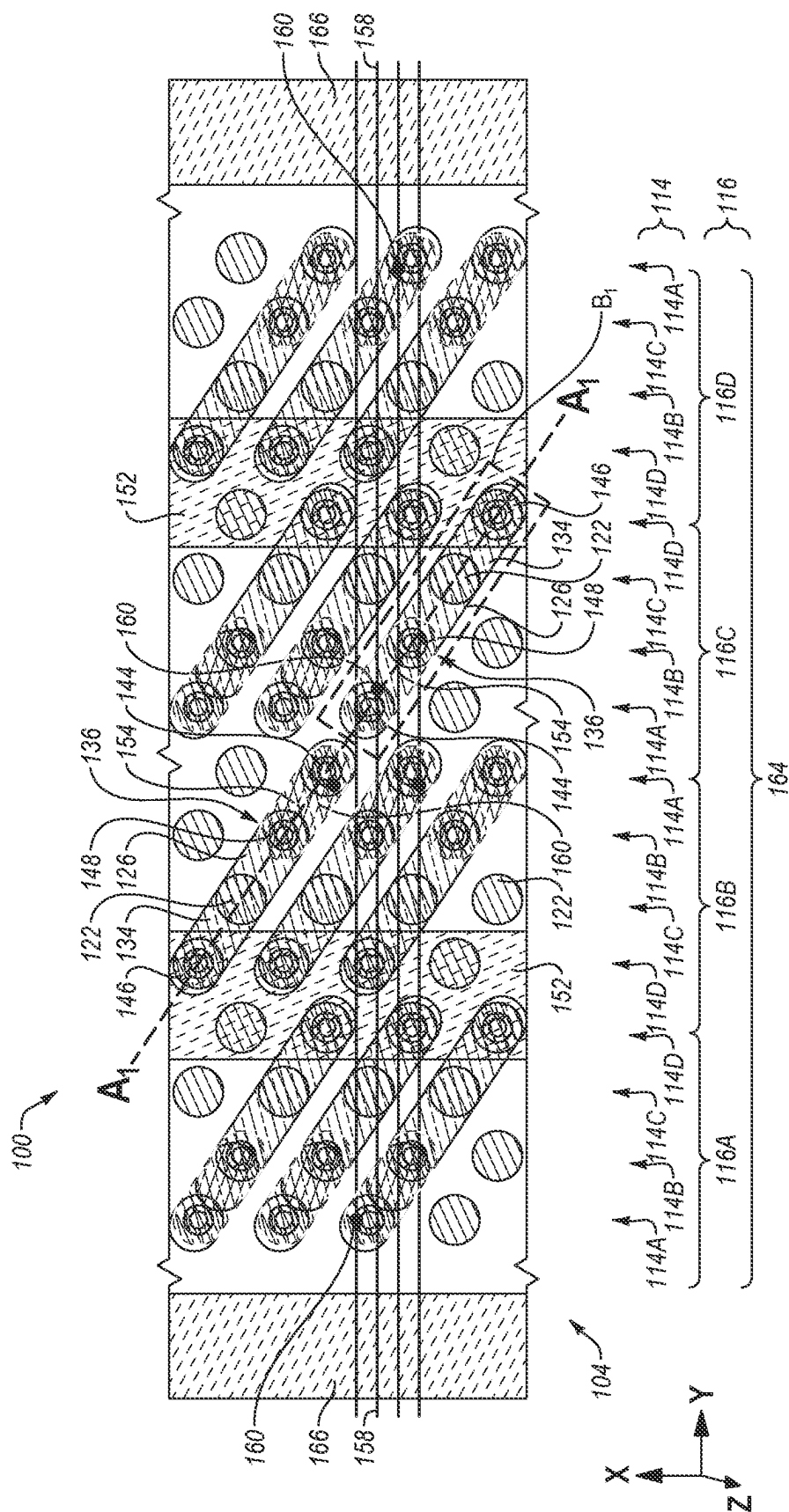
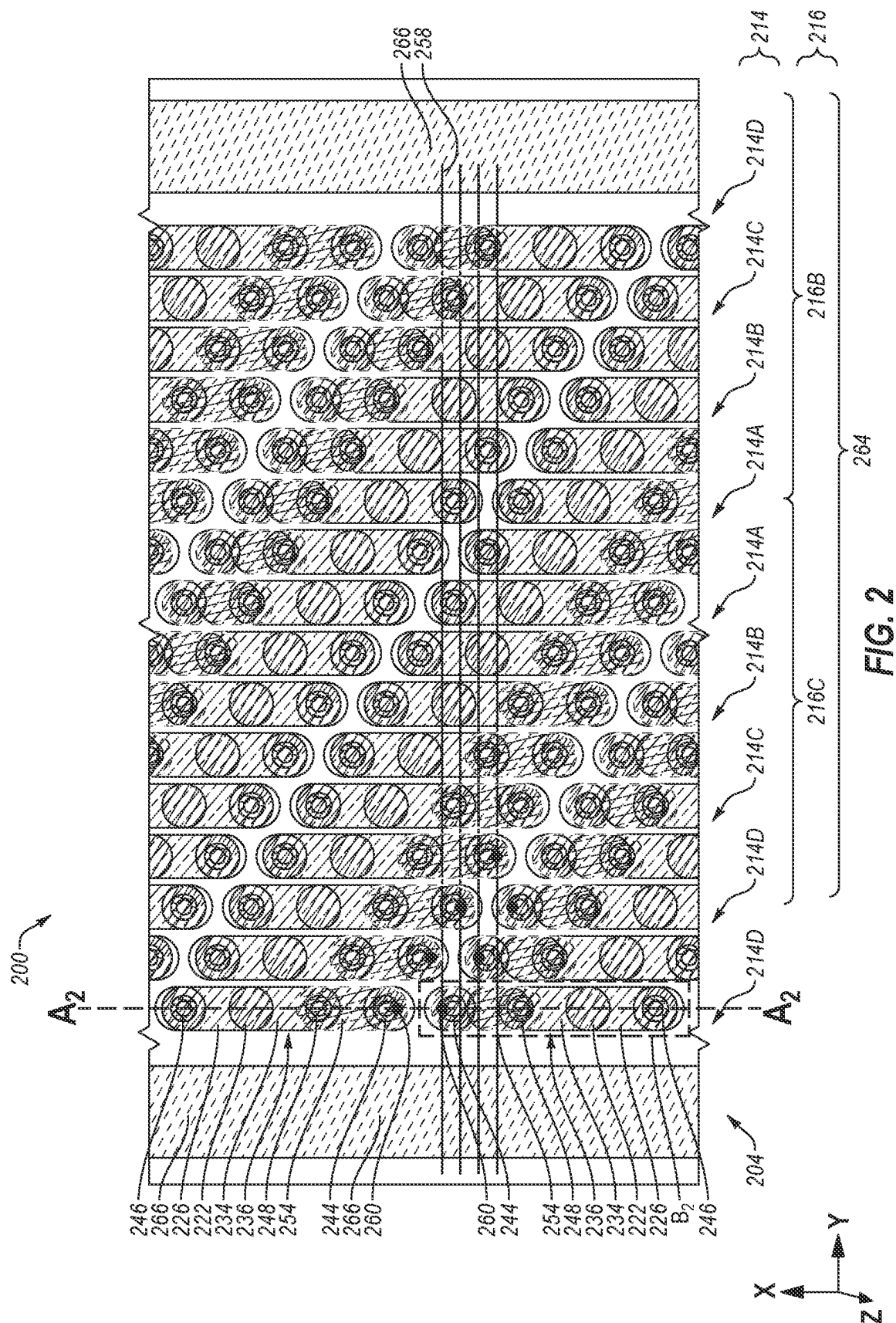


FIG. 1B



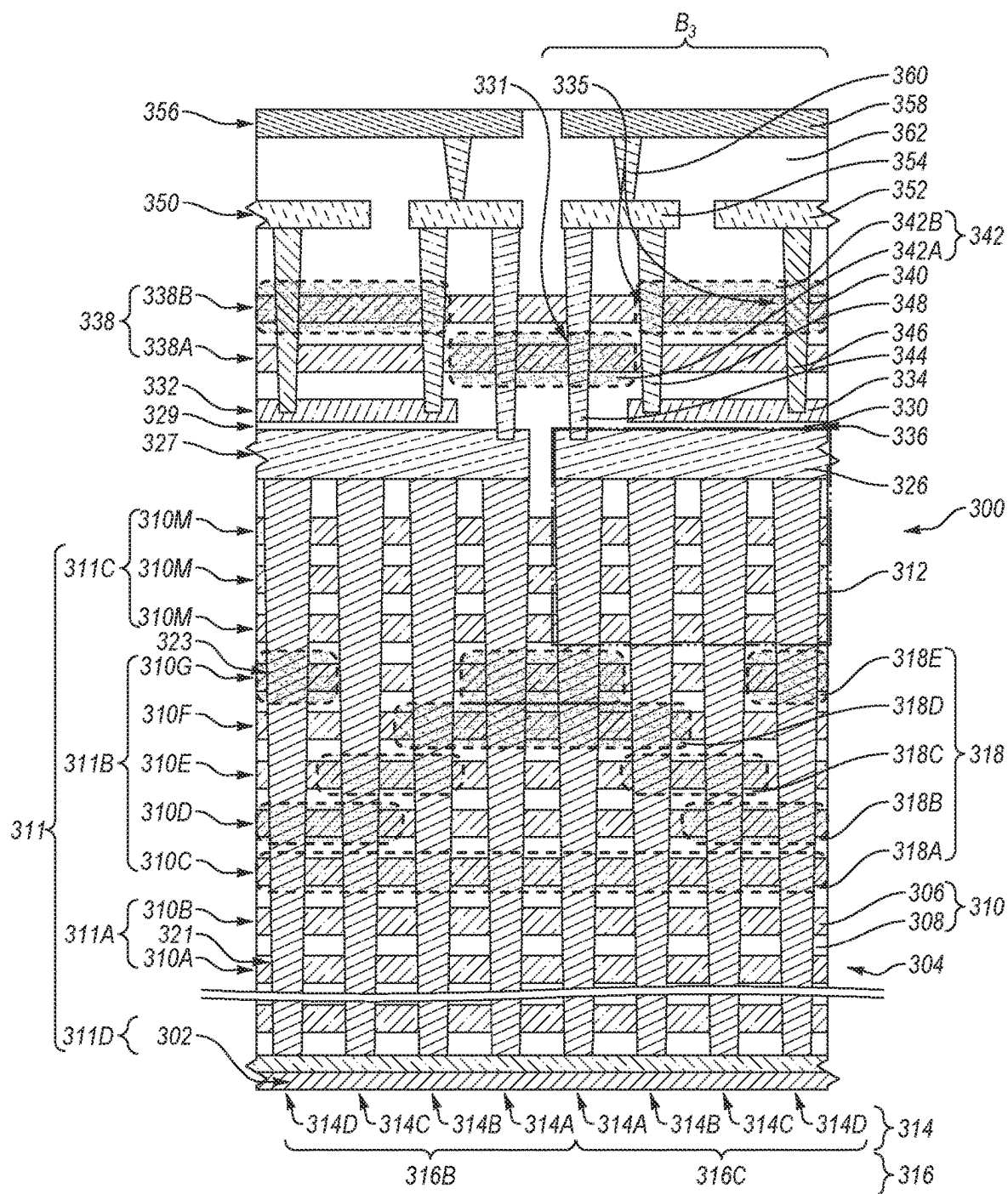
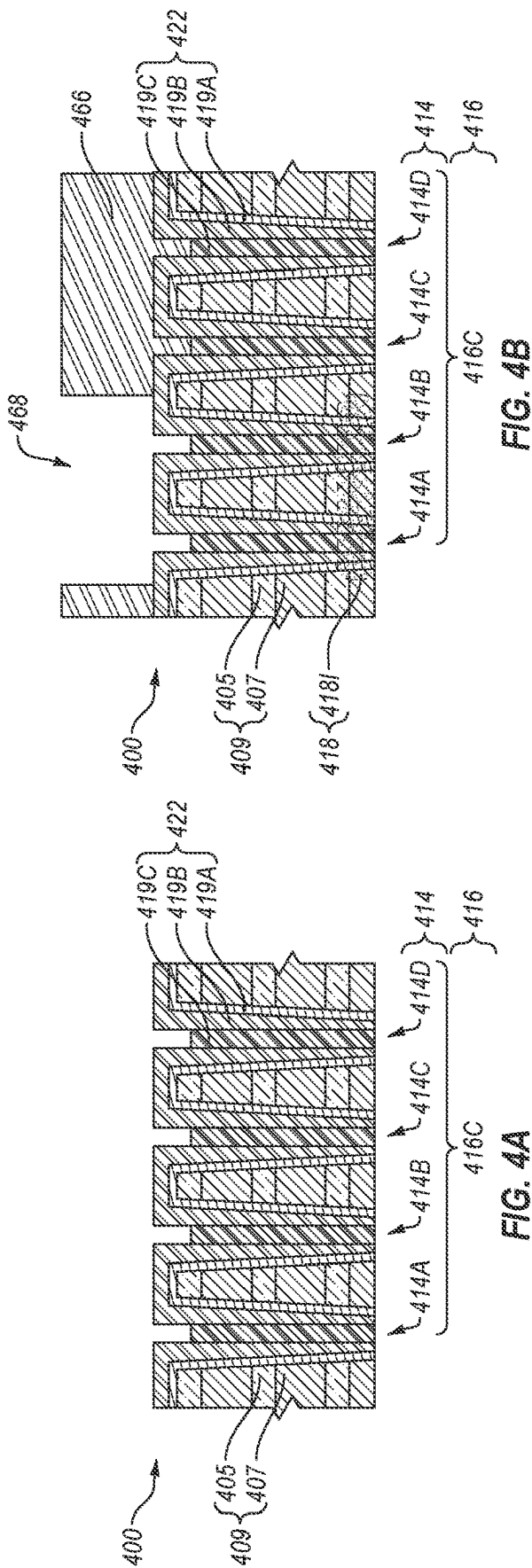


FIG. 3



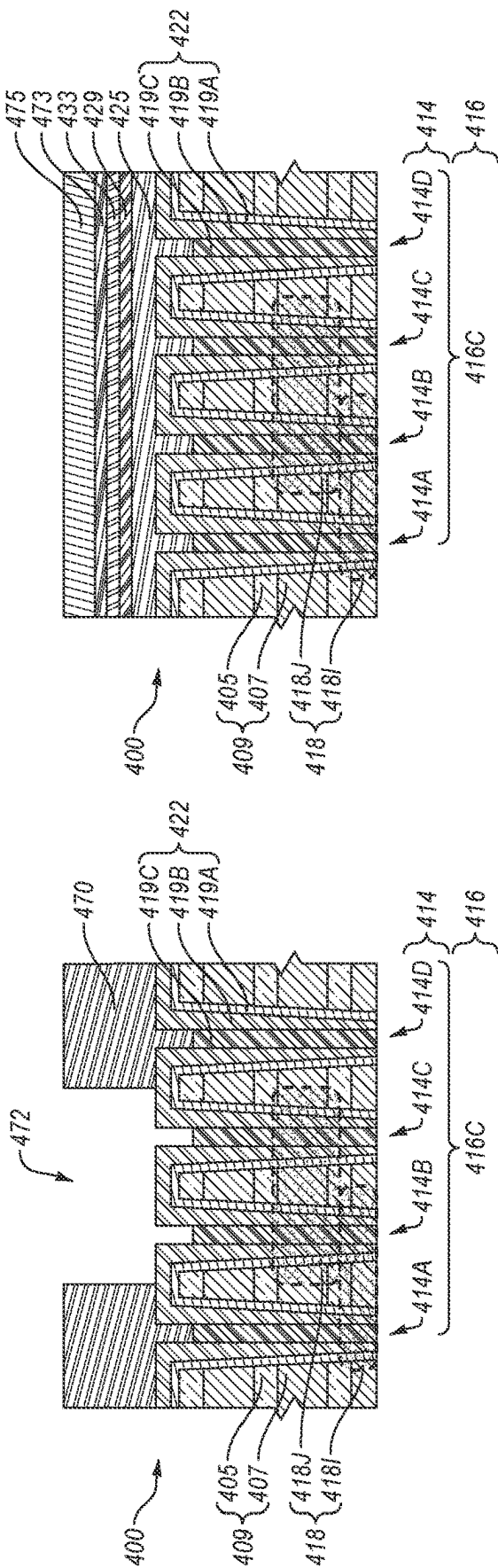


FIG. 4D

FIG. 4C

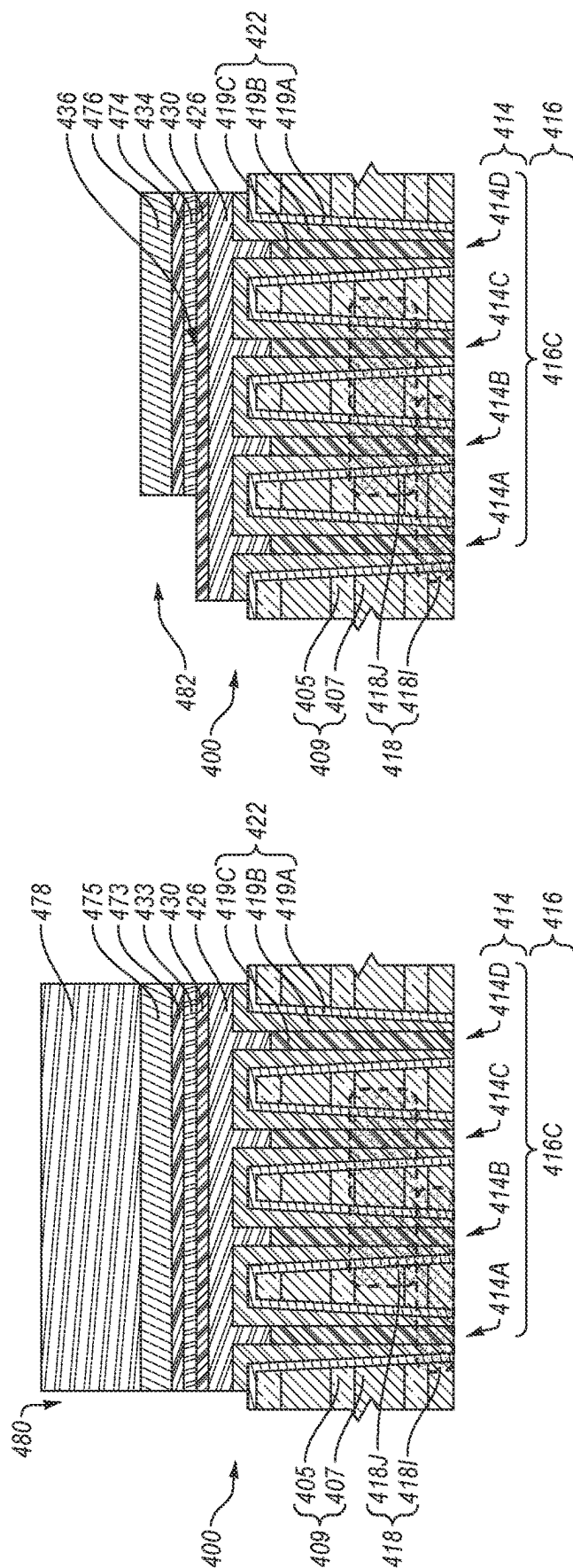
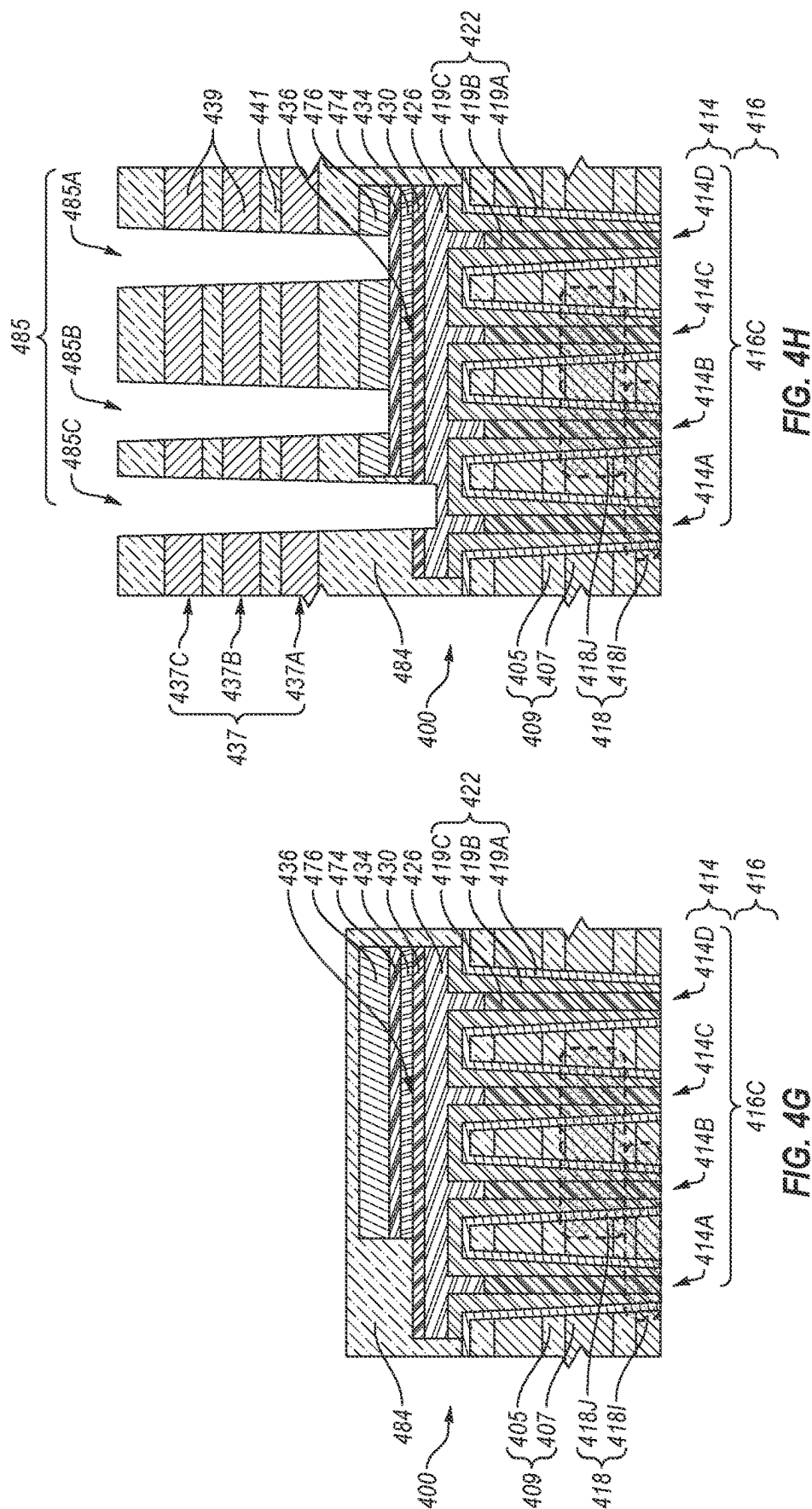
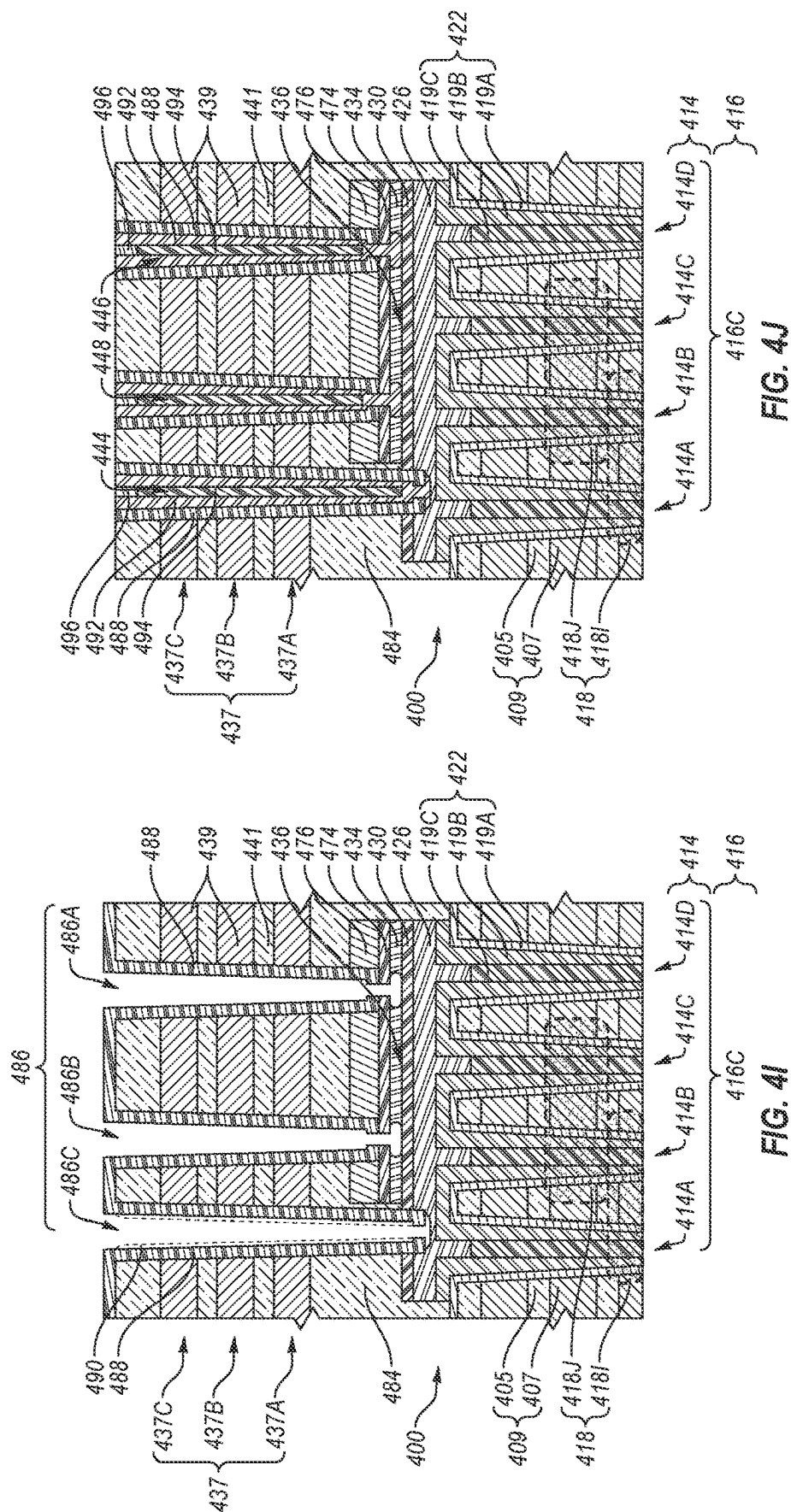


FIG. 4F

FIG. 4E





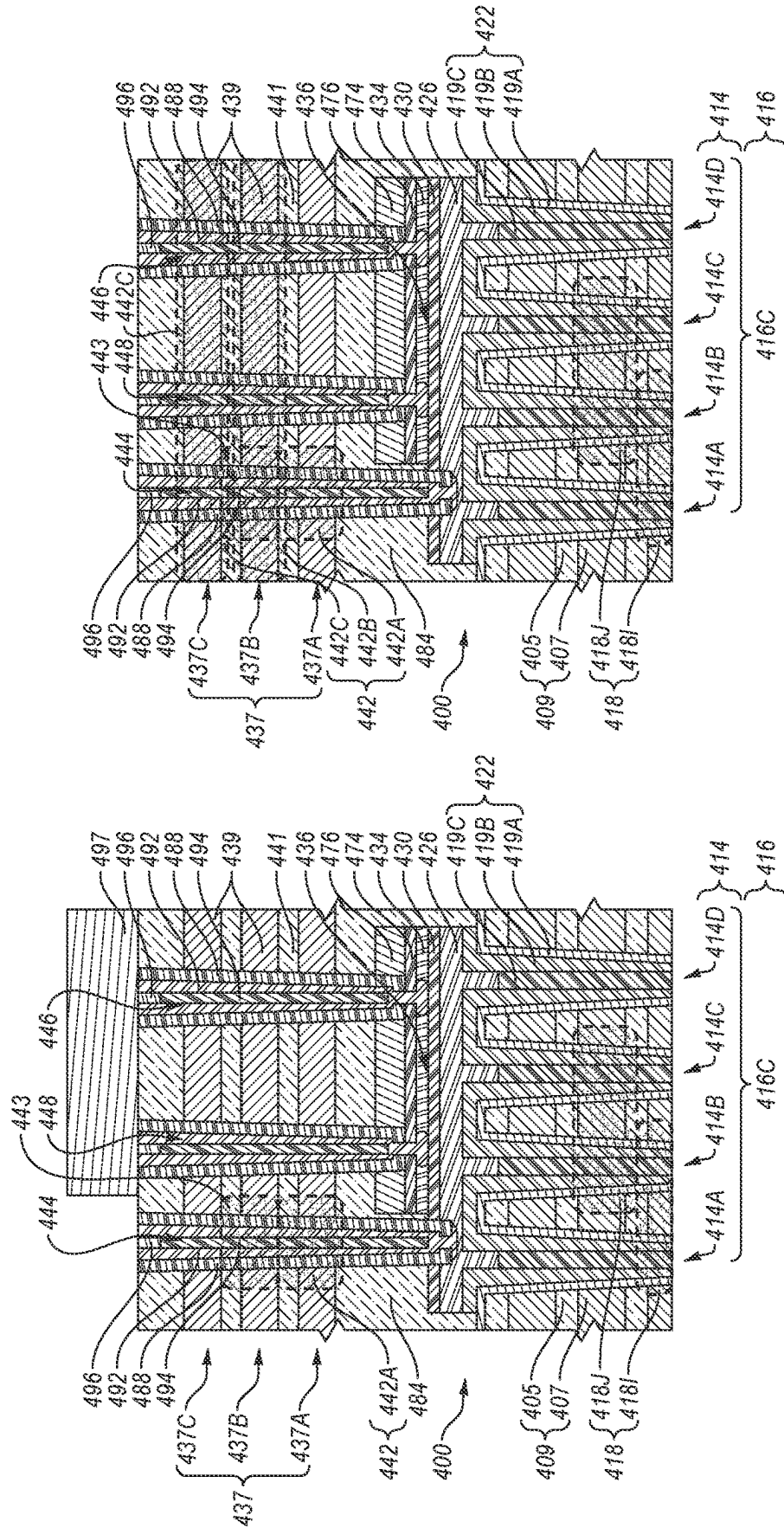


FIG. 4L

FIG. 4K

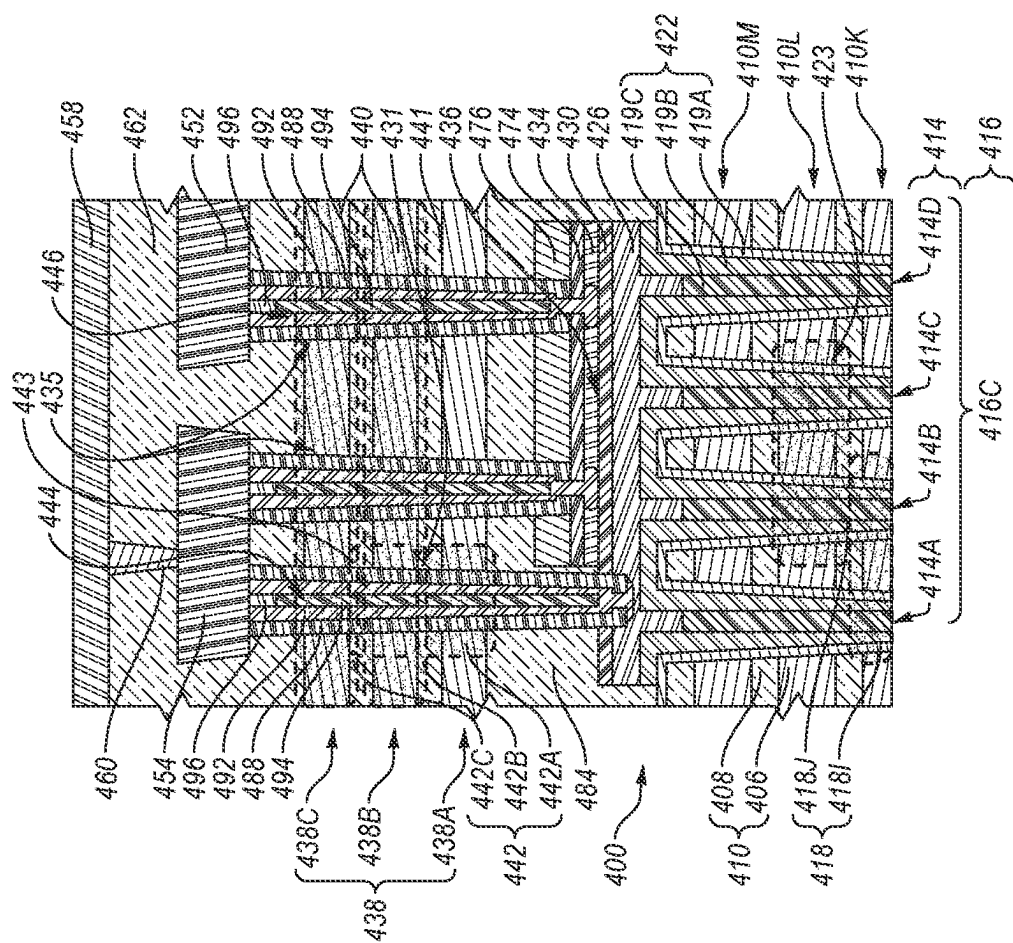
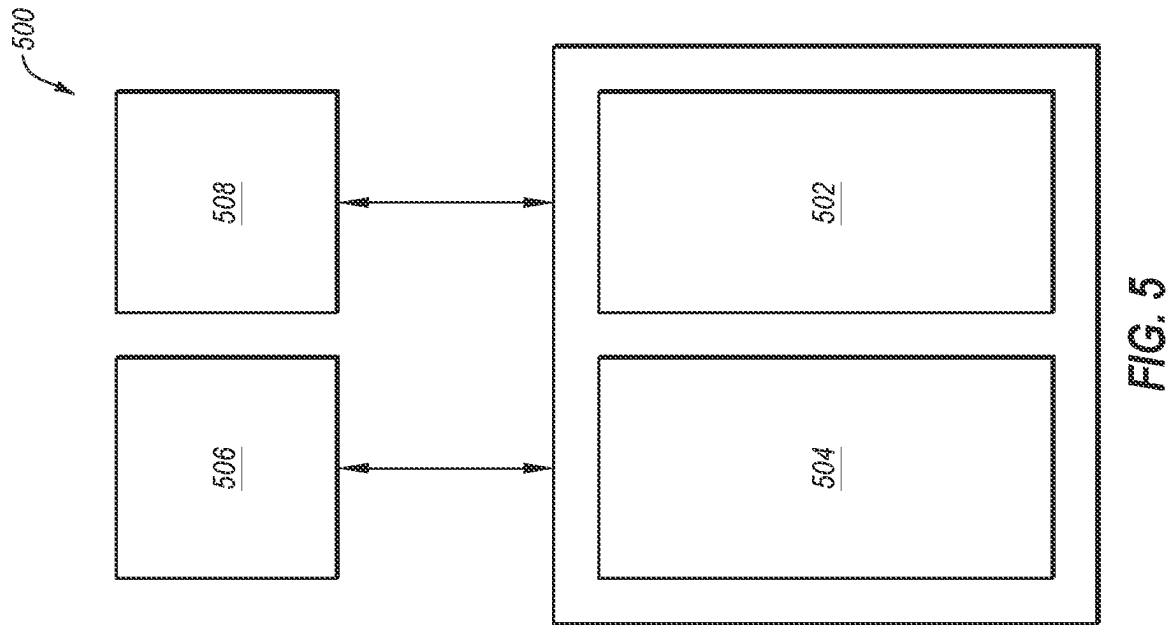


FIG. 4M



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MICROELECTRONIC DEVICES, AND RELATED MEMORY DEVICES AND ELECTRONIC SYSTEMS

TECHNICAL FIELD

The disclosure, in various embodiments, relates generally to the field of microelectronic device design and fabrication. More specifically, the disclosure relates to microelectronic devices including local digit line structures and global digit line structures, and to related memory devices, and electronic systems.

BACKGROUND

Microelectronic device designers often desire to increase the level of integration or density of features within a microelectronic device by reducing the dimensions of the individual features and by reducing the separation distance between neighboring features. In addition, microelectronic device designers often desire to design architectures that are not only compact, but offer performance advantages, as well as simplified, easier and less expensive to fabricate designs.

One example of a microelectronic device is a memory device. Memory devices are generally provided as internal integrated circuits in computers or other electronic devices. There are many types of memory devices including, but not limited to, non-volatile memory devices (e.g., NAND Flash memory devices). One way of increasing memory density in non-volatile memory devices is to utilize vertical memory array (also referred to as a “three-dimensional (3D) memory array”) architectures. A conventional vertical memory array includes strings of memory cells vertically extending through a stack structure including tiers of conductive structures and insulative materials. Each string of memory cells may include at least one select device coupled in series to a serial combination of vertically stacked memory cells. Such a configuration permits a greater number of switching devices (e.g., transistors) to be located in a unit of die area (e.g., length and width of active surface consumed) by building the array upwards (e.g., vertically) on a die, as compared to structures with conventional planar (e.g., two-dimensional (2D)) arrangements of transistors. However, conventional vertical memory array architectures may effectuate can hamper improvements in the performance (e.g., data transfer rates, power consumption) of the non-volatile memory device, and/or can impede reductions to the sizes (e.g., horizontal footprints) of features of the non-volatile memory device.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

FIGS. 1A and 1B are simplified, partial longitudinal cross-sectional (FIG. 1A) and simplified, partial top-down (FIG. 1B) views of a microelectronic device structure for a microelectronic device, in accordance with embodiments of the disclosure.

FIG. 2 is a simplified, partial top-down view of a microelectronic device structures for a microelectronic device, in accordance with additional embodiments of the disclosure.

FIG. 3 is simplified, partial longitudinal cross-sectional view of a microelectronic device structure for a microelectronic device, in accordance with further embodiments of the disclosure.

FIGS. 4A through 4M are simplified, partial longitudinal cross-sectional views of a microelectronic device structure

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at different processing stages of a method of forming a microelectronic device, in accordance with embodiments of the disclosure.

FIG. 5 is a schematic block diagram of an electronic system, in accordance with embodiments of the disclosure.

DETAILED DESCRIPTION

The following description provides specific details, such as material compositions, shapes, and sizes, in order to provide a thorough description of embodiments of the disclosure. However, a person of ordinary skill in the art would understand that the embodiments of the disclosure may be practiced without employing these specific details. Indeed, the embodiments of the disclosure may be practiced in conjunction with conventional microelectronic device fabrication techniques employed in the industry. In addition, the description provided below does not form a complete process flow for manufacturing a microelectronic device (e.g., a memory device, such as 3D NAND Flash memory device). The structures described below do not form a complete microelectronic device. Only those process acts and structures necessary to understand the embodiments of the disclosure are described in detail below. Additional acts to form a complete microelectronic device from the structures may be performed by conventional fabrication techniques.

Drawings presented herein are for illustrative purposes only, and are not meant to be actual views of any particular material, component, structure, device, or system. Variations from the shapes depicted in the drawings as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments described herein are not to be construed as being limited to the particular shapes or regions as illustrated, but include deviations in shapes that result, for example, from manufacturing. For example, a region illustrated or described as box-shaped may have rough and/or nonlinear features, and a region illustrated or described as round may include some rough and/or linear features. Moreover, sharp angles that are illustrated may be rounded, and vice versa. Thus, the regions illustrated in the figures are schematic in nature, and their shapes are not intended to illustrate the precise shape of a region and do not limit the scope of the present claims. The drawings are not necessarily to scale. Additionally, elements common between figures may retain the same numerical designation.

As used herein, a “memory device” means and includes microelectronic devices exhibiting memory functionality, but not necessary limited to memory functionality. Stated another way, and by way of non-limiting example only, the term “memory device” includes not only conventional memory (e.g., conventional non-volatile memory, such as conventional NAND memory; conventional volatile memory, such as conventional DRAM), but also includes an application specific integrated circuit (ASIC) (e.g., a system on a chip (SoC)), a microelectronic device combining logic and memory, and a graphics processing unit (GPU) incorporating memory.

As used herein, the term “configured” refers to a size, shape, material composition, orientation, and arrangement of one or more of at least one structure and at least one apparatus facilitating operation of one or more of the structure and the apparatus in a pre-determined way.

As used herein, the terms “vertical,” “longitudinal,” “horizontal,” and “lateral” are in reference to a major plane of a structure and are not necessarily defined by earth’s gravitational field. A “horizontal” or “lateral” direction is a

direction that is substantially parallel to the major plane of the structure, while a “vertical” or “longitudinal” direction is a direction that is substantially perpendicular to the major plane of the structure. The major plane of the structure is defined by a surface of the structure having a relatively large area compared to other surfaces of the structure. With reference to the figures, a “horizontal” or “lateral” direction may be perpendicular to an indicated “Z” axis, and may be parallel to an indicated “X” axis and/or parallel to an indicated “Y” axis; and a “vertical” or “longitudinal” direction may be parallel to an indicated “Z” axis, may be perpendicular to an indicated “X” axis, and may be perpendicular to an indicated “Y” axis.

As used herein, features (e.g., regions, structures, devices) described as “neighboring” one another means and includes features of the disclosed identity (or identities) that are located most proximate (e.g., closest to) one another. Additional features (e.g., additional regions, additional structures, additional devices) not matching the disclosed identity (or identities) of the “neighboring” features may be disposed between the “neighboring” features. Put another way, the “neighboring” features may be positioned directly adjacent one another, such that no other feature intervenes between the “neighboring” features; or the “neighboring” features may be positioned indirectly adjacent one another, such that at least one feature having an identity other than that associated with at least one of the “neighboring” features is positioned between the “neighboring” features. Accordingly, features described as “vertically neighboring” one another means and includes features of the disclosed identity (or identities) that are located most vertically proximate (e.g., vertically closest to) one another. Moreover, features described as “horizontally neighboring” one another means and includes features of the disclosed identity (or identities) that are located most horizontally proximate (e.g., horizontally closest to) one another.

As used herein, the term “intersection” means and includes a location at which two or more features (e.g., regions, structures, materials, devices) or, alternatively, two or more portions of a single feature meet. For example, an intersection between a first feature extending in a first direction (e.g., an X-direction) and a second feature extending in a second direction (e.g., a Y-direction) different than the first direction may be the location at which the first feature and the second feature meet.

As used herein, spatially relative terms, such as “beneath,” “below,” “lower,” “bottom,” “above,” “upper,” “top,” “front,” “rear,” “left,” “right,” and the like, may be used for ease of description to describe one element’s or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. Unless otherwise specified, the spatially relative terms are intended to encompass different orientations of the materials in addition to the orientation depicted in the figures. For example, if materials in the figures are inverted, elements described as “below” or “beneath” or “under” or “on bottom of” other elements or features would then be oriented “above” or “on top of” the other elements or features. Thus, the term “below” can encompass both an orientation of above and below, depending on the context in which the term is used, which will be evident to one of ordinary skill in the art. The materials may be otherwise oriented (e.g., rotated 90 degrees, inverted, flipped) and the spatially relative descriptors used herein interpreted accordingly.

As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

As used herein, “and/or” includes any and all combinations of one or more of the associated listed items.

As used herein, the phrase “coupled to” refers to structures operatively connected with each other, such as electrically connected through a direct Ohmic connection or through an indirect connection (e.g., by way of another structure).

As used herein, the term “substantially” in reference to a given parameter, property, or condition means and includes to a degree that one of ordinary skill in the art would understand that the given parameter, property, or condition is met with a degree of variance, such as within acceptable tolerances. By way of example, depending on the particular parameter, property, or condition that is substantially met, the parameter, property, or condition may be at least 90.0 percent met, at least 95.0 percent met, at least 99.0 percent met, at least 99.9 percent met, or even 100.0 percent met.

As used herein, “about” or “approximately” in reference to a numerical value for a particular parameter is inclusive of the numerical value and a degree of variance from the numerical value that one of ordinary skill in the art would understand is within acceptable tolerances for the particular parameter. For example, “about” or “approximately” in reference to a numerical value may include additional numerical values within a range of from 90.0 percent to 110.0 percent of the numerical value, such as within a range of from 95.0 percent to 105.0 percent of the numerical value, within a range of from 97.5 percent to 102.5 percent of the numerical value, within a range of from 99.0 percent to 101.0 percent of the numerical value, within a range of from 99.5 percent to 100.5 percent of the numerical value, or within a range of from 99.9 percent to 100.1 percent of the numerical value.

As used herein, “conductive material” means and includes electrically conductive material such as one or more of a metal (e.g., tungsten (W), titanium (Ti), molybdenum (Mo), niobium (Nb), vanadium (V), hafnium (Hf), tantalum (Ta), chromium (Cr), zirconium (Zr), iron (Fe), ruthenium (Ru), osmium (Os), cobalt (Co), rhodium (Rh), iridium (Ir), nickel (Ni), palladium (Pd), platinum (Pt), copper (Cu), silver (Ag), gold (Au), aluminum (Al)), an alloy (e.g., a Co-based alloy, an Fe-based alloy, an Ni-based alloy, an Fe- and Ni-based alloy, a Co- and Ni-based alloy, an Fe- and Co-based alloy, a Co- and Ni- and Fe-based alloy, an Al-based alloy, a Cu-based alloy, a magnesium (Mg)-based alloy, a Ti-based alloy, a steel, a low-carbon steel, a stainless steel), a conductive metal-containing material (e.g., a conductive metal nitride, a conductive metal silicide, a conductive metal carbide, a conductive metal oxide), and a conductively-doped semiconductor material (e.g., conductively-doped polysilicon, conductively-doped germanium (Ge), conductively-doped silicon germanium (SiGe)). In addition, a “conductive structure” means and includes a structure formed of and including conductive material.

As used herein, “insulative material” means and includes electrically insulative material, such one or more of at least one dielectric oxide material (e.g., one or more of a silicon oxide (SiO_x), phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, an aluminum oxide (AlO_x), a hafnium oxide (HfO_x), a niobium oxide (NbO_x), a titanium oxide (TiO_x), a zirconium oxide (ZrO_x), a tantalum oxide (TaO_x), and a magnesium oxide (MgO_x)), at least one dielectric nitride material (e.g., a silicon nitride (SiN_y)), at least one dielectric oxynitride material (e.g., a silicon oxynitride (SiO_xN_y)), at least one dielectric oxycarbide material (e.g., silicon oxycarbide (SiO_xC_y)), at least one hydrogenated dielectric oxycarbide material (e.g., hydroge-

nated silicon oxycarbide ($\text{SiC}_x\text{O}_y\text{H}_z$), and at least one dielectric carboxynitride material (e.g., a silicon carboxynitride ($\text{SiO}_x\text{C}_y\text{N}_z$)). Formulae including one or more of “x,” “y,” and “z” herein (e.g., SiO_x , AlO_x , HfO_x , NbO_x , TiO_x , SiN_y , SiO_xN_y , SiO_xC_y , $\text{SiC}_x\text{O}_y\text{H}_z$, $\text{SiO}_x\text{C}_y\text{N}_z$) represent a material that contains an average ratio of “x” atoms of one element, “y” atoms of another element, and “z” atoms of an additional element (if any) for every one atom of another element (e.g., Si, Al, Hf, Nb, Ti). As the formulae are representative of relative atomic ratios and not strict chemical structure, an insulative material may comprise one or more stoichiometric compounds and/or one or more non-stoichiometric compounds, and values of “x,” “y,” and “z” (if any) may be integers or may be non-integers. As used herein, the term “non-stoichiometric compound” means and includes a chemical compound with an elemental composition that cannot be represented by a ratio of well-defined natural numbers and is in violation of the law of definite proportions. In addition, an “insulative structure” means and includes a structure formed of and including insulative material.

As used herein, the term “semiconductor material” refers to a material having an electrical conductivity between those of insulative materials and conductive materials. For example, a semiconductor material may have an electrical conductivity of between about 10^{-8} Siemens per centimeter (S/cm) and about 10^4 S/cm (10^6 S/m) at room temperature. Examples of semiconductor materials include elements found in column IV of the periodic table of elements such as silicon (Si), germanium (Ge), and carbon (C). Other examples of semiconductor materials include compound semiconductor materials such as binary compound semiconductor materials (e.g., gallium arsenide (GaAs)), ternary compound semiconductor materials (e.g., $\text{Al}_x\text{Ga}_{1-x}\text{As}$), and quaternary compound semiconductor materials (e.g., $\text{Ga}_x\text{In}_{1-x}\text{As}_y\text{P}_{1-y}$), without limitation. Compound semiconductor materials may include combinations of elements from columns III and V of the periodic table of elements (III-V semiconductor materials) or from columns II and VI of the periodic table of elements (II-VI semiconductor materials), without limitation. Further examples of semiconductor materials include oxide semiconductor materials such as zinc tin oxide ($\text{Zn}_x\text{Sn}_y\text{O}$, commonly referred to as “ZTO”), indium zinc oxide ($\text{In}_x\text{Zn}_y\text{O}$, commonly referred to as “IZO”), zinc oxide (Zn_xO), indium gallium zinc oxide ($\text{In}_x\text{Ga}_y\text{Zn}_z\text{O}$, commonly referred to as “IGZO”), indium gallium silicon oxide ($\text{In}_x\text{Ga}_y\text{Si}_z\text{O}$, commonly referred to as “IGSO”), indium tungsten oxide ($\text{In}_x\text{W}_y\text{O}$, commonly referred to as “IWO”), indium oxide (In_xO), tin oxide (Sn_xO), titanium oxide (Ti_xO), zinc oxide nitride (Zn_xON_z), magnesium zinc oxide ($\text{Mg}_x\text{Zn}_y\text{O}$), zirconium indium zinc oxide ($\text{Zr}_x\text{In}_y\text{Zn}_z\text{O}$), hafnium indium zinc oxide ($\text{Hf}_x\text{In}_y\text{Zn}_z\text{O}$), tin indium zinc oxide ($\text{Sn}_x\text{In}_y\text{Zn}_z\text{O}$), aluminum tin indium zinc oxide ($\text{Al}_x\text{Sn}_y\text{In}_z\text{Zn}_d\text{O}$), silicon indium zinc oxide ($\text{Si}_x\text{In}_y\text{Zn}_z\text{O}$), aluminum zinc tin oxide ($\text{Al}_x\text{Zn}_y\text{Sn}_z\text{O}$), gallium zinc tin oxide ($\text{Ga}_x\text{Zn}_y\text{Sn}_z\text{O}$), zirconium zinc tin oxide ($\text{Zr}_x\text{Zn}_y\text{Sn}_z\text{O}$), and other similar materials.

As used herein, the term “homogeneous” means relative amounts of elements included in a feature (e.g., a region, a structure, a material) do not vary throughout different portions (e.g., different horizontal portions, different vertical portions) of the feature. Conversely, as used herein, the term “heterogeneous” means relative amounts of elements included in a feature (e.g., a material, a structure) vary throughout different portions of the feature. If a feature is heterogeneous, amounts of one or more elements included in the feature may vary stepwise (e.g., change abruptly), or

may vary continuously (e.g., change progressively, such as linearly, parabolically) throughout different portions of the feature. The feature may, for example, be formed of and include a stack of at least two different materials.

Unless the context indicates otherwise, the materials described herein may be formed by any suitable technique including, but not limited to, spin coating, blanket coating, chemical vapor deposition (CVD), plasma enhanced CVD (PECVD), atomic layer deposition (ALD), plasma enhanced ALD (PEALD), physical vapor deposition (PVD) (e.g., sputtering), or epitaxial growth. Depending on the specific material to be formed, the technique for depositing or growing the material may be selected by a person of ordinary skill in the art. In addition, unless the context indicates otherwise, removal of materials described herein may be accomplished by any suitable technique including, but not limited to, etching (e.g., dry etching, wet etching, vapor etching), ion milling, abrasive planarization (e.g., chemical-mechanical planarization (CMP)), or other known methods.

FIG. 1A is a simplified, partial longitudinal cross-sectional view of a microelectronic device structure **100** (e.g., a memory device structure, such as a 3D NAND Flash memory device structure) for a microelectronic device (e.g., a memory device, such as a 3D NAND Flash memory device), in accordance with embodiments of the disclosure. FIG. 1B is simplified, partial top-down view of the microelectronic device structure **100** depicted in FIG. 1A, wherein the view depicted in FIG. 1A is about dashed line A_1 - A_1 depicted in FIG. 1B. For clarity and ease of understanding of the drawings and related description, not all features (e.g., regions, structures, materials, devices) of the microelectronic device structure **100** depicted in one of FIGS. 1A and 1B are depicted in the other of FIGS. 1A through 1B.

Referring to FIG. 1A, the microelectronic device structure **100** may be formed to include a base structure **102**; a stack structure **104** vertically overlying the base structure **102**, and including pillar structures **122** vertically extending there-through; a select gate drain (SGD) plug tier **127** vertically overlying the stack structure **104**; a digit line tier **156** vertically overlying the SGD plug tier **127**; selector tiers **138** vertically interposed between the SGD plug tier **127** and the digit line tier **156**; and a conductive routing tier **150** vertically interposed between the digit line tier **156** and the selector tiers **138**. As described in further detail below, the microelectronic device structure **100** includes various features (e.g., regions, structures, materials, devices) individually operatively associated with (e.g., within; extending to, into, through, and/or between; physically and/or electrically connected to additional features of) one or more of the base structure **102**, the stack structure **104**, the SGD plug tier **127**, the digit line tier **156**, the selector tiers **138**, and the conductive routing tier **150**.

The base structure **102** may comprise a base material or construction upon which additional features (e.g., materials, structures, devices) of the microelectronic device structure **100** are formed. The base structure **102** may comprise one or more of semiconductor material, conductive material, and insulative material. The base structure **102** may include an arrangement of different materials, different structures, and/or different regions. In some embodiments, the base structure **102** includes various circuitry (e.g., logic circuitry) therein.

The stack structure **104** of the microelectronic device structure **100** may be formed to include a vertically alternating sequence of conductive structures **106** and insulative structures **108** arranged in tiers **110**. The conductive struc-

tures **106** may be vertically interleaved with the insulative structures **108**. Each of the tiers **110** of the stack structure **104** may include at least one of the conductive structures **106** vertically neighboring at least one of the insulative structures **108**. The stack structure **104** may be formed to include any desired quantity of the tiers **110**, such as greater than or equal to sixteen (16) of the tiers **110**, greater than or equal to thirty-two (32) of the tiers **110**, greater than or equal to sixty-four (64) of the tiers **110**, greater than or equal to one hundred twenty-eight (128) of the tiers **110**, or greater than or equal to two hundred fifty-six (256) of the tiers **110**.

The conductive structures **106** of the tiers **110** of the stack structure **104** may be formed of and include conductive material. By way of non-limiting example, the conductive structures **106** may each individually be formed of and include a metallic material comprising one or more of at least one metal, at least one alloy, and at least one conductive metal-containing material (e.g., a conductive metal nitride, a conductive metal silicide, a conductive metal carbide, a conductive metal oxide). In some embodiments, the conductive structures **106** are formed of and include W. Each of the conductive structures **106** may individually be substantially homogeneous, or one or more of the conductive structures **106** may individually be substantially heterogeneous. In some embodiments, each of the conductive structures **106** is formed to be substantially homogeneous.

Optionally, one or more liner materials(s) (e.g., insulative liner material(s), conductive liner material(s)) may also be formed around the conductive structures **106**. The liner material(s) may, for example, be formed of and include one or more a metal (e.g., titanium, tantalum), an alloy, a metal nitride (e.g., tungsten nitride, titanium nitride, tantalum nitride), and a metal oxide (e.g., aluminum oxide). In some embodiments, the liner material(s) comprise at least one conductive material employed as a seed material for the formation of the conductive structures **106**. In some embodiments, the liner material(s) comprise titanium nitride. In further embodiments, the liner material(s) further include aluminum oxide. As a non-limiting example, aluminum oxide may be formed directly adjacent the insulative structures **108**, titanium nitride may be formed directly adjacent the aluminum oxide, and tungsten may be formed directly adjacent the titanium nitride. For clarity and ease of understanding the description, the liner material(s) are not illustrated in FIGS. 1A and 1B, but it will be understood that the liner material(s) may be disposed around the conductive structures **106**.

The insulative structures **108** of the tiers **110** of the stack structure **104** may be formed of and include insulative material, such one or more of at least one dielectric oxide material (e.g., one or more of SiO_x , phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, AlO_x , HfO_x , NbO_x , TiO_x , ZrO_x , TaO_x , and MgO_x), at least one dielectric nitride material (e.g., SiN_y), at least one dielectric oxynitride material (e.g., SiO_xN_y), and at least one dielectric carboxynitride material (e.g., $\text{SiO}_x\text{C}_z\text{N}_y$). In some embodiments, each of the insulative structures **108** is formed of and includes a dielectric oxide material, such as SiO_x (e.g., SiO_2). Each of the insulative structures **108** may individually be substantially homogeneous, may be substantially heterogeneous. In some embodiments, each of the insulative structures **108** is substantially homogeneous.

Referring to FIG. 1B, the stack structure **104** may be divided (e.g., separated, partitioned) into blocks **164** separated from one another by dielectric slot structures **166** (e.g., dielectric-filled slots, dielectric-filled openings). The dielectric slot structures **166** may vertically extend (e.g., in the

Z-direction) completely through the stack structure **104**. The blocks **164** of the stack structure **104** may be formed to horizontally extend parallel in an X-direction. As used herein, the term “parallel” means substantially parallel. Horizontally neighboring blocks **164** of the stack structure **104** may be separated from one another in a Y-direction orthogonal to the X-direction by the dielectric slot structures **166**. The dielectric slot structures **166** may also horizontally extend parallel in the X-direction. Each of the blocks **164** of the stack structure **104** may exhibit substantially the same geometric configuration (e.g., substantially the same dimensions and substantially the same shape) as each other of the blocks **164**, or one or more of the blocks **164** may exhibit a different geometric configuration (e.g., one or more different dimensions and/or a different shape) than one or more other of the blocks **164**. In addition, each pair of horizontally neighboring blocks **164** of the stack structure **104** may be horizontally separated from one another by substantially the same distance (e.g., corresponding to a width in the Y-direction of each of the dielectric slot structures **166**) as each other pair of horizontally neighboring blocks **164** of the stack structure **104**, or at least one pair of horizontally neighboring blocks **164** of the stack structure **104** may be horizontally separated from one another by a different distance than that separating at least one other pair of horizontally neighboring blocks **164** of the stack structure **104**. In some embodiments, the blocks **164** of the stack structure **104** are substantially uniformly (e.g., substantially non-variably, substantially equally, substantially consistently) sized, shaped, and spaced relative to one another.

Still referring to FIG. 1B, each block **164** of the stack structure **104** may be sub-divided into multiple (e.g., a plurality of, more than one) sub-blocks **114**. The sub-blocks **114** of an individual block **164** may horizontally extend parallel in the X-direction. Each of the sub-blocks **114** may individually be operatively associated with at least one row of the pillar structures **122** extending the X-direction. For example, as depicted in FIG. 1B, an individual block **164** of the stack structure **104** may include sixteen (16) sub-blocks **114** operatively associated with sixteen (16) rows of the pillar structures **122**. In addition, for an individual block **164**, multiple (e.g., a plurality of, more than one) sub-blocks **114** thereof may be grouped together with one another within sub-block groups **116**. As depicted in FIG. 1B, an individual block **164** of the stack structure **104** may, for example, include four (4) sub-block groups **116** each including four (4) of the sub-blocks **114**. By way of non-limiting example, the sub-block groups **116** may include a first sub-block group **116A**, a second sub-block group **116B**, a third sub-block group **116C**, and a fourth sub-block group **116D**; and each of the sub-block groups **116** may individually include a first sub-block **114A**, a second sub-block **114B**, a third sub-block **114C**, and a fourth sub-block **114D**. Sub-block groups **116** horizontally neighboring one another in the Y-direction within an individual block **164** may exhibit an inverse horizontal arrangement of the different sub-blocks **114** thereof relative to one another. For example, the fourth sub-block **114D** of the first sub-block group **116A** may be most horizontally proximate the fourth sub-block **114D** of the second sub-block group **116B** horizontally neighboring the first sub-block group **116A**; the first sub-block **114A** of the second sub-block group **116B** may be most horizontally proximate to the first sub-block **114A** of the third sub-block group **116C** horizontally neighboring the second sub-block group **116B**; and the fourth sub-block **114D** of the third sub-block group **116C** may be most

horizontally proximate to the fourth sub-block **114D** of the fourth sub-block group **116D** horizontally neighboring the third sub-block group **116C**.

While FIG. 1B depicts an individual block **164** of the stack structure **104** as including sixteen (16) sub-blocks **114** and four (4) sub-block groups **116** each including a different four (4) of the sixteen (16) sub-blocks **114**, in additional embodiments one or more (e.g., each) of the blocks **164** of the stack structure **104** includes a different quantity of sub-blocks **114**, a different quantity of sub-block groups **116**, and/or a different quantity of sub-blocks **114** within an individual sub-block group **116** thereof. For example, an individual block **164** may include greater than sixteen (16) sub-blocks **114** or less than sixteen (16) sub-blocks **114**; may include greater than four (4) sub-block groups **116** or less than four (4) sub-block groups **116**; and/or may include greater than four (4) sub-blocks **114** in an individual sub-block group **116** or less than four (4) sub-blocks **114** in an individual sub-block group **116**. For an individual block **164** of the stack structure **104**, the quantity of sub-blocks **114**, the quantity of sub-block groups **116**, and the quantity of sub-blocks **114** per sub-block group **116** may be selected, at least in part, based on the horizontal area of the block **164** as well as the horizontal areas and horizontal positions of the pillar structures **122** located within the block **164**.

Referring collectively to FIGS. 1A and 1B, it will be recognized that the dashed line A_1 - A_1 depicted in FIG. 1B, which identifies the position and orientation of the simplified, partial longitudinal cross-sectional view of the microelectronic device structure **100** shown in FIG. 1A, horizontally extends diagonal to each of the X-direction (e.g., first horizontal direction) and the Y-direction (e.g., second horizontal direction orthogonal to the first horizontal direction) shown in FIG. 1A. Put another way, the horizontal orientation of the dashed line A_1 - A_1 depicted in FIG. 1B (and, hence, the longitudinal cross-sectional view of FIG. 1A), is acutely angled relative to each of the X-direction and the Y-direction shown in FIG. 1B. Accordingly, it will be appreciated that the horizontal orientations some features (e.g., some structures, some devices, some regions) of the microelectronic device structure **100** depicted in FIG. 1A and described in further details below, such as (but without limitation) some features within a sub-section B_1 shown in FIGS. 1A and 1B, are also acutely angled relative to each of the X-direction and the Y-direction shown in FIG. 1B.

Referring to FIG. 1A, the tiers **110** of the stack structure **104** may be grouped into different tier sections **111**. The tier sections **111** may include an access line tier section **111A**, a stacked select gate drain (SGD) tier section **111B** overlying the access line tier section **111A**, and a sense node tier section **111C** overlying the stacked SGD tier section **111B**. In addition, the tier sections **111** of the stack structure **104** may further include a source side select (SGS) gate tier section **111D** underlying the access line tier section **111A**. As described in further details below, during use and operation of a microelectronic device including the microelectronic device structure **100**, at least some of the tiers **110** within the stacked SGD tier section **111B** may be employed to select different sub-blocks **114** within individual blocks **164** (FIG. 1B) of the stack structure **104**; and at least some other of the tiers **110** within the sense node tier section **111C** may be employed for capacitive-sense operations (amongst other operations) for the individual blocks **164** (FIG. 1B) of the stack structure **104**.

As shown in FIG. 1A, a first group of the tiers **110** of the stack structure **104** within the access line tier section **111A** may include an active access line tier **110A**, and a dummy

access line tier **110B** overlying the active access line tier **110A**. For an individual block **164** (FIG. 1B) of the stack structure **104**, a conductive structure **106** of the active access line tier **110A** may be employed as a so-called “active” access line structure facilitating electrical communication between two or more components (e.g., memory cells, string drivers) of a microelectronic device including the microelectronic device structure **100**; and a conductive structure **106** of the dummy access line tier **110B** may be employed as a so-called “dummy” access line structure that does not facilitate electrical communication between two or more components (e.g., memory cells, string drivers) of a microelectronic device including the microelectronic device structure **100**. While FIG. 1A only depicts the access line tier section **111A** as including one (1) active access line tier **110A** and one (1) dummy access line tier **110B** the disclosure is not so limited. Rather, the access line tier section **111A** may include greater than one active access line tier **110A** (e.g., greater than or equal to eight (8) active access line tiers **110A**, greater than or equal to sixteen (16) active access line tiers **110A**, greater than or equal to thirty-two (32) active access line tiers **110A**, greater than or equal to sixty-four (64) active access line tiers **110A**, greater than or equal to one-hundred twenty-eight (128) active access line tiers **110A**, greater than or equal to two-hundred fifty-six (256) active access line tiers **110A**); and/or greater than one dummy access line tier **110B** (e.g., greater than or equal to two (2) dummy access line tiers **110B**).

Still referring to FIG. 1A, a second group of the tiers **110** of the stack structure **104** within the stacked SGD tier section **111B** may include a program-inhibit SGD tier **110C** overlying the dummy access line tier **110B**, a first SGD bar tier **110D** (e.g., a first “complementary” SGD tier) overlying the program-inhibit SGD tier **110C**, a second SGD bar tier **110E** (e.g., a second “complementary” SGD tier) overlying the first SGD bar tier **110D**, a first SGD tier **110F** (e.g., a first “true” SGD tier) overlying the second SGD bar tier **110E**, a second SGD tier **110G** (e.g., a second “true” SGD tier) overlying the first SGD tier **110F**, and a read-amplification SGD tier **110H** overlying the second SGD tier **110G**. For an individual block **164** (FIG. 1B) of the stack structure **104**, a conductive structure **106** of the program-inhibit SGD tier **110C** may be employed as a program-inhibit SGD structure; a conductive structure **106** of the first SGD bar tier **110D** may be employed as a first SGD bar structure; a conductive structure **106** of the second SGD bar tier **110E** may be employed as a second SGD bar structure; a conductive structure **106** of the first SGD tier **110F** may be employed as a first SGD structure; a conductive structure **106** of the second SGD tier **110G** may be employed as a second SGD structure; and a conductive structure **106** of the read-amplification SGD tier **110H** may be employed as a read-amplification SGD structure.

With continued reference to FIG. 1A, a third group of the tiers **110** of the stack structure **104** within the sense node tier section **111C** may include a first select gate programming (SGP) bar tier **110I** (e.g., a first “complementary” SGP tier) overlying the read-amplification SGD tier **110H**, a second SGP bar tier **110J** (e.g., a second “complementary” SGP tier) overlying the first SGP bar tier **110I**, a first SGP tier **110K** (e.g., a first “true” SGP tier) overlying the second SGP bar tier **110J**, a second SGP tier **110L** (e.g., a second “true” SGP tier) overlying the first SGP tier **110K**, and a gate-induced drain-leakage (GIDL) generation (GG) tier **110M** overlying the second SGP tier **110L**. For an individual block **164** (FIG. 1B) of the stack structure **104**, a conductive structure **106** of the first SGP bar tier **110I** may be employed as a first SGP

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bar structure; a conductive structure **106** of the second SGP bar tier **110J** may be employed as a second SGD bar structure; a conductive structure **106** of the first SGP tier **110K** may be employed as a first SGP structure; a conductive structure **106** of the second SGP tier **110L** may be employed as a second SGP structure; and a conductive structure **106** of the GG tier **110M** may be employed as a GG structure. As described in further detail below, portions of different tiers **110** (e.g., the first SGP bar tier **110I**, the second SGP bar tier **110J**, the first SGP tier **110K**, the second SGP tier **110L**) within the stacked SGD tier section **111B** may be doped with one or more conductivity-enhancing species (e.g., N-type dopant, P-type dopant) that permit horizontally neighboring select transistors within vertical boundaries of an individual tier **110** of the sense node tier section **111C** to have different threshold voltage (V_t) properties than one another. The third group of the tiers **110** of the stack structure **104** within the sense node tier section **111C** facilitate associating different pillar structures **122** vertically extending through an individual block **164** (FIG. 1B) with different sub-blocks **114** of the block **164** (FIG. 1B) by programming selected transistors at vertical elevations of different tiers **110** within the sense node tier section **111C** to desired V_t levels, as described in further detail below.

Optionally, one or more additional “dummy” tiers (e.g., in addition to the dummy access line tiers **110B**) may be included within the stack structure **104**, vertically between two or more other of the tiers **110**. A conductive structure **106** of an individual additional dummy tier may be employed as an additional so-called “dummy” structure that does not facilitate electrical communication between two or more components of a microelectronic device including the microelectronic device structure **100**. If included, additional dummy tier(s) may be utilized to mitigate so-called “trap-up” and “downshift” of V_t by relaxing the electric field between devices in ON and OFF state. An additional dummy tier may be biased at intermediate potential to mitigate (e.g., avoid) a too abrupt potential difference between the tiers at HIGH and LOW biases. As a non-limiting example, an additional dummy tier may be vertically interposed between the program-inhibit SGD tier **110C** and the first SGD bar tier **110D**. As another non-limiting example, an additional dummy tier may be vertically interposed between the second SGD tier **110G** and the read-amplification SGD tier **110H**. As a further non-limiting example, an additional dummy tier may be vertically interposed between the read-amplification SGD tier **110H** and the first SGP bar tier **110I**.

Still referring to FIG. 1A, the pillar structures **122** may vertically extend completely through the tiers **110** of the stack structure **104**. For example, the pillar structures **122** may individually extend through each of the sense node tier section **111C**, the stacked SGD tier section **111B**, the access line tier section **111A**, and the SGS gate tier section **111D**, and to or into the base structure **102**. As described in further detail below, the pillar structures **122** may each individually be formed of and include a stack of materials. By way of non-limiting example, each of the pillar structures **122** may be formed to include a charge-blocking material, such as first dielectric oxide material (e.g., SiO_x , such as SiO_2 ; AlO_x , such as Al_2O_3); a charge-trapping material, such as a dielectric nitride material (e.g., SiN_x , such as Si_3N_4); a gate dielectric material, such as a second dielectric oxide material (e.g., SiO_x , such as SiO_2); a channel material, such as a semiconductor material (e.g., silicon, such as polycrystalline silicon); and a dielectric fill material (e.g., a dielectric oxide, a dielectric nitride, air). The charge-blocking material may be formed on or over surfaces of the conductive structures

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106 and the insulative structures **108** of the tiers **110** of stack structure **104** at least partially defining horizontal boundaries of the pillar structures **122**; the charge-trapping material may be horizontally surrounded by the charge-blocking material; the gate dielectric material may be horizontally surrounded by the charge-trapping material; the channel material may be horizontally surrounded by the gate dielectric material; and the dielectric fill material may be horizontally surrounded by the channel material.

The pillar structures **122** may individually exhibit a desirable geometric configuration (e.g., dimensions, shape), and may also be distributed relative to one another in a desirable manner within horizontal areas of the blocks **164** (FIG. 1B), of the stack structure **104**. The pillar structures **122** may individually exhibit a critical dimension (CD) (e.g., maximum horizontal dimension) less than about 120 nanometers (nm), such as less than or equal to about 110 nm, less than or equal to about 100 nm, less than or equal to about 90 nm, or less than or equal to about 80 nm. In some embodiments, the pillar structures **122** individually exhibit a CD within a range of from about 70 nm to about 80 nm. A pitch between pillar structures **122** horizontally neighboring one another may be less than about 140 nm, such as less than or equal to about 130 nm, less than or equal to about 120 nm, less than or equal to about 110 nm, or less than or equal to about 100 nm. In some embodiments, a pitch between pillar structures **122** horizontally neighboring one another is within a range of from about 90 nm to about 100 nm. A taper of each pillar structure **122**, as defined by the difference between a largest horizontal dimension of the pillar structure **122** (e.g., at an upper vertical boundary thereof) and a smallest horizontal dimension of the pillar structure **122** (e.g., at a lower vertical boundary thereof), may be less than or equal to about 20 nm, such as less than or equal to about 19 nm, or less than or equal to about 18 nm. In some embodiments, a taper of each pillar structure **122** is within a range of from about 16 nm to about 18 nm. In addition, as shown in FIG. 1B, in some embodiments, each block **164** of the stack structure **104** exhibits a hexagonal distribution (e.g., a hexagonal pattern) of the pillar structures **122**. As described in further detail below, the horizontal dimensions and pitch of the pillar structures **122** may respectively be relatively smaller than those for conventional pillar structures as a result of lower sense current (I_{sense}) requirements facilitated by the configuration of the microelectronic device structure **100** relative to conventional configurations.

Referring to FIG. 1A, intersections of the pillar structures **122** and the conductive structures **106** of the active access line tiers (e.g., the active access line tier **110A**) within the access line tier section **111A** of the stack structure **104** may define vertically extending strings of memory cells **121** coupled in series with one another within the stack structure **104**. As shown in FIG. 1A, intersections of the pillar structures **122** and the conductive structure **106** of the active access line tier **110A** may define memory cells **121** at the horizontal positions of the pillar structures **122** and at the vertical position of the conductive structure **106**. Intersections of the pillar structures **122** and the conductive structures **106** of additional active access line tiers vertically underling the active access line tier **110A** may define more of the memory cells **121** at the horizontal positions of the pillar structures **122** and at the vertical positions of the conductive structures **106** of the additional active access line tiers. In some embodiments, the memory cells **121** formed at the intersections of the conductive structures **106** of the active access line tiers (e.g., the active access line tier **110A**) and the pillar structures **122** within access line tier section

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111A comprise so-called “MONOS” (metal-oxide-nitride-oxide-semiconductor) memory cells. In additional embodiments, the memory cells 121 comprise so-called “TANOS” (tantalum nitride-aluminum oxide-nitride-oxide-semiconductor) memory cells, or so-called “BETANOS” (band/barrier engineered TANOS) memory cells, each of which are subsets of MONOS memory cells. In further embodiments, the memory cells 121 comprise so-called “floating gate” memory cells including floating gates (e.g., metallic floating gates) as charge storage structures. The floating gates may horizontally intervene between central structures of the pillar structures 122 and the conductive structures 106 of the different active access line tiers (e.g., the active access line tier 110A) of the stack structure 104. The vertically extending strings of memory cells 121 together form at least one memory array within the stack structure 104.

Intersections of the pillar structures 122 and the conductive structures 106 within the tiers 110 of the sense node tier section 111C, the stacked SGD tier section 111B, and the SGS gate tier section 111D of the stack structure 104 may define different select transistors 123 coupled in series with the vertically extending strings of memory cells 121. As described in further detail below, the select transistors 123 within the sense node tier section 111C, the stacked SGD tier section 111B, and the SGS gate tier section 111D of the stack structure 104 may be employed for various memory array operations (e.g., program operations, erase operations, read operations) during use and operation of a microelectronic device including the microelectronic device structure 100.

Still referring to FIG. 1A, the microelectronic device structure 100 includes different threshold voltage (V_t)-enhanced sections 118 at different vertical and horizontal positions therein. By way of non-limiting example, within horizontal areas of the blocks 164 (FIG. 1B), the microelectronic device structure 100 may include V_t -enhanced sections 118 at vertical positions of at least some of the tiers 110 of the stack structure 104 within the stacked SGD tier section 111B and the sense node tier section 111C. Some of the V_t -enhanced sections 118 may include portions of a tier 110 of the stack structure 104 doped with at least one conductivity-enhancing species (e.g., at least one P-type dopant), as well as portions of one or more pillar structures 122 doped with the at least one conductivity-enhancing species. Some other of the V_t -enhanced sections 118 may include portions of a tier 110 of the stack structure 104 not doped with at least one conductivity-enhancing species, and portions of one or more pillar structures 122 also not doped with the at least one conductivity-enhancing species. Even in the absence of the conductivity-enhancing species within boundaries thereof, an individual V_t -enhanced section 118 may be considered to be “ V_t -enhanced” as a result of device (e.g., transistor) V_t characteristics facilitated within the boundaries thereof by way of configurations of additional features (e.g., SGD plug structures) of the microelectronic device structure 100 in conjunction with programming operations performed for a microelectronic device including microelectronic device structure 100, as described in further detail below. For example, a select transistor 123 associated with a portions of pillar structure 122 within vertical and horizontal boundaries of an individual V_t -enhanced section 118 substantially free of conductivity-enhancing species may be provided with enhanced V_t characteristics due, in part, to a material composition (e.g., N-type doped semiconductive material) of an SGD plug structure coupled to the pillar structure 122, as described in further detail below. Within a horizontal area of an individual block 164 (FIG. 1B) of the stack structure 104, each V_t -enhanced section 118

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may individually vertically overlap at least the conductive structure 106 of one (1) of the tiers 110 of the stack structure 104, and may encompass at least two (2) horizontally neighboring pillar structures 122 within different sub-blocks 114 than one another. An individual V_t -enhanced section 118 may be substantially confined within horizontal boundaries of an individual sub-block group 116 of the block 164 (FIG. 1B), or may horizontally extend at least partially across and between at least two (2) horizontally neighboring sub-block groups 116 of the block 164 (FIG. 1B).

The V_t -enhanced sections 118 of the microelectronic device structure 100 may be used to provide different select transistors 123 within the stacked SGD tier section 111B and the sense node tier section 111C of the stack structure 104 with desired V_t characteristics. For example, within an individual block 164 (FIG. 1B) of the stack structure 104, the V_t -enhanced sections 118 may be configured such that some horizontally neighboring select transistors 123 substantially vertically aligned with one another (e.g., within the same tier 110 as one another) and operatively associated with pillar structures 122 in different sub-blocks 114 of the block 164 (FIG. 1B) than one another have different V_t characteristics (e.g., are set to different V_t levels) than one another; and such that some other horizontally neighboring select transistors 123 substantially vertically aligned with one another and operatively associated with pillar structures 122 in different sub-blocks 114 of the block 164 (FIG. 1B) than one another have substantially the same V_t characteristics (e.g., are set to the same V_t levels) as one another. As another example, within an individual block 164 (FIG. 1B) of the stack structure 104, the V_t -enhanced sections 118 may be configured such that some vertically neighboring select transistors 123 operatively associated with the same pillar structure 122 as one another have substantially the same V_t characteristics (e.g., are set to the same V_t levels) as one another; and such that some other vertically neighboring select transistors 123 operatively associated with the same pillar structure 122 as one another have different V_t properties (e.g., are set to different V_t levels) than one another. As an additional example, within an individual block 164 (FIG. 1B) of the stack structure 104, the V_t -enhanced sections 118 may be configured such that some select transistors 123 vertically and horizontally neighboring one another and operatively associated with pillar structures 122 in different sub-blocks 114 of the block 164 (FIG. 1B) than one another have different V_t properties (e.g., are set to different V_t levels) than one another; and such that some other select transistors 123 vertically and horizontally neighboring one another and operatively associated with pillar structures 122 in different sub-blocks 114 of the block 164 (FIG. 1B) than one another have substantially the same V_t properties (e.g., are set to the same V_t levels) as one another.

As shown in FIG. 1A, in some embodiments, the V_t -enhanced sections 118 of the microelectronic device structure 100 include first V_t -enhanced sections 118A at a vertical elevation of the program-inhibit SGD tier 110C; second V_t -enhanced sections 118B at a vertical elevation of the first SGD bar tier 110D; third V_t -enhanced sections 118C at a vertical elevation the second SGD bar tier 110E; fourth V_t -enhanced sections 118D at a vertical elevation of the first SGD tier 110F; fifth V_t -enhanced sections 118E at a vertical elevation of the second SGD tier 110G; sixth V_t -enhanced sections 118E at a vertical elevation of the read-amplification SGD tier 110H; seventh V_t -enhanced sections 118G at a vertical elevation of the first SGP bar tier 110I; eighth V_t -enhanced sections 118H at a vertical elevation of the

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second SGP bar tier 110J; ninth V_f -enhanced sections 118I at a vertical elevation of the first SGP tier 110K; and tenth V_f -enhanced sections 118J at a vertical elevation of the second SGP tier 110L. Within an individual sub-block group 116 of an individual block 164 (FIG. 1B), different V_f -enhanced sections 118 may encompass different sub-blocks 114 of the sub-block group 116. Non-limiting examples of configurations of some of the V_f -enhanced sections 118 within a horizontal area of the sub-section B₁ of the micro-electronic device structure 100 shown in FIGS. 1A and 1B are described in further detail below.

Referring to FIG. 1A, within a horizontal area of the sub-section B₁ of the microelectronic device structure 100, one first V_f -enhanced section 118A may encompass portions of the first sub-block 114A, the second sub-block 114B, the third sub-block 114C, and the fourth sub-block 114D of the third sub-block group 116C. The first V_f -enhanced section 118A may comprise portions of the program-inhibit SGD tier 110C, as well as portions of four (4) of the pillar structures 122 at the vertical position of the program-inhibit SGD tier 110C. The four (4) of the pillar structures 122 may include a first pillar structure 122A, a second pillar structure 122B, a third pillar structure 122C, and a fourth pillar structure 122D. The four (4) of the pillar structures 122 may be included in one (1) of multiple pillar groups 124 of within horizontal boundaries of the third sub-block group 116C, wherein each of the pillar groups 124 individually includes four (4) pillar structures 122 that are different than four (4) other pillar structures 122 of each other of the pillar groups 124. Each of the four (4) of the pillar structures 122 within the horizontal area of the sub-section B₁ may be positioned in a different one of the first sub-block 114A, the second sub-block 114B, the third sub-block 114C, and the fourth sub-block 114D of the third sub-block group 116C than each other of the four (4) of the pillar structures 122. The first V_f -enhanced section 118A may be substantially free of conductivity-enhancing species (e.g., P-type dopants, N-type dopants) therein. Select transistors 123 within the first V_f -enhanced section 118A may be configured (e.g., programmed) to have substantially the same V_f characteristics as one another.

Within the horizontal area of the sub-section B₁ of the microelectronic device structure 100, one second V_f -enhanced section 118B may encompass portions of the third sub-block 114C and the fourth sub-block 114D of the third sub-block group 116C. The second V_f -enhanced section 118B may not encompass portions of either of the first sub-block 114A and the second sub-block 114B of the third sub-block group 116C. The second V_f -enhanced section 118B may comprise portions of the first SGD bar tier 110D, as well as portions of two (2) of the pillar structures 122 of the pillar group 124 at the vertical position of the first SGD bar tier 110D. In some embodiments, the two (2) of the pillar structures 122 include the third pillar structure 122C and the fourth pillar structure 122D. The second V_f -enhanced section 118B may not include portions of the other two (2) of the pillar structures 122 of the pillar group 124. In some embodiments, the other two (2) of the pillar structures 122 include the first pillar structure 122A and the second pillar structure 122B. The second V_f -enhanced section 118B may be substantially free of conductivity-enhancing species (e.g., P-type dopants, N-type dopants) therein. Select transistors 123 within the second V_f -enhanced section 118B may be configured (e.g., programmed) to have substantially the same V_f characteristics as one another.

Within the horizontal area of the sub-section B₁ of the microelectronic device structure 100, one third V_f -enhanced

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section 118C may encompass portions of the second sub-block 114B and the third sub-block 114C of the third sub-block group 116C. The third V_f -enhanced section 118C may not encompass portions of either of the first sub-block 114A and the fourth sub-block 114D of the third sub-block group 116C. The third V_f -enhanced section 118C may comprise portions of the second SGD bar tier 110E doped with at least one conductivity-enhancing species, as well as portions of two (2) of the pillar structures 122 of the pillar group 124 at the vertical position of the second SGD bar tier 110E. In some embodiments, the two (2) of the pillar structures 122 include the second pillar structure 122B and the third pillar structure 122C. The third V_f -enhanced section 118C may not include portions of the other two (2) of the pillar structures 122 of the pillar group 124. In some embodiments, the other two (2) of the pillar structures 122 include the first pillar structure 122A and the fourth pillar structure 122D. The third V_f -enhanced section 118C may be substantially free of conductivity-enhancing species (e.g., P-type dopants, N-type dopants) therein. Select transistors 123 within the third V_f -enhanced section 118C may be configured (e.g., programmed) to have substantially the same V_f characteristics as one another.

Within the horizontal area of the sub-section B₁ of the microelectronic device structure 100, one fourth V_f -enhanced section 118D may encompass portions of the first sub-block 114A and the second sub-block 114B of the third sub-block group 116C. The fourth V_f -enhanced section 118D may not encompass portions of either of the third sub-block 114C and the fourth sub-block 114D of the third sub-block group 116C. The fourth V_f -enhanced section 118D may comprise portions of the first SGD tier 110F doped with at least one conductivity-enhancing species, as well as portions of two (2) of the pillar structures 122 of the pillar group 124 at the vertical position of the first SGD tier 110F. In some embodiments, the two (2) of the pillar structures 122 include the first pillar structure 122A and the second pillar structure 122B. The fourth V_f -enhanced section 118D may not include portions of the other two (2) of the pillar structures 122 of the pillar group 124. In some embodiments, the other two (2) of the pillar structures 122 include the third pillar structure 122C and the fourth pillar structure 122D. The fourth V_f -enhanced section 118D may be substantially free of conductivity-enhancing species (e.g., P-type dopants, N-type dopants) therein. Select transistors 123 within the fourth V_f -enhanced section 118D may be configured (e.g., programmed) to have substantially the same V_f characteristics as one another.

Within the horizontal area of the sub-section B₁ of the microelectronic device structure 100, one fifth V_f -enhanced section 118E may encompass a portion of the first sub-block 114A and another fifth V_f -enhanced section 118E may encompass a portion of the fourth sub-block 114D. Neither of the fifth V_f -enhanced sections 118E may encompass portions of either of the second sub-block 114B and the third sub-block 114C of the third sub-block group 116C. The fifth V_f -enhanced sections 118E may comprise portions of the second SGD tier 110G doped with at least one conductivity-enhancing species, as well as portions of two (2) of the pillar structures 122 of the pillar group 124 at the vertical position of the second SGD tier 110G. In some embodiments, the two (2) of the pillar structures 122 include the first pillar structure 122A and the fourth pillar structure 122D. The fifth V_f -enhanced sections 118E may not include portions of the other two (2) of the pillar structures 122 of the pillar group 124. In some embodiments, the other two (2) of the pillar structures 122 include the second pillar structure 122B and

the third pillar structure 122C. The fifth V_t -enhanced sections 118E may be substantially free of conductivity-enhancing species (e.g., P-type dopants, N-type dopants) therein. Select transistors 123 within the fifth V_t -enhanced sections 118E may be configured (e.g., programmed) to have substantially the same V_t characteristics as one another.

Within the horizontal area of the sub-section B_1 of the microelectronic device structure 100, one sixth V_t -enhanced section 118F may encompass portions of the first sub-block 114A, the second sub-block 114B, the third sub-block 114C, and the fourth sub-block 114D of the third sub-block group 116C. The sixth V_t -enhanced section 118F may comprise portions of the read-amplification SGD tier 110H doped with at least one conductivity-enhancing species, as well as portions of all four (4) of the pillar structures 122 of the pillar structures 122 of the pillar group 124 at the vertical position of the read-amplification SGD tier 110H. The sixth V_t -enhanced section 118F may be substantially free of conductivity-enhancing species (e.g., P-type dopants, N-type dopants) therein. Select transistors 123 within the sixth V_t -enhanced section 118F may be configured (e.g., programmed) to have substantially the same V_t characteristics as one another.

Within the horizontal area of the sub-section B_1 of the microelectronic device structure 100, one seventh V_t -enhanced section 118G may encompass portions of the third sub-block 114C and the fourth sub-block 114D of the third sub-block group 116C. The seventh V_t -enhanced section 118G may not encompass portions of either of the first sub-block 114A and the second sub-block 114B of the third sub-block group 116C. The seventh V_t -enhanced section 118G may comprise portions of the first SGP bar tier 110I doped with at least one conductivity-enhancing species, as well as portions of two (2) of the pillar structures 122 of the pillar group 124 at the vertical position of the first SGP bar tier 110I. In some embodiments, the two (2) of the pillar structures 122 include the third pillar structure 122C and the fourth pillar structure 122D. The seventh V_t -enhanced section 118G may not include portions of the other two (2) of the pillar structures 122 of the pillar group 124. In some embodiments, the other two (2) of the pillar structures 122 include the first pillar structure 122A and the second pillar structure 122B. The seventh V_t -enhanced section 118G may be substantially free of conductivity-enhancing species (e.g., P-type dopants, N-type dopants) therein. Select transistors 123 within the seventh V_t -enhanced section 118G may be configured (e.g., programmed) to have substantially the same V_t characteristics as one another.

Within the horizontal area of the sub-section B_1 of the microelectronic device structure 100, one eighth V_t -enhanced section 118H may encompass portions of the second sub-block 114B and the third sub-block 114C of the third sub-block group 116C. The eighth V_t -enhanced section 118H may not encompass portions of either of the first sub-block 114A and the fourth sub-block 114D of the third sub-block group 116C. The eighth V_t -enhanced section 118H may comprise portions of the second SGP bar tier 110J doped with at least one conductivity-enhancing species, as well as portions of two (2) of the pillar structures 122 of the pillar group 124 at the vertical position of the second SGP bar tier 110J. In some embodiments, the two (2) of the pillar structures 122 include the second pillar structure 122B and the third pillar structure 122C. The eighth V_t -enhanced section 118H may not include portions of the other two (2) of the pillar structures 122 of the pillar group 124. In some embodiments, the other two (2) of the pillar structures 122 include the first pillar structure 122A and the fourth pillar

structure 122D. The eighth V_t -enhanced section 118H may be substantially free of conductivity-enhancing species (e.g., P-type dopants, N-type dopants) therein. Select transistors 123 within the eighth V_t -enhanced section 118H may be configured (e.g., programmed) to have substantially the same V_t characteristics as one another.

Within the horizontal area of the sub-section B_1 of the microelectronic device structure 100, one ninth V_t -enhanced section 118I may encompass portions of the first sub-block 114A and the second sub-block 114B of the third sub-block group 116C. The ninth V_t -enhanced section 118I may not encompass portions of either of the third sub-block 114C and the fourth sub-block 114D of the third sub-block group 116C. The ninth V_t -enhanced section 118I may comprise portions of the first SGP tier 110K doped with at least one conductivity-enhancing species, as well as portions of two (2) of the pillar structures 122 of the pillar group 124 at the vertical position of the first SGP tier 110K. In some embodiments, the two (2) of the pillar structures 122 include the first pillar structure 122A and the second pillar structure 122B. The ninth V_t -enhanced section 118I may not include portions of the other two (2) of the pillar structures 122 of the pillar group 124. In some embodiments, the other two (2) of the pillar structures 122 include the third pillar structure 122C and the fourth pillar structure 122D. The ninth V_t -enhanced section 118I may be doped with one or more conductivity-enhancing species (e.g., at least one P-type dopant, such as one or more of boron (B), aluminum (Al), and gallium (Ga)). In some embodiments, the ninth V_t -enhanced section 118I is doped with B. The ninth V_t -enhanced section 118I may permit select transistors 123 associated with the two (2) of the pillar structures 122 at the vertical elevation of the first SGP tier 110K to have substantially the same V_t characteristics as one another; and may also permit select transistors 123 associated with the other two (2) of the pillar structures 122 to have substantially the same V_t characteristics as one another that are different than the V_t characteristics of the two (2) of the pillar structures 122. In some embodiments, the select transistors 123 associated with the two (2) of the pillar structures 122 at the vertical elevation of the first SGP tier 110K comprise relatively higher V_t transistors; and the select transistors 123 associated with the other two (2) of the pillar structures 122 at the vertical elevation of the first SGP tier 110K comprise relatively lower V_t transistors.

Within the horizontal area of the sub-section B_1 of the microelectronic device structure 100, one tenth V_t -enhanced section 118J may encompass a portion of the first sub-block 114A and another tenth V_t -enhanced section 118J may encompass a portion of the fourth sub-block 114D. Neither of the tenth V_t -enhanced sections 118J may encompass portions of either of the second sub-block 114B and the third sub-block 114C of the third sub-block group 116C. The tenth V_t -enhanced sections 118J may comprise portions of the second SGP tier 110L doped with at least one conductivity-enhancing species, as well as portions of two (2) of the pillar structures 122 of the pillar group 124 at the vertical position of the second SGP tier 110L. In some embodiments, the two (2) of the pillar structures 122 include the first pillar structure 122A and the fourth pillar structure 122D. The tenth V_t -enhanced sections 118J may not include portions of the other two (2) of the pillar structures 122 of the pillar group 124. In some embodiments, the other two (2) of the pillar structures 122 include the second pillar structure 122B and the third pillar structure 122C. The tenth V_t -enhanced sections 118J may be doped with one or more conductivity-enhancing species (e.g., at least one P-type dopant, such as

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one or more of B, Al, and Ga). In some embodiments, the tenth V_f -enhanced sections 118J are doped with B. The tenth V_f -enhanced sections 118J may permit select transistors 123 associated with the two (2) of the pillar structures 122 at the vertical elevation of the second SGP tier 110L to have substantially the same V_f characteristics as one another; and may also permit select transistors 123 associated with the other two (2) of the pillar structures 122 to have substantially the same V_f characteristics as one another that are different than the V_f characteristics of the two (2) of the pillar structures 122. In some embodiments, the select transistors 123 associated with the two (2) of the pillar structures 122 at the vertical elevation of the second SGP tier 110L comprise relatively higher V_f transistors; and the select transistors 123 associated with the other two (2) of the pillar structures 122 at the vertical elevation of the second SGP tier 110L comprise relatively lower V_f transistors.

Collectively referring to FIGS. 1A and 1B, the SGD plug tier 127 may include multiple (e.g., more than one, a plurality of) SGD plug structures 126. The SGD plug structures 126 may vertically overlie the pillar structures 122, and may individually horizontally extend across and between four (4) of the pillar structures 122 of an individual pillar group 124. An individual SGD plug structure 126 may be in electrical communication with each of the four (4) of the pillar structures 122 of an individual pillar group 124. For example, as shown in FIG. 1A, within the horizontal area of the sub-section B_1 of the microelectronic device structure 100, one (1) of the SGD plug structures 126 may vertically overlie and be coupled to each of the first pillar structure 122A, the second pillar structure 122B, the third pillar structure 122C, and the fourth pillar structure 122D of an individual pillar group 124 within the third sub-block group 116C of an individual block 164 (FIG. 1B) of the stack structure 104. The first pillar structure 122A may be located within the first sub-block 114A of the third sub-block group 116C; the second pillar structure 122B may be located within the second sub-block 114B of the third sub-block group 116C; the third pillar structure 122C may be located within the third sub-block 114C of the third sub-block group 116C; and the fourth pillar structure 122D may be located within the fourth sub-block 114D of the third sub-block group 116C. Accordingly, an individual SGD plug structure 126 may be coupled to each of four (4) pillar structures 122 (and, hence, the select transistors 123 and the strings of memory cells 121 that are operatively associated with each of the four (4) of the pillar structures 122) in different sub-blocks 114 than one another within an individual block 164 (FIG. 1B).

Referring to FIG. 1B, some of the SGD plug structures 126 may horizontally extend in parallel with one another, and some other of the SGD plug structures 126 may horizontally extend in series with one another. SGD plug structures 126 within a horizontal area of the same sub-block group 116 (e.g., the first sub-block group 116A, the second sub-block group 116B, the third sub-block group 116C, or the fourth sub-block group 116D) as one another may horizontally extend in parallel with one another in a direction acutely angled relative to each of the X-direction and the Y-direction shown in FIG. 1B. In addition, at least some SGD plug structures 126 within horizontal areas of different sub-block groups 116 than one another may horizontally extend in series with one another in the direction acutely angled relative to each of the X-direction and the Y-direction shown in FIG. 1B.

The SGD plug structures 126 may individually be formed of and include conductive material. By way of non-limiting

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example, the SGD plug structures 126 may each individually formed of and include semiconductor material (e.g., polycrystalline silicon) doped with doped with at least one conductivity-enhancing dopant (e.g., at least one N-type dopant, such as one or more of P, As, Sb, and Bi; at least one P-type dopant, such as one or more of B, Al, and Ga). In some embodiments, the SGD plug structures 126 may each individually formed of and include N-type polycrystalline silicon (e.g., polycrystalline silicon doped with at least one N-type dopant, such as polycrystalline silicon doped with P). Each of the SGD plug structures 126 may individually be substantially homogeneous, or one or more of the SGD plug structures 126 may individually be substantially heterogeneous. In some embodiments, each of the SGD plug structures 126 is substantially homogeneous.

Referring to FIG. 1A, the SGD plug structures 126, portions of the sense node tier section 111C within horizontal areas of the SGD plug structures 126, and portions of the pillar structures 122 within horizontal areas of the SGD plug structures 126 and within vertical boundaries of the sense node tier section 111C may together form sense nodes 112 of the microelectronic device structure 100. Each sense node 112 may individually be positioned within a horizontal area of an individual sub-block group 116 of an individual block 164 (FIG. 1B) of the stack structure 104, and may vertically overlie the stacked SGD tier section 111B (and, hence, the access line tier section 111A, and the strings of memory cells 121) of the stack structure 104. An individual sense node 112 may include a group of the select transistors 123 within a horizontal area of the SGD plug structure 126 of sense node 112 and within vertical boundaries of the sense node tier section 111C. For example, an individual sense node 112 within the horizontal area of the sub-section B_1 of the microelectronic device structure 100 shown in FIG. 1A may include one (1) of the SGD plug structures 126, as well as the select transistors 123 formed at the intersections of the four (4) pillar structures 122 (e.g., the first pillar structure 122A, the second pillar structure 122B, the third pillar structure 122C, and the fourth pillar structure 122D) coupled to the SGD plug structures 126 and the conductive structures 106 of tiers 110 (e.g., the first SGP bar tier 110I, the second SGP bar tier 110J, the first SGP tier 110K, the second SGP tier 110L) of the sense node tier section 111C and the four (4) pillar structures 122 (e.g., the first pillar structure 122A, the second pillar structure 122B, the third pillar structure 122C, and the fourth pillar structure 122D). The sense nodes 112 within horizontal areas of blocks 164 (FIG. 1B) of the stack structure 104 may facilitate capacitive-sense on block functionality for the microelectronic device structure 100, as described in further detail below.

Still referring to FIG. 1A, the microelectronic device structure 100 further includes a channel tier 132 including channel structures 134 vertically interposed between the SGD plug tier 127 and the selector tiers 138. The channel structures 134 may horizontally overlap the SGD plug structures 126. In some embodiments, the channel structures 134 are substantially confined within horizontal areas of the SGD plug structures 126, and exhibit horizontal areas relatively smaller than the horizontal areas of the SGD plug structures 126. For example, an individual channel structure 134 may horizontally extend across a portion of an individual SGD plug structure 126 that covers less than all of the pillar structures 122 coupled to the SGD plug structure 126, such as only three (3) of the four (4) pillar structures 122 coupled to the SGD plug structure 126. In some embodiments, an individual channel structure 134 horizontally overlaps three (3) of the pillar structures 122 (e.g., the

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second pillar structure 122B, the third pillar structure 122C, the fourth pillar structure 122D) coupled to an individual SGD plug structure 126, but does not substantially horizontally overlap a final one (1) of the pillar structures 122 (e.g., the first pillar structure 122A) coupled to the SGD plug structure 126.

Referring to FIG. 1B, some of the channel structures 134 may horizontally extend in parallel with one another, and some other of the channel structures 134 may horizontally extend in series with one another. Channel structures 134 within a horizontal area of the same sub-block group 116 (e.g., the first sub-block group 116A, the second sub-block group 116B, the third sub-block group 116C, or the fourth sub-block group 116D) as one another may horizontally extend in parallel with one another in a direction acutely angled relative to each of the X-direction and the Y-direction shown in FIG. 1B. In addition, at least some channel structures 134 within horizontal areas of different sub-block groups 116 than one another may horizontally extend in series with one another in the direction acutely angled relative to each of the X-direction and the Y-direction shown in FIG. 1B.

The channel structures 134 of the channel tier 132 may individually be formed of and include at least one semiconductor material, such as one or more of silicon, germanium, at least one compound semiconductor material, and at least one oxide semiconductor material. In some embodiments, the channel structures 134 are formed of and include polycrystalline silicon. In additional embodiments, the channel structures 134 are formed of and include oxide semiconductor material. Each of the channel structures 134 may individually be substantially homogeneous, or one or more of the channel structures 134 may individually be substantially heterogeneous. In some embodiments, each of the channel structures 134 is substantially homogeneous. The semiconductor material of the channel structures 134 may be doped with one or more conductivity-enhancing species (e.g., P-type dopant, N-type dopant), or the semiconductor material of the channel structures 134 may not be doped one or more conductivity-enhancing species. In some embodiments, the channel structures 134 are each individually formed of and include P-type polycrystalline silicon (e.g., polycrystalline silicon doped with at least one P-type dopant, such as polycrystalline silicon doped with B).

Still referring to FIG. 1A, a gate dielectric tier 129 including gate dielectric structures 130 vertically interposed between the channel tier 132 and the SGD plug tier 127. The gate dielectric structures 130 may horizontally extend across at least the channel structures 134 of the channel tier 132 and portions of the SGD plug structures 126 of the SGD plug tier 127, and may vertically extend from and between the channel structures 134 and the SGD plug structures 126. The gate dielectric structures 130 may be formed of and include dielectric material, such as one or more of at least one dielectric oxide material (e.g., one or more of SiO_x , AlO_x , phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass), at least one dielectric nitride material (e.g., SiN_y), and at least one low-K dielectric material (e.g., one or more of silicon oxycarbide (SiO_xC_y), silicon oxynitride (SiO_xN_y), hydrogenated silicon oxycarbide ($\text{SiC}_x\text{O}_y\text{H}_z$), and silicon oxycarbonitride ($\text{SiO}_x\text{C}_y\text{N}_z$)). The gate dielectric structures 130 may be substantially homogeneous, or the gate dielectric structures 130 may be heterogeneous. In some embodiments, the gate dielectric structure 130 is formed of and includes SiO_x (e.g., SiO_2).

The SGD plug structures 126, the gate dielectric structures 130, and the channel structures 134 may together form

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horizontal sense transistors 136 (e.g., planar sense transistors, laterally oriented sense transistors) of the microelectronic device structure 100. Each horizontal sense transistor 136 may be a thin film transistor (TFT). The SGD plug structures 126 may serve as gate electrodes for the horizontal sense transistors 136. Each horizontal sense transistor 136 may individually include a SGD plug structure 126, a channel structure 134 horizontally overlapping the SGD plug structure 126, and gate dielectric structures 130 vertically interposed between the SGD plug structure 126 and the channel structure 134.

Still referring to FIG. 1A, the selector tiers 138 may vertically overlie the channel tier 132 (and, hence, the horizontal sense transistors 136). The selector tiers 138 may include a first selector tier 138A, a second selector tier 138B vertically overlying the first selector tier 138A, and a third selector tier 138C vertically overlying the second selector tier 138B. Each of the selector tiers 138 may individually include an additional conductive structure 140 (e.g., gate electrode). Insulative material (e.g., dielectric oxide material, such as SiO_x) may be vertically interleaved with the additional conductive structures 140 of the selector tiers 138. The additional conductive structures 140 of the selector tiers 138 may be confined within horizontal areas of the blocks 164 (FIG. 1B) of the stack structure 104. In some embodiments, the additional conductive structures 140 of the selector tiers 138 are formed substantially simultaneously with the formation of the conductive structures 106 of the tiers 110 of the stack structure 104 (e.g., during so-called “replacement gate” or “gate last” processes), as described in further detail below.

The additional conductive structures 140 of the selector tiers 138 may individually be formed of and include conductive material. By way of non-limiting example, the additional conductive structures 140 may individually be formed of and include one or more of at least one metal, at least one alloy, and at least one conductive metal-containing material (e.g., a conductive metal nitride, a conductive metal silicide, a conductive metal carbide, a conductive metal oxide). In some embodiments, the additional conductive structures 140 are individually formed of and include one or more of W, Ru, Mo, and TiN_y . Each of the additional conductive structures 140 may individually be substantially homogeneous, or one or more of the additional conductive structures 140 may individually be substantially heterogeneous. In some embodiments, each of the additional conductive structures 140 is substantially homogeneous.

Within a horizontal area of an individual block 164 (FIG. 1B), one (1) of the selector tiers 138 (e.g., the first selector tier 138A) may be employed as a write selector tier, another one (1) of the selector tiers 138 (e.g., the second selector tier 138B) may be employed as a read selector tier, and an additional one (1) of the selector tiers 138 (e.g., the third selector tier 138C) may be employed as a block selector tier. In some embodiments, the first selector tier 138A is employed as a write selector tier, and an individual additional conductive structure 140 thereof is employed as write select gate electrode; the second selector tier 138B is employed as a read selector tier, and an individual additional conductive structure 140 thereof is employed as read select gate electrode; and the third selector tier 138C is employed as a block selector tier, and an individual additional conductive structure 140 thereof is employed as block select gate electrode.

With continued reference to FIG. 1A, the microelectronic device structure 100 further includes write select pillar structures 144 vertically extending through the selector tiers

138 and to or into the SGD plug structures 126. The write select pillar structures 144 may be positioned with horizontal areas of the SGD plug structures 126, and may be horizontally offset from the channel structures 134. Each of the SGD plug structures 126 may individually include one (1) of the write select pillar structures 144 in contact (e.g., physical contact, electrical contact) therewith. As shown in FIG. 1A, in some embodiments, an individual write select pillar structure 144 horizontally overlaps (e.g., is substantially horizontally centered about) one (1) of the pillar structures 122 (e.g., the first pillar structure 122A) coupled to the SGD plug structure 126 in contact with the write select pillar structure 144.

Still referring to FIG. 1A, the microelectronic device structure 100 further includes first read select pillar structures 146 (e.g., source-side read select pillars) and second read select pillar structures 148 (e.g., drain-side read select pillars) vertically extending through the selector tiers 138 and to or into the channel structures 134. The first read select pillar structures 146 and the second read select pillar structures 148 may be positioned with horizontal areas of the channel structures 134, and may be horizontally offset from one another and the channel structures 134. Each of the channel structures 134 may individually include one (1) of the first read select pillar structures 146 and one (1) of the second read select pillar structures 148 in contact (e.g., physical contact, electrical contact) therewith. Accordingly, each of the horizontal sense transistors 136 may individually include one (1) of the first read select pillar structures 146 and one (1) of the second read select pillar structures 148 in electrical communication therewith. In some embodiments, for an individual horizontal sense transistor 136, a first read select pillar structure 146 is employed a source-side read select pillar that contacts a source region of the channel structure 134; and a second read select pillar structure 148 is employed a drain-side read select pillar that contacts a drain region of the channel structure 134. Within a horizontal area of an individual SGD plug structure 126, a horizontal distance between the second read select pillar structure 148 and the write select pillar structure 144 may be less than a horizontal distance between the first read select pillar structure 146 and the write select pillar structure 144. As shown in FIG. 1A, in some embodiments, an individual first read select pillar structure 146 horizontally overlaps (e.g., is substantially horizontally centered about) one (1) of the pillar structures 122 (e.g., the fourth pillar structure 122D) coupled to the SGD plug structure 126 horizontally overlapping the first read select pillar structure 146; and an individual second read select pillar structure 148 horizontally overlaps (e.g., is substantially horizontally centered about) an additional one (1) of the pillar structures 122 (e.g., the second pillar structure 122B) coupled to the SGD plug structure 126 horizontally overlapping the second read select pillar structure 148.

The write select pillar structures 144, the first read select pillar structures 146, and the second read select pillar structures 148 may each individually be formed of and include a stack of materials. By way of non-limiting example, the write select pillar structures 144, the first read select pillar structures 146, and the second read select pillar structures 148 may each individually be formed of and include a gate dielectric material (e.g., SiO_x , such as SiO_2), a channel material (e.g., silicon, such as polycrystalline silicon), and a dielectric fill (e.g., SiO_x , such as SiO_2 , SiN_x , such as Si_3N_4 ; air). The gate dielectric material may be formed on or over surfaces of the additional conductive structures 140 and the insulative material of the selector tiers

138 at least partially defining horizontal boundaries of the write select pillar structures 144, the first read select pillar structures 146, and the second read select pillar structures 148; the channel material may be horizontally surrounded by the gate dielectric material; and the dielectric fill material may be horizontally surrounded by the channel material.

Still referring to FIG. 1A, intersections of the write select pillar structures 144 and the additional conductive structures 140 of individual selector tiers 138 may define write selector transistors 131 of the microelectronic device structure 100. By way of non-limiting example, intersections of the write select pillar structures 144 and the additional conductive structure 140 (e.g., serving as a write select gate) of first selector tier 138A may define write selector transistors 131 at the vertical elevation of the first selector tier 138A. An individual write selector transistor 131 may comprise a vertical transistor (e.g., a vertically oriented transistor) including a channel region vertically offset from source/drain regions. Channel regions of the write selector transistors 131 may be positioned within vertical boundaries of the additional conductive structure 140 of an individual selector tier 138; and source/drain regions of the write selector transistors 131 may vertically neighbor the channel regions, and may be vertically offset from the additional conductive structure 140. In some embodiments, the write selector transistors 131 comprise metal-oxide-semiconductor (MOS) transistors. Write selector transistors 131 within vertical boundaries of selector tiers 138 employed as write selector tiers (e.g., the first selector tier 138A) may have different V_t characteristics than write selector transistors 131 of selector tiers 138 employed as read selector tiers (e.g., the second selector tier 138B). For example, write selector transistors 131 within vertical boundaries of the first selector tier 138A may comprise relatively higher V_t transistors, and write selector transistors 131 within vertical boundaries of the second selector tier 138B may comprise relatively lower V_t transistors.

Intersections of the additional conductive structures 140 of individual selector tiers 138 and each of the first read select pillar structures 146 and the second read select pillar structures 148 may define read selector transistors 135 of the microelectronic device structure 100. By way of non-limiting example, intersections of the additional conductive structure 140 of the second selector tier 138B and each of the first read select pillar structures 146 and the second read select pillar structures 148 may define read selector transistors 135 at the vertical elevation of the second selector tier 138B. An individual read selector transistor 135 may comprise a vertical transistor (e.g., a vertically oriented transistor) including a channel region vertically offset from source/drain regions. Channel regions of the read selector transistors 135 may be positioned within vertical boundaries of the additional conductive structure 140 of an individual selector tier 138; and source/drain regions of the read selector transistors 135 may vertically neighbor the channel regions, and may be vertically offset from the additional conductive structure 140. In some embodiments, the read selector transistors 135 comprise MOS transistors. Read selector transistors 135 within vertical boundaries of selector tiers 138 employed as read selector tiers (e.g., the second selector tier 138B) may have different V_t characteristics than read selector transistors 135 of selector tiers 138 employed as write selector tiers (e.g., the first selector tier 138A). For example, read selector transistors 135 within vertical boundaries of the second selector tier 138B may comprise relatively higher V_t transistors, and read selector transistors 135

within vertical boundaries of the first selector tier **138A** may comprise relatively lower V_t transistors.

Write selector transistors **131** and read selector transistors **135** within vertical boundaries of an individual selector tier **138** employed as a block selector tier may be employed for block select operations for a microelectronic device including the microelectronic device structure **100**. By way of non-limiting example, if the third selector tier **138C** is employed as a block selector tier of the microelectronic device structure **100**, the write selector transistors **131** and read selector transistors **135** within vertical boundaries of the third selector tier **138C** may be employed for block select operations for a microelectronic device including the microelectronic device structure **100**. Within a horizontal area of an individual block **164** (FIG. 1B), write selector transistors **131** and read selector transistors **135** within vertical boundaries of an individual selector tier **138** employed as a block selector tier (e.g., the third selector tier **138C**) may have substantially the same V_t characteristics as one another. For example, within a horizontal area of an individual block **164** (FIG. 1B), write selector transistors **131** and read selector transistors **135** within vertical boundaries of the third selector tier **138C** may all comprise relatively higher V_t transistors.

Still referring to FIG. 1A, different portions of the selector tiers **138** (e.g., the first selector tier **138A**, the second selector tier **138B**, the third selector tier **138C**), the write select pillar structures **144**, the first read select pillar structures **146**, and the second read select pillar structures **148** may be doped with one or more conductivity-enhancing species (e.g., one or more P-type dopants, such as one or more of B, Al, and Ga; one or more N-type dopants, such as one or more of P, As, Sb, and Bi). By way of non-limiting example, within horizontal areas of the blocks **164** (FIG. 1B), the microelectronic device structure **100** may include doped sections **142** at vertical positions of at least some of the selector tiers **138**. Each doped section **142** may individually include a portion of a selector tier **138** doped with at least one conductivity-enhancing species (e.g., at least one P-type dopant, at least one N-type dopant), as well as portions of one or more a write select pillar structure **144**, a first read select pillar structure **146**, and a second read select pillar structure **148** doped with the at least one conductivity-enhancing species. An individual doped section **142** may be substantially confined within horizontal boundaries of an individual sub-block group **116** of the block **164** (FIG. 1B), or may horizontally extend at least partially across and between at least two (2) horizontally neighboring sub-block groups **116** of the block **164** (FIG. 1B). The doped sections **142** of the microelectronic device structure **100** may be used to provide different select transistors (e.g., write selector transistors **131**, read selector transistors **135**, block select transistors) within the selector tiers **138** with desired V_t characteristics, as described in further detail below. For example, as a result of the doped sections **142**, some of the write selector transistors **131** and read selector transistors **135** within boundaries (e.g., vertical boundaries, horizontal boundaries) of doped sections **142** may be relatively higher V_t transistors, while some other of the write selector transistors **131** and read selector transistors **135** outside of the boundaries of doped sections **142** may be relatively lower V_t transistors.

Within an individual block **164** (FIG. 1B) of the stack structure **104**, the doped sections **142** may be configured such that the write selector transistors **131**, the read selector transistors **135**, and the block select transistors have desired V_t characteristics for use and operation of a microelectronic

device including the microelectronic device structure **100**. For example, the doped sections **142** may be configured such that read selector transistors **135** substantially vertically aligned with one another (e.g., within the same selector tier **138** as one another) have substantially the same V_t characteristics (e.g., are set to the same V_t levels) as one another, and different V_t characteristics (e.g., are set to different V_t levels) than write selector transistors **131** substantially vertically aligned therewith (e.g., within the same selector tier **138** as one another). As another example, the doped sections **142** may be configured such that some vertically neighboring transistors operatively associated with the same write select pillar structure **144**, the same first read select pillar structure **146**, or the same second read select pillar structure **148** as one another have different V_t characteristics (e.g., different V_t levels) than one another.

As shown in FIG. 1A, in some embodiments, the doped sections **142** of the microelectronic device structure **100** include first doped sections **142A** at a vertical elevation of the first selector tier **138A**; second doped sections **142B** at a vertical elevation of the second selector tier **138B**; and third doped sections **142C** at a vertical elevation of the third selector tier **138C**. Within a horizontal area of an individual SGD plug structure **126**, different doped sections **142** may have horizontal positions than one another and/or may have different horizontal dimensions than one another. Non-limiting examples of configurations of some of the doped sections **142** within a horizontal area of the sub-section B_1 of the microelectronic device structure **100** shown in FIGS. 1A and 1B are described in further detail below.

Referring to FIG. 1A, within a horizontal area of the sub-section B_1 of the microelectronic device structure **100**, a first doped section **142A** may comprise a portion of at least the additional conductive structure **140** of the first selector tier **138A** doped with at least one conductivity-enhancing species, as well as a portion of the write select pillar structure **144** at the vertical position of the additional conductive structure **140** of the first selector tier **138A**. The first doped section **142A** may not include portions of the first read select pillar structure **146** and the second read select pillar structure **148** at the vertical position of the additional conductive structure **140** of the first selector tier **138A**. For example, the first doped section **142A** may not substantially horizontally overlap the channel structure **134** in contact with the first read select pillar structure **146** and the second read select pillar structure **148**. The first doped section **142A** may permit a write selector transistor **131** at the vertical position of the first selector tier **138A** to have different V_t characteristics than additional transistors (e.g., read selector transistors **135**) at the vertical position of the first selector tier **138A**. In some embodiments, the conductivity-enhancing species of the first doped section **142A** includes P-type dopant (e.g., B).

Within a horizontal area of the sub-section B_1 of the microelectronic device structure **100**, a second doped section **142B** may comprise portions of at least the additional conductive structure **140** of the second selector tier **138B** doped with at least one conductivity-enhancing species, as well as a portion of the first read select pillar structure **146** and the second read select pillar structure **148** at the vertical position of the additional conductive structure **140** of the second selector tier **138B**. The second doped section **142B** may not include a portion of the write select pillar structure **144** at the vertical position of the additional conductive structure **140** of the second selector tier **138B**. For example, the second doped section **142B** may substantially horizontally confined within the horizontal area of the channel

structure **134** in contact with the first read select pillar structure **146** and the second read select pillar structure **148**. The second doped section **142B** may permit read selector transistors **135** at the vertical position of the second selector tier **138B** to have substantially the same V_t characteristics as one another and different V_t characteristics than additional transistors (e.g., a write selector transistor **131**) at the vertical position of the second selector tier **138B**. In some embodiments, the conductivity-enhancing species of the second doped section **142B** includes P-type dopant (e.g., B). Optionally, an additional portion of at least the additional conductive structure **140** of the second selector tier **138B** and a portion of the write select pillar structure **144** may be doped with an additional conductivity-enhancing species (e.g., N-type dopant, such as P) having a conductivity type different than the conductivity-enhancing species (e.g., P-type dopant, such as B) included in the second doped section **142B**.

Within a horizontal area of the sub-section B_1 of the microelectronic device structure **100**, a third doped section **142C** may comprise portions of at least the additional conductive structure **140** of the third selector tier **138C** doped with at least one conductivity-enhancing species, as well as a portion of the write select pillar structure **144**, the first read select pillar structure **146**, and the second read select pillar structure **148** at the vertical position of the additional conductive structure **140** of the third selector tier **138C**. For example, the third doped section **142C** may continuously horizontally extend across and beyond the horizontal area of the SGD plug structure **126** within the sub-section B_1 of the microelectronic device structure **100**. The third doped section **142C** may permit additional transistors (e.g., write selector transistors **131**, read selector transistors **135**, block select transistors) at the vertical position of the third selector tier **138C** to have substantially the same V_t characteristics as one another. In some embodiments, the conductivity-enhancing species of the third doped section **142C** includes P-type dopant (e.g., B).

Referring collectively to FIGS. 1A and 1B, the conductive routing tier **150** (FIG. 1A) includes read source line structures **152** and local strap structures **154** vertically overlying the selector tiers **138**. The read source line structures **152** may individually vertically overlie, horizontally overlap, and contact (e.g., physically contact, electrically contact) multiple of the first read select pillar structures **146** (e.g., source-side read select pillar structures) a horizontal area of and individual block **164** (FIG. 1B) of the stack structure **104**. The local strap structures **154** may individually vertically overlie, horizontally extend from and between, and contact (e.g., physically contact, electrically contact) a write select pillar structure **144** and a second read select pillar structure **148** within a horizontal area of an individual SGD plug structure **126**. The read source line structures **152** and local strap structures **154** may be substantially vertically aligned with one another within the conductive routing tier **150**.

As shown in FIG. 1B, the read source line structures **152** may horizontally extend in parallel within one another and rows of the first read select pillar structures **146** (as well as rows of the second read select pillar structures **148**, rows of the write select pillar structures **144**, and rows of the pillar structures **122**) in the X-direction. Each of the read source line structures **152** may individually be coupled to the first read select pillar structures **146** of at least one (1) row of the first read select pillar structures **146** extending in the X-direction. As depicted in FIG. 1B, in some embodiments, each source line structure **152** is individually coupled to two (2)

rows of the first read select pillar structures **146**. A first of the two (2) rows of the first read select pillar structures **146** may be positioned within a horizontal area of one (1) of the sub-block groups **116** of an individual block **164**; and a second of the two (2) rows of the first read select pillar structures **146** may be positioned within a horizontal area of another one (1) of the sub-block groups **116** of block **164** horizontally neighboring the one (1) of the sub-block groups **116**.

Referring to FIGS. 1A and 1B, the local strap structures **154** may individually couple a write select pillar structure **144** within a horizontal area of an individual SGD plug structure **126** to a second read select pillar structure **148** within the horizontal area of SGD plug structure **126**. As shown FIG. 1B, some of the local strap structures **154** may horizontally extend in parallel with one another (and some of the SGD plug structures **126** and some of the channel structures **134**), and some other of the local strap structures **154** may horizontally extend in series with one another (and some other of the SGD plug structures **126** and some other of the channel structures **134**). Local strap structures **154** within a horizontal area of the same sub-block group **116** (e.g., the first sub-block group **116A**, the second sub-block group **116B**, the third sub-block group **116C**, or the fourth sub-block group **116D**) as one another may horizontally extend in parallel with one another in a direction acutely angled relative to each of the X-direction and the Y-direction shown in FIG. 1B. In addition, at least some local strap structures **154** within horizontal areas of different sub-block groups **116** than one another may horizontally extend in series with one another in the direction acutely angled relative to each of the X-direction and the Y-direction shown in FIG. 1B.

The read source line structures **152** and the local strap structures **154** may each individually be formed of and include conductive material. By way of non-limiting example, the read source line structures **152** and the local strap structures **154** may each individually be formed of and include one or more of at least one metal, at least one alloy, and at least one conductive metal-containing material (e.g., a conductive metal nitride, a conductive metal silicide, a conductive metal carbide, a conductive metal oxide). In some embodiments, the read source line structures **152** and the local strap structures **154** are each individually formed of and include one or more of W, Ru, Mo, and TiN_x . Each of the read source line structures **152** and the local strap structures **154** may individually be substantially homogeneous, or one or more of the read source line structures **152** and/or one or more of the local strap structures **154** may individually be substantially heterogeneous. In some embodiments, each of the read source line structures **152** and each of the local strap structures **154** is substantially homogeneous.

Referring to FIG. 1B, in some embodiments, within boundaries of an individual sub-block group **116**, all of the channel structures **134** are substantially aligned with one another in the Y-direction, all of the write select pillar structures **144** are substantially aligned with one another in the Y-direction, all of the first read select pillar structures **146** are substantially aligned with one another in the Y-direction, all of the second read select pillar structures **148** are substantially aligned with one another in the Y-direction, and all of the local strap structures **154** are substantially aligned with one another in the Y-direction. In additional embodiments, within boundaries of an individual sub-block group **116**, some of the channel structures **134** are horizontally offset from one another in the Y-direction, some of the write

select pillar structures **144** are horizontally offset from one another in the Y-direction, some of the first read select pillar structures **146** are horizontally offset from one another in the Y-direction, some of the second read select pillar structures **148** are horizontally offset from one another in the Y-direction, and/or some of the local strap structure **154** are horizontally offset from one another in the Y-direction. For example, horizontal positions in the Y-direction of channel structures **134**, write select pillar structures **144**, first read select pillar structures **146**, second read select pillar structures **148**, and local strap structure **154** operatively associated with SGD plug structures **126** (and, hence, horizontal sense transistors **136**) horizontally neighboring one another in the X-direction within boundaries of an individual sub-block group **116** may be horizontally inverted in the Y-direction relative to another. By way of non-limiting example, within boundaries of an individual sub-block group **116**, a write select pillar structure **144** operatively associated with a first SGD plug structure **126** of two (2) horizontally neighboring SGD plug structures **126** may be substantially horizontally aligned in the Y-direction with a first read select pillar structure **146** operatively associated with a second SGD plug structure **126** of the two (2) horizontally neighboring SGD plug structures **126**; a write select pillar structure **144** operatively associated with the second SGD plug structure **126** may be substantially horizontally aligned in the Y-direction with a first read select pillar structure **146** operatively associated with the first SGD plug structure **126**; a second read select pillar structure **148** operatively associated with the first SGD plug structure **126** may be completely horizontally offset in the Y-direction from a second read select pillar structure **148** operatively associated with the second SGD plug structure **126**; a channel structure **134** operatively associated with the first SGD plug structure **126** may be partially horizontally offset in the Y-direction from a channel structure **134** operatively associated with the second SGD plug structure **126**; and a local strap structure **154** operatively associated with the first SGD plug structure **126** may be completely horizontally offset in the Y-direction from a local strap structure **154** operatively associated with the second SGD plug structure **126**.

Referring collectively to FIGS. **1A** and **1B**, the digit line tier **156** (FIG. **1A**) includes digit line structures **158** vertically overlying the conductive routing tier **150**. As shown in FIG. **1B**, the digit line structures **158** may horizontally extend in parallel with one another in the Y-direction. The digit line structures **158** may horizontally extend perpendicular to the read source line structures **152**. The digit line structures **158** may be coupled to logic circuitry (e.g., page buffer circuitry) of a microelectronic device including the microelectronic device structure **100**. The logic circuitry may, for example, be included within the base structure **102** (FIG. **1A**) vertically underlying the stack structure **104**. In some embodiments, the digit line structures **158** are coupled to page buffer devices each individually including an arrangement of data cache circuitry (e.g., dynamic data cache (DDC) circuitry, primary data cache (PDC) circuitry, secondary data cache (SDC) circuitry, temporary data cache (TDC) circuitry), sense amplifier (SA) circuitry, and digit line pre-charge circuitry. Optionally, isolation devices (e.g., isolation transistors) may be interposed between the digit line structures **158** and the page buffer devices at desirable locations along conductive paths extending from and between the digit line structures **158** and the page buffer devices. In some embodiments, the isolation devices comprise high-voltage-isolation (HVISO) transistors configured and operated to pass voltages greater than or equal to about

18V, such as within a range of from about 18V to about 25V. In additional embodiments, the isolation devices comprise low-voltage-isolation (LVISO) transistors configured and operated to substantially block applied voltages less than about 18 V while in an OFF state (e.g., an inactive state, a depletion state, a deselected state).

The digit line structures **158** may each individually be formed of and include conductive material. By way of non-limiting example, the digit line structures **158** may each individually be formed of and include one or more of at least one metal, at least one alloy, and at least one conductive metal-containing material (e.g., a conductive metal nitride, a conductive metal silicide, a conductive metal carbide, a conductive metal oxide). In some embodiments, the digit line structures **158** are each individually formed of and include one or more of W, Ru, Mo, and TiN_x. Each of the digit line structures **158** may individually be substantially homogeneous, or one or more of the digit line structures **158** may individually be substantially heterogeneous. In some embodiments, each of the digit line structures **158** is substantially homogeneous.

Referring to FIG. **1A**, the microelectronic device structure **100** further includes digit line contact structures **160** vertically interposed between the digit line tier **156** and the conductive routing tier **150**. The digit line contact structures **160** may be configured to couple individual digit line structures **158** to individual local strap structures **154**. Each local strap structure **154** within a horizontal area of an individual block **164** (FIG. **1B**) of the stack structure **104** may individually be coupled to an individual digit line structure **158** by way of an individual digit line contact structure **160**. The digit line contact structure **160** may vertically extend from and between the local strap structure **154** and the digit line structure **158**.

The digit line contact structures **160** may each individually be formed of and include conductive material. By way of non-limiting example, the digit line contact structures **160** may each individually be formed of and include one or more of at least one metal, at least one alloy, and at least one conductive metal-containing material (e.g., a conductive metal nitride, a conductive metal silicide, a conductive metal carbide, a conductive metal oxide). In some embodiments, the digit line contact structures **160** are each individually formed of and include one or more of W, Ru, Mo, and TiN_x. Each of the digit line contact structures **160** may individually be substantially homogeneous, or one or more of the digit line contact structures **160** may individually be substantially heterogeneous.

Still referring to FIG. **1A**, the microelectronic device structure **100** further includes at least one isolation material **162** covering and surrounding various features (e.g., materials, structures, devices) thereof. For example, the isolation material **162** may vertically overlie the stack structure **104**, and may at least partially cover, at least partially surround, and/or at least partially be interposed (e.g., horizontally interposed, vertically interposed) between additional features of the microelectronic device structure **100** vertically overlying the stack structure **104** (e.g., the SGD plug structures **126**, the channel structures **134**, the additional conductive structures **140**, the write select pillar structures **144**, the first read select pillar structures **146**, the second read select pillar structures **148**, the read source line structures **152**, the local strap structures **154**, the digit line contact structures **160**, the digit line structures **158**).

The isolation material **162** may be formed of and include insulative material. By way of non-limiting example, the isolation material **162** may be formed of and include one or

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more of at least one dielectric oxide material (e.g., one or more of SiO_x , phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, AlO_x , HfO_x , NbO_x , and TiO_x), at least one dielectric nitride material (e.g., SiN_y), at least one dielectric oxynitride material (e.g., SiO_xN_y), at least one dielectric carboxynitride material (e.g., $\text{SiO}_x\text{C}_z\text{N}_y$), and amorphous carbon. In some embodiments, the isolation material **162** is formed of and includes SiO_x (e.g., SiO_2). The isolation material **162** may be substantially homogeneous, or the isolation material **162** may be hetero-

During use and operation of a microelectronic device (e.g., a memory device, such as a 3D NAND Flash memory device) including the microelectronic device structure **100**, programming operations may be performed to set (e.g.,

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SGP tier **110K** and second SGP tier **110L**. The fourth pillar structure **122D** may be identified by the horizontal overlap of one (1) of the tenth V_t -enhanced sections **118J** with the second pillar structure **122B**, along with the absence of horizontal overlap of one (1) of the ninth V_t -enhanced sections **118I** with the second pillar structure **122B**, due to the impact thereof on the characteristics (e.g., channel types) of the select transistors **123** at the intersections of the fourth pillar structure **122D** and the first SGP tier **110K** and second SGP tier **110L**.

A non-limiting example of SGPGM operations to set V_t levels for different select transistors **123** within the SGD tier section **111B** and the sense node tier section **111C** of the stack structure **104** is described below by and with reference to Table 1 and Table 2.

TABLE 1

	SB	L	K	J	I	H	G	F	E	D	C
J	C + B	L	H	Pgm	H	H	H	H	H	H	H
I	B + A	L	H	H	Pgm	H	H	H	H	H	H
H	By SB*	L/H	L/H	L/H	L/H	Pgm	H	H	H	H	H
G	By SB*	L/H	L/H	L/H	L/H	Inh	Pgm	Pass	Pass	Pass	Pass
F	By SB*	L/H	L/H	L/H	L/H	Inh	Pass	Pgm	Pass	Pass	Pass
E	By SB*	L/H	L/H	L/H	L/H	Inh	Pass	Pass	Pgm	Pass	Pass
D	By SB*	L/H	L/H	L/H	L/H	Inh	Pass	Pass	Pass	Pgm	Pass
C	By SB*	L/H	L/H	L/H	L/H	Inh	Pass	Pass	Pass	Pass	Pgm

tier section **111B** and the sense node tier section **111C** of the stack structure **104**. Such operations are referred to herein as “SGPGM” operations. The SGPGM operations may be facilitated, in part, by the V_t -enhanced sections **118** within the sense node tier section **111C**. The V_t -enhanced sections **118** within the sense node tier section **111C** may, for example, be used to identify different pillar structures **122** within the horizontal area of an individual block **164** (FIG. 1B) of the stack structure. As a non-limiting example, within a horizontal area of the sub-section B_1 of the microelectronic device structure **100**, the tenth V_t -enhanced sections **118J** of the second SGP tier **110L** and the ninth V_t -enhanced sections **118I** of the first SGP tier **110K** may be used to identify each of the first pillar structure **122A**, the second pillar structure **122B**, the third pillar structure **122C**, and the first pillar structure **122A**. The first pillar structure **122A** may be identified by the horizontal overlap of each of one (1) of the tenth V_t -enhanced sections **118J** and one (1) of the ninth V_t -enhanced sections **118I** with the first pillar structure **122A**, due to the impact thereof on the characteristics (e.g., channel types) of the select transistors **123** at the intersections of the first pillar structure **122A** and the first SGP tier **110K** and second SGP tier **110L**. The second pillar structure **122B** may be identified by the horizontal overlap of one (1) of the ninth V_t -enhanced sections **118I** with the second pillar structure **122B**, along with the absence of horizontal overlap (and select transistor **123** V_t properties) of one (1) of the tenth V_t -enhanced sections **118J** within the second pillar structure **122B**, due to the impact thereof on the characteristics (e.g., channel types) of the select transistors **123** at the intersections of the second pillar structure **122B** and the first SGP tier **110K** and second SGP tier **110L**. The third pillar structure **122C** may be identified by the absence of horizontal overlap of the ninth V_t -enhanced sections **118I** and the tenth V_t -enhanced sections **118J** with the third pillar structure **122C**, due to the impact thereof on the characteristics (e.g., channel types) of the select transistors **123** at the intersections of the third pillar structure **122C** and the first

TABLE 2

	SB	L	K	J	I
H, G, F, E,	A	H	H	L	L
D, C	B	L	H	H	L
	C	L	L	H	H
	D	H	L	L	H

In Table 1, the farthest column to the left represents the tier **110** of the stack structure **104** of the microelectronic device structure **400** in which select transistors **123** are being programmed, wherein J, I, H, G, F, E, D, and C represent programming operations for select transistors **123** within the second SGP bar tier **110J**, the first SGP bar tier **110I**, the read-amplification SGD tier **110H**, the second SGD tier **110G**, the first SGD tier **110F**, the second SGD bar tier **110E**, the first SGD bar tier **110D**, and the program-inhibit SGD tier **110C**, respectively. In addition, in the uppermost row in Table 1, SB, L, K, J, I, H, G, F, E, D, and C represent the sub-blocks **114** being acted upon, the second SGP tier **110L**, the first SGP tier **110K**, the second SGP bar tier **110J**, the first SGP bar tier **110I**, the read-amplification SGD tier **110H**, the second SGD tier **110G**, the first SGD tier **110F**, the second SGD bar tier **110E**, the first SGD bar tier **110D**, and the program-inhibit SGD tier **110C**, respectively. Furthermore, within the “SB” (sub-block) column, “C+B” represents the combination of a third sub-block **114C** and a second sub-block **114B** of the stack structure **104**; “B+A” represents the combination of a second sub-block **114B** and a first sub-block **114A** of the stack structure **104**; and “By SB*” represents a sub-block selection scheme further described by and with reference to Table 2. In addition, at intersections of the columns and the rows of Table 1, “L” represents a LOW drive state, “H” represents HIGH drive state; “L/H” represents a LOW drive state or a HIGH drive state, according to sub-block address; “Pgm” represents an fulfilled program action for associated select transistors **123**; “Inh” represents a program inhibit action; and “Pass” represents a pass through action.

In Table 2, the farthest column to the left represents the tiers 110 of the stack structure 104 of the microelectronic device structure 400 subject to the "By SB*" sub-block selection scheme identified in Table 1, wherein H, G, F, E, D, and C represent operations for select transistors 123 within the read-amplification SGD tier 110H, the second SGD tier 110G, the first SGD tier 110F, the second SGD bar tier 110E, the first SGD bar tier 110D, and the program-inhibit SGD tier 110C, respectively. In addition, in the uppermost row in Table 2, SB, L, K, J, and I represent the sub-blocks 114 being acted upon, the second SGP tier 110L, the first SGP tier 110K, the second SGP bar tier 110J, and the first SGP bar tier 110I. Furthermore, A, B, C, and D within the "SB" (sub-block) column represent the first sub-block 114A, the second sub-block 114B, the third sub-block 114C, and the fourth sub-block 114D, respectively. In addition, at intersections of the columns and the rows of Table 2, "L" represents a LOW drive state, and "H" represents HIGH drive state.

As a non-limiting example, if it is desired to program select transistors 123 at the vertical elevation of the second SGP bar tier 110J to desired V_t levels, 0 volts (V) may be supplied by a digit line structure 158 to the pillar structures 122 within an individual pillar group 124 by way of a horizontal sense transistor 136 while also biasing the conductive structure 106 of the second SGP bar tier 110J to a relatively HIGH (H) programming voltage (V_{pgm}) to facilitate a V_t potential difference. However, if the conductive structure 106 of the second SGP tier 110L is biased to 0 V, the 0 V supplied by the digit line structure 158 cannot be transferred to the tiers 110 thereunder. Accordingly, the select transistors 123 at intersections of the second SGP tier 110L and the first pillar structures 122A within the first sub-block 114A and the fourth pillar structures 122D within the fourth sub-block 114D may be set to an OFF (e.g., inactive) state. Through the combination of 0 V supplied from the digit line structure 158, the relatively HIGH V_{pgm} applied to the conductive structure 106 of the second SGP bar tier 110J, and the OFF state of the select transistors 123 associated with the first pillar structures 122A and the second pillar structures 122B at the vertical elevation of the second SGP tier 110L, select transistors 123 at intersections of the second SGP bar tier 110J and the second pillar structures 122B within the second sub-block 114B and the third pillar structures 122C within the third sub-block 114C be programmed to desired V_t levels. In addition, since the select transistors 123 at intersections of the second SGP bar tier 110J and the first pillar structures 122A within the first sub-block 114A and the fourth pillar structures 122D within the fourth sub-block 114D have not been programmed through the foregoing process, the second SGP tier 110L may be utilized in the reverse (e.g., select transistors 123 at intersections of the second SGP tier 110L and the first pillar structures 122A within the first sub-block 114A and the fourth pillar structures 122D within the fourth sub-block 114D may be set to an ON state, while select transistors 123 at intersections of the second SGP tier 110L and the second pillar structures 122B within the second sub-block 114B and the third pillar structures 122C within the third sub-block 114C may be set to an OFF state) to program the select transistors 123 at intersections of the second SGP bar tier 110J and the first pillar structures 122A within the first sub-block 114A and the fourth pillar structures 122D within the fourth sub-block 114D. The second SGP bar tier 110J may be employed to program V_t levels of the select transistors 123 at the vertical elevation of the second SGP tier 110L in a similar manner. In addition, the first SGP bar tier

110I and the first SGP tier 110K may be utilized in combination with one another to program V_t levels of the select transistors 123 and the vertical elevations thereof in a similar manner. Once the select transistors 123 within the sense node tier section 111C of the stack structure 104 are programmed to desired V_t levels, the pillar structures 122 may be individually accessed, and V_t levels of select transistors 123 within tiers 110 of the stack structure 104 underlying the sense node tier section 111C may be programmed as desired.

During use and operation of a microelectronic device (e.g., a memory device, such as a 3D NAND Flash memory device) including the microelectronic device structure 100, memory cell programming operations may also be performed. Such operations are referred to herein as "program operations." As a non-limiting example, still referring to FIG. 1A, if it is desired to program a memory cell 121 operatively associated with the second pillar structure 122B within the second sub-block 114B in the sub-section B_1 of the microelectronic device structure 100, the conductive structures 106 of the second SGD tier 110G and the first SGD bar tier 110D may be grounded, and the conductive structures 106 of the GG tier 110M, the first SGD tier 110F, the second SGD bar tier 110E, and the read-amplification SGD tier 110H may be biased HIGH. As a result, the second transistors 123 at vertical elevations of the second SGD tier 110G, the first SGD bar tier 110D, the first SGD tier 110F, and the second SGD bar tier 110E may facilitate access to second pillar structure 122B within the second sub-block 114B alone. In addition, the conductive structure 106 of the program-inhibit SGD tier 110C may be biased to a program-inhibit voltage (V_{SGD_Inh}). Depending on digit line size potential, if the digit line structure 158 supplies 0 V bias to the vertical position of the second SGD bar tier 110E, back-bias effect may effectuate the discharge of the second pillar structure 122B. Thus, the second pillar structure 122B may be biased to 0 V, and if the conductive structure 106 of an active access line tier 110A intersecting the second pillar structure 122B is being driven to V_{pgm} , the memory cell 121 defined at the intersection will be programmed. Conversely, if the digit line structure 158 is biased to supply voltage (V_{cc}), the select transistor 123 at the intersection of the second pillar structure 122B and the program-inhibit SGD tier 110C is not activated due to relatively higher source side potential, and memory cell 121 programming is inhibited.

During use and operation of a microelectronic device (e.g., a memory device, such as a 3D NAND Flash memory device) including the microelectronic device structure 100, memory cell read operations may also be performed. Such operations are referred to herein as "read operations." As a non-limiting example, during a read operation, write selector transistors 131 at the vertical elevation of the first selector tier 138A (e.g., employed as a write selector tier) may be set to an OFF state and a flow of current may be directed along a path extending from a source line structure 152, through each of a first read select pillar structure 146, a horizontal sense transistor 136, a second read select pillar structure 148, a local strap structure 154, and a digit line structure 158, and to a digit line structure 158. During the read operation, the horizontal sense transistor 136 may be used to sense the potential of the SGD plug structure 126, and the potential of the sense node 112 may change (e.g., from 4 V to 0 V) depending on the status of selected memory cells 121. The conductive structures 106 of all of the tiers 110 (e.g., the first SGP bar tier 110I, the second SGP bar tier 110J, the first SGP tier 110K, the second SGP tier 110L, and the GG tier 110M) within the sense node tier section 111C may be pre-charged to a desired level, such as 4 V (without limitation), and the

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select transistors **123** at vertical positions thereof may be provided in an ON state. Accordingly, the sense node **112** may act as a capacitive node having a desired potential, such as 4 V (without limitation). If, for example, it is desired to read a memory cell **121** operatively associated with a second pillar structure **122B** within the second sub-block **114B** in the sub-section B_1 of the microelectronic device structure **100**, the conductive structures **106** of the second SGD tier **110G** and the first SGD bar tier **110D** may be grounded and conductive structures **106** of the first SGD tier **110F** and the second SGD bar tier **110E** may be biased HIGH. By doing so, the select transistors **123** at vertical elevations of the second SGD tier **110G**, the first SGD bar tier **110D**, the first SGD tier **110F**, and the second SGD bar tier **110E** may facilitate access to the second pillar structure **122B** within the second sub-block **114B** alone. In addition, the conductive structure **106** of the read-amplification SGD tier **110H** may be utilized to provide a bias, and the select transistor **123** defined at the intersection of the second pillar structure **122B** and the read-amplification SGD tier **110H** may be employed as a clamp transistor and may facilitate a back bias effect. A lower portion of the second pillar structure **122B** may be pre-charged to a desired level, such as 1 V (without limitation). If 1 V reaches the vertical elevation of the second SGD tier **110G**, the relatively higher potential below the read-amplification SGD tier **110H** results in an OFF state of the select transistor **123** at the intersection of the second pillar structure **122B** and the read-amplification SGD tier **110H** and the 4 V potential of the sense node **112** is maintained. However, once the memory cell **121** associated with the second pillar structure **122B** is ramped down to 0 V, depending on the status of the associated active access line tier **110A**, the second pillar structure **122B** will discharge. If the access transistor of the read-amplification SGD tier **110H** is in an OFF state, then 1 V is maintained at the top portion of the second pillar structure **122B**, and the select transistor **123** at the intersection of the second pillar structure **122B** and the read-amplification SGD tier **110H** is in an OFF state such that the 4 V potential of the sense node **112** is maintained. However, if the access transistor of the read-amplification SGD tier **110H** is in an ON state, then after ramping the source to 0 V, a portion of the second pillar structure **122B** below the read-amplification SGD tier **110H** will be discharged to close to 0 V, resulting in an ON state of the select transistor **123** at the intersection of the second pillar structure **122B** and the read-amplification SGD tier **110H** to leak charge from the sense node **112** and bring the potential thereof to close to 0 V. By sensing the potential difference between 0 V and 4 V, horizontal sense transistor **136** facilitates read out the data stored in the memory cell **121**.

Having pillar structures **122** within multiple different sub-blocks **114** share an individual sense node **112** with one another provides efficiency and performance benefits relative to conventional configurations wherein sensing circuitry is simply shared by pillar structures within the same sub-block as one another as it alleviates a need for sequential reading of all sub-blocks **114** to access specific sub-block data. In addition, the continuity of the conductive structures **106** of the tiers **110** within the sense node tier section **111C** and the stacked SGD tier section **111B** across and between different sub-blocks **114** of an individual block **164** alleviates the need for the complex and costly patterning conventionally required to form segmented SGD structures (e.g., discrete SGD structures, each discrete SGD structure within a different sub-block than each other discrete SGD structure) within an individual SGD tier, as well as undesirable resis-

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tivity characteristics (e.g., relatively high resistivity) that may be associated with segmented SGD structures within an individual block. Furthermore, as sense node capacitance is orders of magnitude lower than digit line capacitance, discharge can be achieved using a relatively small amount of current, promoting relatively faster sense and read operations. Moreover, as the current path through a horizontal sense transistor **136** may be shorter than a current path through relatively long pillar structure **122**, current drive rating may be relatively higher and may facilitate faster read operation and simpler page buffer design as compared to conventional configurations.

Thus, a microelectronic device according to embodiments of the disclosure comprises a microelectronic device comprising a stack structure, pillar structures, a conductive plug structure, a sense transistor, and selector transistors. The stack structure comprises a vertically alternating sequence of conductive material and insulative material. The stack structure is divided into blocks separated by dielectric slot structures. The blocks individually include sub-blocks horizontally extending in parallel with one another. The pillar structures vertically extend through one of the blocks of the stack structure. Each pillar structure of a group of the pillar structures is horizontally positioned within a different one of the sub-blocks of the one of the blocks than each other pillar structure of the group of the pillar structures. The conductive plug structure is coupled to and horizontally extends across and between multiple of the pillar structures of the group of the pillar structures. The sense transistor is gated by the conductive plug structure. The selector transistors couple the sense transistor to a read source line structure and a digit line structure.

Furthermore, a memory device according to embodiments of the disclosure comprises a stack structure divided into blocks separated by dielectric slot structures, pillar groups within horizontal areas of the blocks, horizontal sense transistors vertically overlying and coupled to the pillar groups, vertical read selector transistors vertically overlying and coupled to the horizontal sense transistors, vertical write selector transistors vertically overlying and coupled to the horizontal sense transistors, and vertical write selector transistors vertically overlying and coupled to the horizontal sense transistors. Each of the each of the block of the stack structure comprises an access line section comprising tiers including access line structures, a select gate section overlying the access line section and comprising additional tiers including drain side select gate (SGD) structures, and a sense node section overlying the select gate section and comprising further tiers including select gate programming (SGP) structures. The pillar groups individually comprise multiple pillar structures vertically extending completely through one of the blocks, each pillar structure of the multiple pillar structures horizontally positioned within a different sub-block of the one of the blocks than each other pillar structure of the multiple pillar structures. The vertical write selector transistors are horizontally offset from the vertical read selector transistors.

In additional embodiments, the microelectronic device structure **100** may be formed to have a different configuration than that previously described with reference to FIGS. **1A** and **1B**. The microelectronic device structure **100** may, for example, be formed to exhibit a configuration such as one of the configurations depicted in FIGS. **2** and **3** and described in further detail below. With the description provided below, it will be readily apparent to one of ordinary

skill in the art that the structures and devices described herein may be included in relatively larger structures, devices, and systems.

Before referring to FIG. 2, it will be understood that throughout FIGS. 2, 3, and 4A through 4M and the associated description, features (e.g., regions, materials, structures, devices) functionally similar previously described features (e.g., previously described materials, structures, devices) are referred to with similar reference numerals incremented by 100. To avoid repetition, not all features shown in FIGS. 2, 3, and 4A through 4M are described in detail herein. Rather, unless described otherwise below, a feature in one or more of FIGS. 2, 3, and 4A through 4M designated by a reference numeral that is a 100 increment of the reference numeral of a feature previously described with reference to one or more of FIGS. 1A and 1B will be understood to be substantially similar to the previously described feature. As a non-limiting example, unless described otherwise below, features designated by the reference numerals 236, 336, and 436 in FIGS. 2, 3, and 4F (and subsequent FIGS. 4G through 4M), respectively, will be understood to respectively be substantially similar to the horizontal sense transistors 136 previously described herein with reference to FIGS. 1A and 1B.

FIG. 2 is simplified, partial top-down view of a microelectronic device structure 200 for a microelectronic device (e.g., a memory device, such as a NAND Flash memory device), in accordance with additional embodiments of the disclosure. The microelectronic device structure 200 may be similar to the microelectronic device structure 100 previously described with reference to FIGS. 1A and 1B, except that, for example, relative horizontal orientations and/or relative horizontal positions of some features of the microelectronic device structure 200 within horizontal areas of individual blocks 264 of a stack structure 204 are different than relative horizontal orientations and/or relative horizontal positions of corresponding features of the microelectronic device structure 100. For instance, as shown in FIG. 2, horizontal orientations of SGD plug structures 226 within a horizontal area of an individual block 264 of the stack structure 204 may be different than horizontal orientations of the SGD plug structures 126 of the microelectronic device structure 100 previously described with reference to FIG. 1B. As a result, horizontal orientations and/or horizontal positions of additional features of the microelectronic device structure 200 operatively associated with the SGD plug structures 226 may be different than horizontal orientations and/or horizontal positions of additional features of the microelectronic device structure 100 operatively associated with the SGD plug structures 126, as described in further detail below. It will be understood that a simplified, partial longitudinal cross-sectional view of the microelectronic device structure 200 about dashed line A₂-A₂ depicted in FIG. 2 is substantially similar to the simplified, partial longitudinal cross-sectional view of the microelectronic device structure 100 shown in FIG. 1A.

As shown in FIG. 2, the SGD plug structures 226 of the microelectronic device structure 200 may individually horizontally extend in the X-direction. Within a horizontal area of an individual block 264 of the stack structure 204, the SGD plug structures 226 may horizontally extend in parallel in the X-direction with the slot structures 266 horizontally neighboring the block 264 in the Y-direction; and may horizontally extend perpendicular to the digit line structures 258 horizontally extending in the Y-direction. Some of the SGD plug structures 226 may horizontally extend in parallel with one another in the X-direction; and other of the SGD

plug structures 226 may horizontally extend in series with one another in the X-direction.

Similar to the SGD plug structures 126 (FIGS. 1A and 1B), each of the SGD plug structures 226 may individually be operatively associated with a pillar group including four (4) pillar structures 222 vertically extending through the stack structure 204. An individual SGD plug structure 226 may be coupled to and may horizontally extend in the X-direction across and between the four (4) of the pillar structures 222 of the pillar group associated therewith. The four (4) pillar structures 222 of an individual pillar group operatively associated with an individual SGD plug structure 226 may substantially horizontally aligned with one another in the Y-direction.

Still referring to FIG. 2, each block 264 of the stack structure 204 may be sub-divided into multiple (e.g., a plurality of, more than one) sub-blocks 214. The sub-blocks 214 may horizontally extend parallel with one another in a direction acutely angled relative to each of the X-direction and the Y-direction shown in FIG. 2. Each of the sub-blocks 214 may individually be operatively associated with at least one set of the pillar structures 222 horizontally extending in the direction acutely angled relative to each of the X-direction and the Y-direction. In addition, for an individual block 264, multiple (e.g., a plurality of, more than one) sub-blocks 214 thereof may be grouped together with one another within sub-block groups 216. In FIG. 2, two (2) sub-block groups 216 are depicted, a second sub-block group 216B and a third sub-block group 216C. Each of the sub-block groups 216 may individually include a first sub-block 214A, a second sub-block 214B, a third sub-block 214C, and a fourth sub-block 214D. Sub-block groups 216 horizontally neighboring one another within an individual block 264 may exhibit an inverse horizontal arrangement of the different sub-blocks 214 thereof relative to one another. For example, the first sub-block 214A of the second sub-block group 216B may be most horizontally proximate the first sub-block 214A of the third sub-block group 216C horizontally neighboring the second sub-block group 216B.

Similar to the configuration of the microelectronic device structure 100 previously described with reference to FIGS. 1A and 1B, pillar structures 222 of the microelectronic device structure 200 operatively associated with an individual SGD plug structure 226 may be located within different sub-blocks 214 of an individual block 264 than one another. For example, within the horizontal area of a sub-section B₂ of the microelectronic device structure 200, one (1) of the SGD plug structures 226 may vertically overlie and be coupled to each of four (4) pillar structures 222 located in different sub-blocks 214 within the third sub-block group 216C than one another. A first of the pillar structures 222 may be located within the first sub-block 214A of the third sub-block group 216C; a second of the pillar structure 222 may be located within the second sub-block 214B of the third sub-block group 216C; a third of the pillar structures 222C may be located within the third sub-block 214C of the third sub-block group 216C; and a fourth of the pillar structure 222 may be located within the fourth sub-block 214D of the third sub-block group 216C.

With continued reference to FIG. 2, channel structures 234 of the microelectronic device structure 200 may be positioned within horizontal areas of the SGD plug structures 226, and may individually horizontally extend in the X-direction. The SGD plug structures 226, the channel structures 234, and gate dielectric structures (e.g., corresponding to the gate dielectric structures 130 previously described with reference to FIG. 1A) may together form

horizontal sense transistors **236** substantially similar to the horizontal sense transistors **136** previously described with reference to FIGS. **1A** and **1B**, except for the horizontal orientations thereof. In addition, the microelectronic device structure **200** may include tiers, memory cells, select transistors, V_t -enhanced sections, sense nodes, write pillar structures **244**, first read pillar structures **246**, second read pillar structures **248**, selector tiers, doped sections, write selector transistors, read selector transistors, read source line structures, local strap structures **254**, digit line contact structures **260**, digit line structures **258**, and isolation material respectively substantially similar to and utilized in substantially the same manner as the tiers **110**, the memory cells **121**, the select transistors **123**, the V_t -enhanced sections **118**, the sense nodes **112**, the write pillar structures **144**, the first read pillar structures **146**, the second read select pillar structures **148**, the selector tiers **138**, the doped sections **142**, the write selector transistors **131**, the read selector transistors **135**, the read source line structures **152**, the local strap structures **154**, the digit line contact structures **160**, the digit line structures **158**, and the isolation material **162** previously described with reference to FIGS. **1A** and **1B**.

FIG. **3** is simplified, partial longitudinal cross-sectional view of a microelectronic device structure **300** for a microelectronic device (e.g., a memory device, such as a NAND Flash memory device), in accordance with further embodiments of the disclosure. The microelectronic device structure **300** may be similar to the microelectronic device structure **100** previously described with reference to FIGS. **1A** and **1B**, except that, for example, the microelectronic device structure **300** may have a reduced quantity of tiers **310** and/or some of the tiers **310** may be utilized differently for various operations (e.g., SGPGM operations, program operations, read operations, erase operations) of a microelectronic device (e.g., a memory device, such as a 3D NAND Flash memory device) including the microelectronic device structure **300**. For instance, as shown in FIG. **3**, the microelectronic device structure **300** may be free of (e.g., may not include) tiers **310** corresponding to the read-amplification SGD tier **110H**, the first SGP bar tier **110I**, the second SGP bar tier **110J**, the first SGP tier **110K**, and the second SGP tier **110L** of the microelectronic device structure **100** previously described with reference to FIG. **1A**. Accordingly, a sense node tier section **311C** of a stack structure **304** of the microelectronic device structure **300** may be modified relative to the sense node tier section **111C** (FIG. **1A**) of the microelectronic device structure **100** (FIG. **1A**), and may include one or more (e.g., two or more, three or more) GG tiers **310M**. Sense nodes **312** of the microelectronic device structure **300** may be defined by and include SGD plug structures **326** and portions of the GG tiers **310M**. The sense nodes **312** may not include portions of SGP bar tiers and SGP tiers corresponding to the first SGP bar tier **110I**, the second SGP bar tier **110J**, the first SGP tier **110K**, and the second SGP tier **110L** of the microelectronic device structure **100** previously described with reference to FIG. **1A**.

In some embodiments, the microelectronic device structure **300** includes multiple (e.g., two or more, three or more) of the GG tiers **310M**. For example, as shown in FIG. **3**, the stack structure **304** of the microelectronic device structure **300** may include three (3) of the GG tiers **310M** in a vertically stacked arrangement with one another. The GG tiers **310M** may be vertically interposed between a second SGD tier **310G** of the stack structure **304** and a SGD plug tier **327** of the microelectronic device structure **300**.

Since the microelectronic device structure **300** does not include tiers **310** corresponding to the read-amplification

SGD tier **110H**, the first SGP bar tier **110I**, the second SGP bar tier **110J**, the first SGP tier **110K**, and the second SGP tier **110L** of the microelectronic device structure **100** previously described with reference to FIG. **1A**, operations (e.g., SGPGM operations, program operations, read operations, erase operations) that may otherwise utilize such omitted (e.g., absent) tiers **310** (e.g., in the manners previously described in relation to use and operation of a microelectronic device including the microelectronic device structure **100**) may be effectuated using different tiers **310** of the microelectronic device structure **300**. For example, for program operations, the first SGD bar tier **110D** and the second SGD bar tier **110E** may be employed to program V_t levels of the first SGD tier **110F** and the second SGD tier **110G**, respectively; and the first SGD bar tier **110D**, the second SGD bar tier **110E**, the first SGD tier **110F**, and the second SGD tier **110G** may be employed to program V_t levels of the program-inhibit SGD tier **310C**. In addition, for read operations, the program-inhibit SGD tier **310C** in a manner similar to that of the read-amplification SGD tier **110H** previously described with reference to FIGS. **1A** and **1B**; and the GG tiers **310M** may all be provided in an ON state, to effectively serve as a coupling capacitor for the sense nodes **312**.

FIGS. **4A** through **4M** are simplified, partial longitudinal cross-sectional views illustrating a microelectronic device structure at different processing stages of a method of forming a microelectronic device (e.g., a memory device, such as a 3D NAND Flash memory device), in accordance with embodiments of the disclosure. The methods described herein with reference to FIGS. **4A** through **4M** may be used to form one or more of the microelectronic device structures (e.g., the microelectronic device structure **100**, the microelectronic device structure **200**, the microelectronic device structure **300**) of the disclosure previously described herein with reference to FIGS. **1A** and **1B**, **2**, and **3**. With the description provided below, it will be readily apparent to one of ordinary skill in the art that the methods described herein may be used for forming various devices. In other words, the methods of the disclosure may be used whenever it is desired to form a microelectronic device.

FIG. **4A** is a simplified, longitudinal cross-sectional view of a portion of a microelectronic device structure **400** at a processing stage of a method of forming a microelectronic device, in accordance with embodiments of the disclosure. A horizontal position of the portion of the microelectronic device structure **400** shown in FIG. **4A** may correspond to the horizontal position of the sub-section B_1 of the microelectronic device structure **100** previously described with reference to FIGS. **1A** and **1B**; and a vertical position of the portion of the microelectronic device structure **400** shown in FIG. **4A** may correspond to that of a portion of the microelectronic device structure **100** vertically extending between the first SGP tier **110K** (FIG. **1A**) of the stack structure **104** (FIG. **1A**) and an upper vertical boundary of the stack structure **104** (FIG. **1A**).

As shown in FIG. **4A**, the microelectronic device structure **400** may be formed to include a preliminary stack structure including a vertically alternating sequence of insulative material **405** and sacrificial material **407** arranged in preliminary tiers **409**. The sacrificial material **407** may be vertically interleaved with the insulative material **405**, and each of the preliminary tiers **409** may include the sacrificial material **407** vertically neighboring the insulative material **405**. In addition, the microelectronic device structure **400** may be formed to include pillar structures **422** vertically extending through the preliminary tiers **409** of the prelimi-

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nary stack structure. The pillar structures **422** may correspond to the pillar structures **122** previously described herein with reference to FIGS. 1A and 1B.

The sacrificial material **407** of each of the preliminary tiers **409** of the preliminary stack structure may be formed of and include at least one material (e.g., at least one insulative material) that may be selectively removed relative to the insulative material **405** during subsequent processing (e.g., subsequent “replacement gate” or “gate last” processing) of the microelectronic device structure **400**, as described in further detail below. A material composition of the sacrificial material **407** is different than a material composition of the insulative material **405**. The sacrificial material **407** may be selectively etchable relative to the insulative material **405** during common (e.g., collective, mutual) exposure to a first etchant, and the insulative material **405** may be selectively etchable to the sacrificial material **407** during common exposure to a second, different etchant. As a non-limiting example, the sacrificial material **407** may be formed of and include additional insulative material, such as one or more of at least one dielectric oxide material (e.g., one or more of SiO_x , phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, AlO_x , HfO_x , NbO_x , TiO_x , ZrO_x , TaO_x , and MgO_x), at least one dielectric nitride material (e.g., SiN_y), at least one dielectric oxynitride material (e.g., SiO_xN_y), and at least one dielectric carboxynitride material (e.g., $\text{SiO}_x\text{C}_z\text{N}_y$). In some embodiments, the sacrificial material **407** is formed of and includes dielectric nitride material, such as SiN_y (e.g., Si_3N_4). The sacrificial material **407** of each of the preliminary tiers **409** may be substantially homogeneous, or the sacrificial material **407** of one or more (e.g., each) of the preliminary tiers **409** may be heterogeneous.

The insulative material **405** of each of the preliminary tiers **409** of the preliminary stack structure may be formed of and include at least one dielectric material, such one or more of at least one dielectric oxide material (e.g., one or more of SiO_x , phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, AlO_x , HfO_x , NbO_x , TiO_x , ZrO_x , TaO_x , and MgO_x), at least one dielectric nitride material (e.g., SiN_y), at least one dielectric oxynitride material (e.g., SiO_xN_y), and at least one dielectric carboxynitride material (e.g., $\text{SiO}_x\text{C}_z\text{N}_y$). In some embodiments, the insulative material **405** of each of the preliminary tiers **409** of the preliminary stack structure is formed of and includes dielectric oxide material, such as SiO_x (e.g., SiO_2). The insulative material **405** of each of the preliminary tiers **409** may be substantially homogeneous, or the insulative material **405** of one or more (e.g., each) of the preliminary tiers **409** may be heterogeneous.

The pillar structures **422** may each individually be formed of and include a stack of materials facilitating the use of the pillar structures **422** to form vertically extending strings of memory cells following subsequent processing acts, as described in further detail below. By way of non-limiting example, each of the pillar structures **422** may be formed to include cell film material **419A**, channel material **419B**, and dielectric fill material **419C**. The cell film material **419A** may be formed on or over surfaces of the sacrificial material **407** and the insulative material **405** of the preliminary tiers **409** of the preliminary stack structure and may be formed of and include a stack of materials, such as a charge-blocking material (e.g., a first dielectric oxide material, such as one or more of SiO_x and AlO_x), a charge-trapping material (e.g., dielectric nitride material, such as SiN_y) on the charge-blocking material, and a gate dielectric material (e.g., a second dielectric oxide material, such as SiO_x) on the

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charge-trapping material. The channel material **419B** may be located on the gate dielectric material of the cell film material **419A**, and may be formed of and include semiconductive material (e.g., silicon, such as polycrystalline silicon). The dielectric fill material **419C** may be located on the channel material **419B**, and may be formed of and include dielectric material (e.g., one or more of dielectric oxide material, dielectric nitride material, and an air gap).

As shown in FIG. 4A, in some embodiments, the cell film material **419A** and the channel material **419B** are formed to substantially continuously extend (e.g., horizontally extend, vertically extend) from and between horizontally neighboring pillar structures **422** within horizontal areas of different sub-blocks **414** (e.g., a first sub-block **414A**, a second sub-block **414B**, a third sub-block **414C**, a fourth sub-block **414D**) within the horizontal area of an individual sub-block group **416** (e.g., a third sub-block group **416C**). For example, portions of the cell film material **419A** and the channel material **419B** may substantially continuously horizontally extend on or over an uppermost surface of the preliminary stack structure (e.g., an uppermost surface of an uppermost preliminary tier **409**). In additional embodiments, the cell film material **419A** and the channel material **419B** do not horizontally extend from and between horizontally neighboring pillar structures **422** within the horizontal areas of horizontally neighboring sub-blocks **414**. For example, the cell film material **419A** and the channel material **419B** forming a portion of individual pillar structure **422** may be substantially confined within a horizontal area of the sub-block **414** in which the pillar structure **422** is located.

Still referring to FIG. 4A, for each pillar structure **422**, an upper vertical boundary (e.g., an upper surface) of the dielectric fill material **419C** thereof may be formed to be vertically recessed (e.g., to vertically underlie) relative to an upper vertical boundary of the channel material **419B** thereof. Accordingly, inner side surfaces (e.g., inner side walls) of the channel material **419B** forming portions of the pillar structures **422** may be exposed by openings (e.g., plug recesses) vertically extending into upper regions of the pillar structures **422**.

Referring next to FIG. 4B, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. 4A following the processing stage previously described with reference to FIG. 4A, at least one first masking structure **466** including openings **468** vertically extending therethrough may be formed on or over an uppermost one of the preliminary tiers **409** of the preliminary stack structure, and then one or more conductivity-enhancing species may be provided (e.g., implanted) into one or more of the preliminary tiers **409** at horizontal positions of the openings **468** in the first masking structure **466** to form one or more V_f -enhanced sections **418** of the microelectronic device structure **400**. As a non-limiting example, as shown in FIG. 4B, ninth V_f -enhanced sections **418I** corresponding to the ninth V_f -enhanced sections **118I** previously described with reference to FIG. 1A may be formed at horizontal positions and vertical locations corresponding to those previously described with reference to FIG. 1A.

In some embodiments, the first masking structure **466** is a photoresist mask formed of and including photoresist material, such as positive tone photoresist material or negative tone photoresist material. Suitable photoresist materials (e.g., positive tone photoresist materials, negative tone photoresist materials) are known in the art, and are, therefore, not described in detail herein. The first masking structure **466** may, for example, be compatible with 13.7 nm, 157 nm,

193 nm, 248 nm, or 365 nm wavelength systems; with 193 nm wavelength immersion systems; and/or with electron beam lithographic systems. In addition, the openings 468 in the first masking structure 466 may be formed using conventional processes (e.g., conventional photolithographic patterning and development processes) and conventional equipment, which are not described in detail herein.

In some embodiments, at least one P-type dopant (e.g., one or more of B, Al, and Ga) is implanted at one or more vertical positions (e.g., at a vertical position corresponding to that of first SGP tier 110K previously described with reference to FIG. 1A) within the microelectronic device structure 400 at the processing stage of FIG. 4B to form some of the V_f -enhanced sections 418 of the microelectronic device structure 400. The conductivity-enhancing species (e.g., P-type dopant) may be implanted into the microelectronic device structure 400 using conventional implantation processes (e.g., a plasma doping (PLAD) implantation process, a beam-line (BL) implantation process) and equipment, which are not described in detail herein.

Referring next to FIG. 4C, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure 400 shown in FIG. 4B following the processing stage previously described with reference to FIG. 4B, the first masking structure 466 may be removed and an second masking structure 470 include additional openings 472 vertically extending therethrough may be formed on or over the uppermost one of the preliminary tiers 409 of the preliminary stack structure. Horizontal positions of the additional openings 472 in the second masking structure 470 may be at least partially horizontally offset from the horizontal positions of the openings 468 (FIG. 4B) in the first masking structure 466 (FIG. 4B). Following the formation of the second masking structure 470, one or more conductivity-enhancing species may be provided (e.g., implanted) into one or more other of the preliminary tiers 409 at the horizontal positions of the additional openings 472 in the second masking structure 470 to form more of the V_f -enhanced sections 418 of the microelectronic device structure 400. As a non-limiting example, as shown in FIG. 4C, tenth V_f -enhanced sections 418J corresponding to the tenth V_f -enhanced sections 118J previously described with reference to FIG. 1A may be formed at horizontal positions and vertical locations corresponding to those previously described with reference to FIG. 1A.

In some embodiments, the second masking structure 470 is an additional photoresist mask formed of and including additional photoresist material, such as additional positive tone photoresist material or additional negative tone photoresist material. The second masking structure 470 may, for example, be compatible with 13.7 nm, 157 nm, 193 nm, 248 nm, or 365 nm wavelength systems; with 193 nm wavelength immersion systems; and/or with electron beam lithographic systems. Furthermore, the additional openings 472 in the second masking structure 470 may be formed using conventional processes (e.g., conventional photolithographic patterning and development processes) and conventional equipment, which are not described in detail herein.

In some embodiments, at least one P-type dopant (e.g., one or more of B, Al, and Ga) is implanted at one or more vertical positions (e.g., at a vertical position corresponding to that of second SGP tier 110L previously described with reference to FIG. 1A) within the microelectronic device structure 400 at the processing stage of FIG. 4C to form some more of the V_f -enhanced sections 418 of the microelectronic device structure 400. The conductivity-enhancing species (e.g., P-type dopant) may be implanted into the

microelectronic device structure 400 using conventional implantation processes (e.g., PLAD implantation process, a BL implantation process) and equipment, which are not described in detail herein.

Referring next to FIG. 4D, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure 400 shown in FIG. 4C following the processing stage previously described with reference to FIG. 4C, the second masking structure 470 (FIG. 4C) may be removed, and then SGD plug material 425, gate dielectric material 429, channel material 433, backside dielectric material 473, and dielectric cap material 475 may be formed in sequence over the pillar structures 422 and the uppermost one of the preliminary tiers 409 of the preliminary stack structure. The SGD plug material 425 may be formed on or over the channel material 419B and the dielectric fill material 419C of the pillar structures 422, and may substantially continuously extend from and between the pillar structures 422. The gate dielectric material 429 may be formed on or over the SGD plug material 425. The channel material 433 may be formed on or over the gate dielectric material 429. The backside dielectric material 473 may be formed on or over the channel material 433. The dielectric cap material 475 may be formed on or over the backside dielectric material 473.

As shown in FIG. 4D, the SGD plug material 425 may be formed to substantially fill the openings (e.g., plug recesses) in the upper regions of the pillar structures 422 partially defined by the upper vertical boundaries of the dielectric fill material 419C of the pillar structures 422. Accordingly, portions of the SGD plug material 425 may contact (e.g., physically contact, electrically contact) inner side surfaces (e.g., inner sidewalls) and upper surfaces of the channel material 419B forming portions of the pillar structures 422, and may also physically contact upper surfaces of the dielectric fill material 419C forming portions of the pillar structures 422. The SGD plug material 425 may vertically overlie the pillar structures 422, and may horizontally extend across and between the pillar structures 422. An upper vertical boundary (e.g., an upper surface) of the SGD plug material 425 may be substantially planar, and a lower vertical boundary (e.g., a lower surface) of the SGD plug material 425 may be non-planar. Within a horizontal area of an individual sub-block group 416 (e.g., the third sub-block group 416C), some of the pillar structures 422 covered by the SGD plug material 425 may be located in the same sub-block 414 as one another, and other of the pillar structures 422 covered by the SGD plug material 425 may be located in different sub-blocks 414 than one another. A material composition of the SGD plug material 425 may correspond to (e.g., be the same as) the material composition of the SGD plug structure 126 previously described with reference to FIGS. 1A and 1B. In some embodiments, the SGD plug material 425 is formed of and includes N-type polycrystalline silicon (e.g., polycrystalline silicon doped with at least one N-type dopant, such as polycrystalline silicon doped with phosphorous (P)).

The gate dielectric material 429 may be formed to vertically overlie and substantially continuously horizontally extend across and cover an upper surface of the SGD plug material 425. A material composition of the gate dielectric material 429 may correspond to (e.g., be the same as) the material composition of the gate dielectric structures 130 previously described with reference to FIG. 1A. In some embodiments, the gate dielectric material 429 is formed of and includes SiO_x (e.g., SiO_2).

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The channel material **433** may be formed to vertically overlie and substantially continuously horizontally extend across and cover an upper surface of the gate dielectric material **429**. A material composition of the channel material **433** may correspond to (e.g., be the same as) the material composition of the channel structures **134** previously described with reference to FIGS. 1A and 1B. In some embodiments, the channel material **433** is formed of and includes lightly P-type doped polycrystalline silicon (e.g., polycrystalline silicon doped with at least one P-type dopant, such as polycrystalline silicon doped with B).

The backside dielectric material **473** may be formed to vertically overlie and substantially continuously horizontally extend across and cover an upper surface of the channel material **433**. The backside dielectric material **473** may be formed of and include dielectric material, such as one or more of at least one dielectric oxide material (e.g., one or more of SiO_x , AlO_x , phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass), at least one dielectric nitride material (e.g., SiN_y), and at least one low-K dielectric material (e.g., one or more of SiO_xC_y , SiO_xN_y , $\text{SiC}_x\text{O}_y\text{H}_z$, and $\text{SiO}_x\text{C}_z\text{N}_y$). The backside dielectric material **473** may be substantially homogeneous, or the backside dielectric material **473** may be heterogeneous. In some embodiments, the backside dielectric material **473** is formed of and includes SiO_x (e.g., SiO_2).

The dielectric cap material **475** may be formed to vertically overlie and substantially continuously horizontally extend across and cover an upper surface of the backside dielectric material **473**. The dielectric cap material **475** may be formed of and include dielectric material, such as one or more of at least one dielectric oxide material (e.g., one or more of SiO_x , AlO_x , phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass), at least one dielectric nitride material (e.g., SiN_y), and at least one low-K dielectric material (e.g., one or more of SiO_xC_y , SiO_xN_y , $\text{SiC}_x\text{O}_y\text{H}_z$, and $\text{SiO}_x\text{C}_z\text{N}_y$). A material composition of the dielectric cap material **475** may be different than a material composition of the backside dielectric material **473**. The dielectric cap material **475** may be substantially homogeneous, or the dielectric cap material **475** may be heterogeneous. In some embodiments, the dielectric cap material **475** is formed of and includes SiN_y (e.g., Si_3N_4).

Referring next to FIG. 4E, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. 4D following the processing stage previously described with reference to FIG. 4D, a third masking structure **478** may be formed on or over the dielectric cap material **475**, and then portions of the dielectric cap material **475**, the backside dielectric material **473**, the channel material **433**, the gate dielectric material **429** (FIG. 4D), the SGD plug material **425** (FIG. 4D), and the channel material **419B**. The material removal process may form first openings **480** vertically extending from an upper boundary (e.g., an upper surface) of a remaining portion of the third masking structure **478** to or beyond an upper boundary of the cell film material **419A**.

The material removal process of FIG. 4E may form SGD plug structures **426** from (e.g., from remaining portions of) the SGD plug material **425** (FIG. 4D), and may form gate dielectric structures **430** from (e.g., from remaining portions of) the gate dielectric material **429** (FIG. 4D). The SGD plug structures **426** may be formed to have configurations corresponding to (e.g., substantially the same as) the configurations of the SGD plug structures **126** previously described with reference to FIGS. 1A and 1B. In addition, the gate dielectric structures **430** may be formed to have configura-

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tions corresponding to (e.g., substantially the same as) the configurations of the gate dielectric structures **130** previously described with reference to FIG. 1A. The first openings **480** horizontally intervene between and separate horizontally neighboring SGD plug structures **126** and horizontally neighboring gate dielectric structures **130**.

In some embodiments, the third masking structure **478** is a photoresist mask formed of and including photoresist material, such as positive tone photoresist material or negative tone photoresist material. The third masking structure **478** may, for example, be compatible with 13.7 nm, 157 nm, 193 nm, 248 nm, or 365 nm wavelength systems; with 193 nm wavelength immersion systems; and/or with electron beam lithographic systems. Furthermore, the first openings **480** may be formed using conventional processes (e.g., conventional photolithographic patterning and development processes, conventional etching processes) and conventional equipment, which are not described in detail herein.

Referring next to FIG. 4F, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. 4E following the processing stage previously described with reference to FIG. 4E, portions of the third masking structure **478** (FIG. 4E) may be removed; additional portions of the dielectric cap material **475** (FIG. 4E), the backside dielectric material **473** (FIG. 4E), the channel material **433** (FIG. 4E) not covered by a remaining portion of the third masking structure **478** (FIG. 4E) may be removed; and the remaining portion of the third masking structure **478** (FIG. 4E) may be removed. The material removal process may form second openings **482** from some of the first openings **480** (FIG. 4E), wherein horizontal dimensions of the second openings **482** at vertical positions above of the gate dielectric structures **430** are relatively greater than horizontal dimensions of the first openings **480** (FIG. 4E) at the vertical positions above of the gate dielectric structures **430**. As shown in FIG. 4E, portions of upper surfaces of the gate dielectric structures **430** may be exposed by the second openings **482**.

The material removal process of FIG. 4F may form channel structures **434** from (e.g., from remaining portions of) the channel material **433** (FIG. 4E), backside dielectric structures **474** from (e.g., from remaining portions of) the backside dielectric material **473** (FIG. 4E), and dielectric cap structures **476** from (e.g., from remaining portions of) dielectric cap material **475** (FIG. 4E). The channel structures **434** may be formed to have configurations corresponding to (e.g., substantially the same as) the configurations of the channel structures **134** previously described with reference to FIGS. 1A and 1B. In addition, the backside dielectric structures **474** and the dielectric cap structures **476** may be formed to have horizontal areas, horizontal shape, and horizontal positions substantially the same as the horizontal areas, the horizontal shapes, and the horizontal positions of the channel structures **434**, respectively.

The material removal process of FIG. 4F may form horizontal sense transistors **436** of the microelectronic device structure **400**. Each of the horizontal sense transistors **436** may individually include one (1) of the SGD plug structures **426**, one (1) of the channel structures **434**, and one (1) of the gate dielectric structures **430** vertically interposed between the one (1) of the SGD plug structures **426** and the one (1) of the channel structures **434**. The horizontal sense transistors **436** may be formed to have configurations corresponding to (e.g., substantially the same as) the configurations of the horizontal sense transistors **136** previously described with reference to FIGS. 1A and 1B. In addition, each of the horizontal sense transistors **436** may be opera-

tively associated with a group of the pillar structures **422** in a manner corresponding to (e.g., substantially the same as) that previously described in relation to operative association of the horizontal sense transistors **136** (FIGS. 1A and 1B) with the pillar groups **124** (FIG. 1A) of the pillar structures **122** (FIGS. 1A and 1B).

Referring next to FIG. 4G, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. 4F following the processing stage previously described with reference to FIG. 4F, isolation material **484** may be formed on or over exposed surfaces of the microelectronic device structure **400**. As shown in FIG. 4G, the isolation material **484** may substantially fill the first openings **480** (FIG. 4E) and the second openings **482** (FIG. 4F). The isolation material **484** may be formed on surfaces of the SGD plug structures **426**, the gate dielectric structures **430**, the channel structures **434**, the backside dielectric structures **474**, and the dielectric cap structures **476**. An upper vertical boundary (e.g., an upper surface) of the isolation material **484** may be substantially planar, and a lower vertical boundary (e.g., a lower surface) of the isolation material **484** may be non-planar.

The isolation material **484** may be formed of and include dielectric material, such as one or more of at least one dielectric oxide material (e.g., one or more of SiO_x , AlO_x , phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass), at least one dielectric nitride material (e.g., SiN_y), and at least one low-K dielectric material (e.g., one or more of SiO_xC_y , SiO_xN_y , $\text{SiC}_x\text{O}_y\text{H}_z$, and $\text{SiO}_x\text{C}_y\text{N}_y$). The isolation material **484** may be substantially homogeneous, or the isolation material **484** may be heterogeneous. In some embodiments, the isolation material **484** is formed of and includes SiO_x (e.g., SiO_2).

Referring next to FIG. 4H, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. 4G following the processing stage previously described with reference to FIG. 4G, an additional preliminary stack structure including preliminary selector tiers **437** each including additional sacrificial material **439** and additional insulative material **441** may be formed on or over the isolation material **484**; and then pillar openings **485** may be formed to vertically extend therethrough and to the horizontal sense transistors **436**. The pillar openings **485** may include first read pillar openings **485A**, second read pillar openings **485B**, and write pillar openings **485C**.

The additional sacrificial material **439** of each of the preliminary selector tiers **437** of the additional preliminary stack structure may be formed of and include at least one material (e.g., at least one insulative material) that may be selectively removed relative to the additional insulative material **441** during subsequent processing (e.g., subsequent "replacement gate" or "gate last" processing) of the microelectronic device structure **400**, as described in further detail below. A material composition of the additional sacrificial material **439** is different than a material composition of the additional insulative material **441**. The material composition of the sacrificial material **407** and may be substantially the same as the material composition of the sacrificial material **407** of the preliminary tiers **409** of the preliminary stack structure vertically underlying the SGD plug structures **426**. The additional sacrificial material **439** may be selectively etchable relative to the additional insulative material **441** during common (e.g., collective, mutual) exposure to a first etchant, and the additional insulative material **441** may be selectively etchable to the additional sacrificial material **439** during common exposure to a second, different etchant. As

a non-limiting example, the additional sacrificial material **439** may be formed of and include additional insulative material, such as one or more of at least one dielectric oxide material (e.g., one or more of SiO_x , phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, AlO_x , HfO_x , NbO_x , TiO_x , ZrO_x , TaO_x , and MgO_x), at least one dielectric nitride material (e.g., SiN_y), at least one dielectric oxynitride material (e.g., SiO_xN_y), and at least one dielectric carboxynitride material (e.g., $\text{SiO}_x\text{C}_y\text{N}_y$). In some embodiments, the additional sacrificial material **439** is formed of and includes dielectric nitride material, such as SiN_y (e.g., Si_3N_4). The additional sacrificial material **439** of each of the preliminary selector tiers **437** may be substantially homogeneous, or the additional sacrificial material **439** of one or more (e.g., each) of the preliminary selector tiers **437** may be heterogeneous.

The additional insulative material **441** of each of the preliminary selector tiers **437** of the additional preliminary stack structure may be formed of and include at least one dielectric material, such one or more of at least one dielectric oxide material (e.g., one or more of SiO_x , phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, AlO_x , HfO_x , NbO_x , TiO_x , ZrO_x , TaO_x , and MgO_x), at least one dielectric nitride material (e.g., SiN_y), at least one dielectric oxynitride material (e.g., SiO_xN_y), and at least one dielectric carboxynitride material (e.g., $\text{SiO}_x\text{C}_y\text{N}_y$). In some embodiments, the additional insulative material **441** of each of the preliminary selector tiers **437** of the additional preliminary stack structure is formed of and includes dielectric oxide material, such as SiO_x (e.g., SiO_2). The additional insulative material **441** of each of the preliminary selector tiers **437** may be substantially homogeneous, or the additional insulative material **441** of one or more (e.g., each) of the preliminary selector tiers **437** may be heterogeneous.

The additional preliminary stack structure may be formed to include any desired quantity of the preliminary selector tiers **437**. As shown in FIG. 4H in some embodiments, the additional preliminary stack structure is formed to include three (3) of the preliminary selector tiers **437**, a first preliminary selector tier **437A**, a second preliminary selector tier **437B**, and a third preliminary selector tier **437C**. In additional embodiments, the additional preliminary stack structure is formed to include a different quantity of the preliminary selector tiers **437**, such as two (2) of the preliminary selector tiers **437**, or greater than or equal to four (4) of the preliminary selector tiers **437**.

Still referring to FIG. 4H, the first read pillar openings **485A** and the second read pillar openings **485B** may be positioned within horizontal areas of the channel structures **434** (and, hence, the dielectric cap structures **476** and the backside dielectric structures **474**), and may vertical extend to or into the backside dielectric structures **474**. The write pillar openings **485C** may be positioned outside of the horizontal areas of the channel structures **434** (and, hence, the dielectric cap structures **476** and the backside dielectric structures **474**) but within horizontal areas of the SGD plug structures **426**, and may vertically extend to or into the SGD plug structures **426**. Within a horizontal area of an individual SGD plug structure **426**, one (1) of the first read pillar openings **485A** and one (1) of the second read pillar openings **485B** may be formed to vertically extend to and expose portions of one (1) of the backside dielectric structures **474**; and one (1) of the write pillar openings **485C** may be formed to vertically extend to and expose a portion of the SGD plug structures **426**. As shown in FIG. 4H, an individual write pillar openings **485C** may be formed to vertically extend through the preliminary selector tiers **437** (including through

the additional sacrificial material **439** and additional insulative material **441** thereof), the isolation material **484**, and an individual gate dielectric structures **430**, and into an individual SGD plug structure **426**. In addition, each of an individual first read pillar opening **485A** and an individual second read pillar openings **485B** may be formed to vertically extend through the preliminary selector tiers **437** (including through the additional sacrificial material **439** and additional insulative material **441** thereof), the isolation material **484**, and an individual dielectric cap structure **476**, and to or into an individual backside dielectric structure **474**.

The first read pillar openings **485A**, the second read pillar openings **485B**, and the write pillar openings **485C** may be formed substantially simultaneously within one another. In some embodiments, the formation of the first read pillar openings **485A**, the second read pillar openings **485B**, and the write pillar openings **485C** utilizing multiple material removal processes. For example, by way of first material removal process (e.g., a first anisotropic etching process), initial portions of the first read pillar openings **485A** and the second read pillar openings **485B** may be formed substantially simultaneously with one another and initial portions of the write pillar openings **485C**. The initial portions of the first read pillar openings **485A** and the second read pillar openings **485B** may vertically terminate at or within the dielectric cap structures **476**. The initial portions of the write pillar openings **485C** may vertically terminate at or within the SGD plug structures **426**. Next, by way of second material removal process (e.g., a second anisotropic etching process), the initial portions of the first read pillar openings **485A** and the second read pillar openings **485B** and the initial portions of the write pillar openings **485C** may be vertically extended substantially simultaneously with one another to form the first read pillar openings **485A**, the second read pillar openings **485B**, and the write pillar openings **485C**, respectively. The second material removal process may remove portions of the dielectric cap structures **476** exposed by the initial portions of the first read pillar openings **485A** and the second read pillar openings **485B**, and may also remove portions of the SGD plug structures **426** exposed by the initial portions of the write pillar openings **485C**.

Referring next to FIG. 4I, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. 4H following the processing stage previously described with reference to FIG. 4H, gate dielectric material **488** may be formed inside and outside of the pillar openings **485** (FIG. 4H), and then portions of the microelectronic device structure **400** vertically underlying and within horizontal areas of the pillar openings **485** (FIG. 4H) may be removed to vertically extend the pillar openings **485** (FIG. 4H) and form extended pillar openings **486**. The extended pillar openings **486** may include first extended read pillar openings **486A**, second extended read pillar openings **486B**, and extended write pillar openings **486C**.

As shown in FIG. 4I, the gate dielectric material **488** may be formed on and may substantially cover side surfaces of the preliminary selector tiers **437** (including side surfaces of the additional sacrificial material **439** and additional insulative material **441** thereof) partially defining the pillar openings **485** (FIG. 4H). The gate dielectric material **488** may also be formed on and may substantially cover an upper surface of an uppermost one of the preliminary selector tiers **437** outside of the horizontal boundaries of the pillar openings **485** (FIG. 4H). For portions of gate dielectric material **488** partially defining the first extended read pillar openings

486A and the second extended read pillar openings **486B**, the gate dielectric material **488** may also be formed on and may substantially cover side surfaces of the dielectric cap structures **476**, and may vertically terminate at or within the backside dielectric structures **474**. Vertical dimensions of the portions of the gate dielectric material **488** partially defining the first extended read pillar openings **486A** and the second extended read pillar openings **486B** may be less than the vertical dimensions of the first extended read pillar openings **486A** and the second extended read pillar openings **486B**. Furthermore, for additional portions of the gate dielectric material **488** partially defining the extended write pillar openings **486C**, the gate dielectric material **488** may vertically terminate at or within the SGD plug structures **426**. Vertical dimensions of the additional portions of the gate dielectric material **488** partially defining the extended write pillar openings **486C** may be less than the vertical dimensions of the extended write pillar openings **486C**.

The gate dielectric material **488** may be formed of and include dielectric material, such as one or more of at least one dielectric oxide material (e.g., one or more of SiO_x , AlO_x , phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass), at least one dielectric nitride material (e.g., SiN_y), and at least one low-K dielectric material (e.g., one or more of SiO_xC_y , SiO_xN_y , $\text{SiC}_x\text{O}_y\text{H}_z$, and $\text{SiO}_x\text{C}_y\text{N}_z$). The gate dielectric material **488** may be substantially homogeneous, or the gate dielectric material **488** may be heterogeneous. In some embodiments, the gate dielectric material **488** is formed of and includes SiO_x (e.g., SiO_2).

Still referring to FIG. 4I, the first extended read pillar openings **486A** and the second extended read pillar openings **486B** may have horizontal boundaries defined by side surfaces of the gate dielectric material **488**, the backside dielectric structures **474**, and the channel structures **434**; and may have lower vertical boundaries defined by upper surfaces of the gate dielectric structures **430**. Horizontal dimensions of portions of the first extended read pillar openings **486A** and the second extended read pillar openings **486B** at vertical positions of the backside dielectric structures **474** and lower portions of the dielectric cap structures **476** may be relatively smaller (e.g., horizontally narrower) than horizontal dimensions of additional portions of the first extended read pillar openings **486A** and the second extended read pillar openings **486B** at vertical positions of the channel structures **434** and the preliminary selector tiers **437**.

Still referring to FIG. 4I, the extended write pillar openings **486C** may have horizontal boundaries defined by side surfaces of the gate dielectric material **488** and the SGD plug structures **426**; and may have lower vertical boundaries defined by surfaces of the SGD plug structures **426**. Horizontal dimensions of portions of the extended write pillar openings **486C** at relatively higher vertical positions within the SGD plug structures **426** may be relatively smaller (e.g., horizontally narrower) than horizontal dimensions of additional portions of the first extended read pillar openings **486A** and the second extended read pillar openings **486B** at relatively lower vertical positions within the SGD plug structures **426** and at vertical positions of isolation material **484** and the preliminary selector tiers **437**.

To form the gate dielectric material **488** and the extended pillar openings **486**, the gate dielectric material **488** may be formed inside and outside of the pillar openings **485** (FIG. 4H) and may partially (e.g., less than completely) fill the pillar openings **485** (FIG. 4H). Thereafter, a sacrificial liner material, such as a semiconductor liner material (e.g., polycrystalline silicon liner) may be formed on or over the

gate dielectric material **488**, and may partially (e.g., less than completely) fill remaining portions of the pillar openings **485** (FIG. 4H). A relative thickness of the sacrificial liner material formed within the pillar openings **485** (FIG. 4H) is represented by way of the dashed line **490** within the extended write pillar opening **486C** shown in FIG. 4I. Following the formation of sacrificial liner material, portions of the semiconductor liner material and the gate dielectric material **488** proximate lower vertical boundaries (e.g., bottoms) of the pillar openings **485** (FIG. 4H) may be removed, such as by way of a so-called “punch through” etch. Within horizontal areas of the first read pillar openings **485A** (FIG. 4H) and the second read pillar openings **485B** (FIG. 4H), the material removal process may expose positions of the channel structures **434**. Within horizontal areas of the write pillar openings **485C** (FIG. 4H), material removal process may expose positions of the SGD plug structures **426**. Thereafter, at least one additional material removal process (e.g., at least one etching process, such as at least one wet etching process) may be performed to remove the semiconductor liner material, as well as exposed portions of the channel structures **434** and expose positions of the SGD plug structures **426**. The additional material removal process may vertically recess and horizontally recess portions of the channel structures **434**, and may complete the formation of the first extended read pillar openings **486A** and the second extended read pillar openings **486B**. The additional material removal process may also vertically recess and horizontally recess the portions of the SGD plug structures **426**, and may complete the formation of the extended write pillar openings **486C**.

Referring next to FIG. 4J, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. 4J following the processing stage previously described with reference to FIG. 4J, additional channel material **492**, additional dielectric fill material **494**, and plug material **496** may be formed inside and outside of the extended pillar openings **486** (FIG. 4I) (including the first extended read pillar openings **486A** (FIG. 4I), the second extended read pillar openings **486B** (FIG. 4I), and the extended write pillar openings **486C** (FIG. 4I)); and then portions of the gate dielectric material **488**, the additional channel material **492**, the additional dielectric fill material **494**, and the plug material **496** outside of the boundaries of the extended pillar openings **486** (FIG. 4I) may be removed (e.g., by way of a CMP process) to form write select pillar structures **444**, first read select pillar structures **446**, and second read select pillar structures **448**. The write select pillar structures **444**, the first read select pillar structures **446**, and the second read select pillar structures **448** may respectively be formed to have configurations corresponding to (e.g., substantially the same as) the configurations of the write select pillar structures **144**, the first read select pillar structures **146**, and the second read select pillar structures **148** previously described with reference to FIGS. 1A and 1B.

As shown in FIG. 4J, the additional channel material **492** may be formed to substantially fill portions of the extended write pillar openings **486C** (FIG. 4I) at the vertical position of the SGD plug structures **426**. Accordingly, the additional channel material **492** of the write select pillar structures **444** may be formed to contact (e.g., physically contact, electrically contact) the SGD plug structures **426** of the horizontal sense transistors **436**. The additional channel material **492** may also substantially fill additional portions of the extended write pillar openings **486C** (FIG. 4I) at the vertical positions of the gate dielectric structures **430**. In addition,

additional channel material **492** may partially fill further portions of the extended write pillar openings **486C** (FIG. 4I) vertically overlying the additional portions of the extended write pillar openings **486C** (FIG. 4I), such as portions extending from the gate dielectric structures **430** to or beyond an upper boundary of an uppermost one of the preliminary selector tiers **437**.

As also shown in FIG. 4J, the additional channel material **492** may be formed to substantially fill portions of the first extended read pillar openings **486A** (FIG. 4I) and the second extended read pillar openings **486B** (FIG. 4I) at the vertical position of the channel structures **434**. Accordingly, the additional channel material **492** of the first read select pillar structures **446** and the second read select pillar structures **448** may be formed to contact (e.g., physically contact, electrically contact) the channel structures **434** of the horizontal sense transistors **436**. The additional channel material **492** may also substantially fill additional portions of the first extended read pillar openings **486A** (FIG. 4I) and the second extended read pillar openings **486B** (FIG. 4I) at the vertical positions of the backside dielectric structures **474** and lower portions of the dielectric cap structures **476**. In addition, additional channel material **492** may partially fill further portions of the first extended read pillar openings **486A** (FIG. 4I) and the second extended read pillar openings **486B** (FIG. 4I) vertically overlying the additional portions of the first extended read pillar openings **486A** (FIG. 4I) and the second extended read pillar openings **486B** (FIG. 4I), such as portions extending from the lower portions of the dielectric cap structures **476** to or beyond an upper boundary of an uppermost one of the preliminary selector tiers **437**.

The additional dielectric fill material **494** may partially fill portions of the extended pillar openings **486** (FIG. 4I) not occupied by the additional channel material **492**. For each of the write select pillar structures **444**, the first read select pillar structures **446**, and the second read select pillar structures **448**, surfaces of the additional channel material **492** may define horizontal boundaries and lower vertical boundaries of the additional dielectric fill material **494**; and an upper vertical boundary (e.g., an upper surface) of the additional dielectric fill material **494** may be formed to be vertically recessed (e.g., to vertically underlie) relative to an upper vertical boundary of the additional channel material **492**. The additional dielectric fill material **494** may be formed on the additional channel material **492**, and may be formed of and include dielectric material (e.g., one or more of dielectric oxide material, dielectric nitride material, and an air gap).

The plug material **496** may fill upper portions of the extended pillar openings **486** (FIG. 4I) not occupied by the additional channel material **492** and the additional dielectric fill material **494**. For each of the write select pillar structures **444**, the first read select pillar structures **446**, and the second read select pillar structures **448**, side surfaces of the additional channel material **492** may define horizontal boundaries of the plug material **496**, and upper surfaces of the additional dielectric fill material **494** may define lower vertical boundaries of the plug material **496**. An upper vertical boundary (e.g., an upper surface) of the plug material **496** may be formed to be substantially coplanar with to the upper vertical boundary (e.g., the upper surface) of the additional channel material **492**. The plug material **496** may be formed on the additional dielectric fill material **494** and the additional channel material **492**. The plug material **496** may be formed of and include conductive material, such as semiconductor material (e.g., polycrystalline silicon) doped with doped with at least one conductivity-enhancing dopant

(e.g., at least one N-type dopant, such as one or more of P, As, Sb, and Bi; at least one P-type dopant, such as one or more of B, Al, and Ga). In some embodiments, the plug material **496** is formed of and includes N-type polycrystalline silicon (e.g., polycrystalline silicon doped with at least one N-type dopant, such as polycrystalline silicon doped with phosphorous (P)).

Referring next to FIG. **4K**, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. **4J** following the processing stage previously described with reference to FIG. **4J**, at least one third masking structure **497** including openings vertically extending therethrough may be formed on or over an uppermost one of the preliminary selector tiers **437** (e.g., on or over the third preliminary selector tier **437C**), and then one or more conductivity-enhancing species may be provided (e.g., implanted) into one or more of the preliminary selector tiers **437** at horizontal positions of the openings in the third masking structure **497**. The doping process may form some doped sections **442** (e.g., first doped sections **442A**) and some further doped sections **443** at desired vertical positions within vertical boundaries of the preliminary selector tiers **437**. As a non-limiting example, first doped sections **442A** corresponding to the first doped sections **142A** previously described with reference to FIG. **1A** may be formed at horizontal positions and vertical locations corresponding to those previously described with reference to FIG. **1A**. As another non-limiting example, further doped sections **443** may be formed to vertically overlie and at least partially horizontally overlap the first doped sections **442A**.

In some embodiments, the third masking structure **497** is a photoresist mask formed of and including photoresist material, such as positive tone photoresist material or negative tone photoresist material. The third masking structure **497** may, for example, be compatible with 13.7 nm, 157 nm, 193 nm, 248 nm, or 365 nm wavelength systems; with 193 nm wavelength immersion systems; and/or with electron beam lithographic systems. In addition, the openings in the third masking structure **497** may be formed using conventional processes (e.g., conventional photolithographic patterning and development processes) and conventional equipment, which are not described in detail herein.

To form the first doped sections **442A**, at least one conductivity-enhancing species (e.g., at least one P-type dopant) may be implanted into portions of the first preliminary selector tier **437A** within horizontal areas of the openings in the third masking structure **497** and into portions of the write select pillar structures **444** at the vertical position of the first preliminary selector tier **437A**. In some embodiments, the conductivity-enhancing species includes at least one P-type dopant (e.g., one or more of B, Al, and Ga). The conductivity-enhancing species (e.g., P-type dopant) may be implanted using conventional implantation processes (e.g., a PLAD implantation process, a BL implantation process) and equipment, which are not described in detail herein.

To form the further doped sections **443**, at least one additional conductivity-enhancing species (e.g., at least one N-type dopant) different than that employed in the first doped sections **442A** may be implanted into portions of the second preliminary selector tier **437B** within the horizontal areas of the openings in the third masking structure **497** and into portions of the write select pillar structures **444** at the vertical position of the second preliminary selector tier **437B**. In some embodiments, the additional conductivity-enhancing species includes at least one N-type dopant (e.g., one or more of P, As, Sb, and Bi). The further doped sections

443 may be formed after the formation of the first doped sections **442A**. The additional conductivity-enhancing species (e.g., N-type dopant) may be implanted using conventional implantation processes (e.g., a PLAD implantation process, a BL implantation process) and equipment, which are not described in detail herein.

Referring next to FIG. **4L**, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure **400** shown in FIG. **4K** following the processing stage previously described with reference to FIG. **4K**, one or more additional conductivity-enhancing species may be provided (e.g., implanted) into one or more of the preliminary selector tiers **437**. The doping process may at least form some more doped sections **442** (e.g., second doped sections **442B**, third doped sections **442C**) at desired vertical positions within vertical boundaries of the preliminary selector tiers **437**. As a non-limiting example, second doped sections **442B** and third doped sections **442C** respectively corresponding to the second doped sections **142B** and the third doped sections **142C** previously described with reference to FIG. **1A** may be formed at horizontal positions and vertical locations corresponding to those previously described with reference to FIG. **1A**.

To form the second doped sections **442B**, at least one conductivity-enhancing species (e.g., at least one P-type dopant) may be implanted into portions of the second preliminary selector tier **437B** and at least into portions of the first read select pillar structures **446** and the second read select pillar structures **448** at the vertical position of the second preliminary selector tier **437B**. In some embodiments, the conductivity-enhancing species includes at least one P-type dopant (e.g., one or more of B, Al, and Ga). Optionally, the conductivity-enhancing species may also be implanted into portions of the write select pillar structures **444** at the vertical position of the second preliminary selector tier **437B**, such that the portions of the write select pillar structures **444** are doped with at least one conductivity-enhancing species (e.g., at least one N-type dopant) as a result of the processing stage of FIG. **4K** and are also doped with at least one different conductivity enhancing species (e.g., at least one P-type dopant) as a result of the processing stage of FIG. **4K**. As a result, the write selector transistors subsequently formed at the portions of the write select pillar structures **444** at the vertical position of the second preliminary selector tier **437B** may have different V_t (e.g., negative V_t) characteristics than V_t (e.g., positive V_t) characteristics of read selector transistors subsequently formed at the portions of the read select pillar structures **446** at the vertical position of the second preliminary selector tier **437B**. The conductivity-enhancing species (e.g., P-type dopant) may be implanted during that processing stage of FIG. **4L** using conventional implantation processes (e.g., a PLAD implantation process, a BL implantation process) and equipment, which are not described in detail herein.

To form the third doped sections **442C**, at least one conductivity-enhancing species (e.g., at least one P-type dopant) may be implanted into portions of the third preliminary selector tier **437C** and into portions of the write select pillar structures **444**, the first read select pillar structures **446**, and the second read select pillar structures **448** at the vertical position of the third preliminary selector tier **437C**. In some embodiments, the conductivity-enhancing species includes at least one P-type dopant (e.g., one or more of B, Al, and Ga). The conductivity-enhancing species (e.g., P-type dopant) may be implanted using conventional

implantation processes (e.g., a PLAD implantation process, a BL implantation process) and equipment, which are not described in detail herein.

Referring next to FIG. 4M, which is a simplified, longitudinal cross-sectional view of the portion of the microelectronic device structure 400 shown in FIG. 4M following the processing stage previously described with reference to FIG. 4M, the preliminary tiers 409 (FIG. 4K) and the preliminary selector tiers 437 (FIG. 4K) may be subjected to so-called “replacement gate” or “gate last” processing to form tiers 410 from the preliminary tiers 409 (FIG. 4K) and selector tiers 428 from the preliminary selector tiers 437 (FIG. 4K). The tiers 410 may each include at least one conductive structure 406 vertically neighboring at least one insulative structure 408; and the selector tiers 438 may each include at least one additional conductive structure 440 vertically neighboring a remaining portion of the additional insulative material 441. Following the replacement gate process, read source line structures 452 may be formed vertically over and in contact (e.g., physical contact, electrical contact) with the first read select pillar structures 446; and local strap structures 454 may be formed vertically over and in contact (e.g., physical contact, electrical contact) with the write select pillar structures 444 and the second read select pillar structures 448. Thereafter, digit line contact structures 460 may be formed vertically over and in contact (e.g., physical contact, electrical contact) with the local strap structures 454; and digit line structures 458 may be formed vertically over and in contact with the digit line contact structures 460. The tiers 410, the selector tiers 438, the read source line structures 452, the local strap structures 454, the digit line contact structures 460, and the digit line structures 458 may respectively be formed to have configurations corresponding to (e.g., substantially the same as) the configurations of the tiers 110, the selector tiers 138, the read source line structures 152, the local strap structures 154, the digit line contact structures 160, and the digit line structures 158 previously described with reference to FIGS. 1A and 1B.

Referring to back to FIG. 4K, the replacement gate process effectuated at the processing stage of FIG. 4M may include forming slots (e.g., slits, openings) vertically extending through the preliminary selector tiers 437 and the preliminary tiers 409, and then treating the microelectronic device structure 400 with at least one wet etchant formulated to selectively remove portions of the sacrificial material 407 of the preliminary tiers 409 and the additional sacrificial material 439 of the preliminary selector tiers 437 through the slots. The wet etchant may be selected to remove the portions of the sacrificial material 407 and the additional sacrificial material 439 without substantially removing portions of the insulative material 405 of the preliminary tiers 409 and additional insulative material 441 of the preliminary selector tiers 437. In some embodiments wherein the sacrificial material 407 and the additional sacrificial material 439 comprise dielectric nitride material (e.g., SiN_x , such as Si_3N_4) and the insulative material 405 and the additional insulative material 441 comprise dielectric oxide material (e.g., SiO_x , such as SiO_2), the sacrificial material 407 and the additional sacrificial material 439 are selectively removed using a wet etchant comprising H_3PO_4 . Following the selective removal of the portions of the sacrificial material 407 and the additional sacrificial material 439, the resulting recesses may be filled with conductive material to form the conductive structures 406 (FIG. 4M) and the additional conductive structures 440 (FIG. 4M). The conductive structures 406 (FIG. 4M) and the additional conductive structures

440 (FIG. 4M) may be formed substantially simultaneously with one another through the replacement gate process.

Referring again to FIG. 4M, the replacement gate process may effectuate the formation of select transistors 423 within vertical boundaries of the tiers 410 of the microelectronic device structure 400, as well as the formation of write selector transistors 431 and read selector transistors 435 within vertical boundaries of the selector tiers 438 of the microelectronic device structure 400. The select transistors 423, the write selector transistors 431, and read selector transistors 435 may respectively have configurations corresponding to (e.g., substantially the same as) the configurations of select transistors 123, the write selector transistors 131, and read selector transistors 135 previously described with reference to FIGS. 1A and 1B. In addition, the replacement gate process may also effectuate the formation of strings of memory cells operatively associated with the pillar structures 422, wherein the strings of memory cells have configurations corresponding to (e.g., substantially the same as) the configurations of the strings of memory cells 121 previously described with reference to FIGS. 1A and 1B.

Following the replacement gate process, the read source line structures 452, the local strap structures 454, the digit line contact structures 460, the digit line structures 458, and different portions of at least one isolation material 462 may be individually formed using conventional processes (e.g., convention material deposition processes, conventional photolithographic patterning processes, conventional material removal processes) and conventional processing equipment, which are not described in detail herein.

Thus, in accordance with embodiments of the disclosure, a method of forming a microelectronic device comprises forming pillar structures to vertically extend through a stack structure comprising tiers each including sacrificial material vertically adjacent insulative material. P-type dopant is implanted into different horizontal portions of some of the tiers and into portions of the pillar structures at vertical positions of the some of the tiers. Sense transistors are formed vertically over the stack structure. The sense transistors are coupled to different groups of the pillar structures than one another. An additional stack structure is formed over the sense transistors. The additional stack structure comprises additional tiers each including additional sacrificial material vertically adjacent additional insulative material. Additional pillar structures are formed to vertically extend through the additional stack structure and to the sense transistors. Additional P-type dopant is implanted into different horizontal portions of two or more of the additional tiers and into portions of the additional pillar structures at vertical elevations of the two or more of the additional tiers. The sacrificial material and the additional sacrificial material are replaced with conductive material. Conductive line structures are formed over and in electrical communication with the additional pillar structures.

Microelectronic devices structures (e.g., the microelectronic device structures 100, 200, 300, 400) and microelectronic devices in accordance with embodiments of the disclosure may be used in embodiments of electronic systems of the disclosure. For example, FIG. 5 is a schematic block diagram of an illustrative electronic system 500 according to embodiments of disclosure. The electronic system 500 may comprise, for example, a computer or computer hardware component, a server or other networking hardware component, a cellular telephone, a digital camera, a personal digital assistant (PDA), portable media (e.g., music) player, a Wi-Fi or cellular-enabled tablet such as, for example, one or more of an iPad® or SURFACE® tablet, an

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electronic book, and a navigation device. The electronic system 500 includes at least one memory device 502. The memory device 502 may comprise, for example, one or more of a microelectronic device structure (e.g., one or more of the microelectronic device structures 100, 200, 300, 400) and a microelectronic device previously described herein. The electronic system 500 may further include at least one electronic signal processor device 504 (often referred to as a “microprocessor”). The electronic signal processor device 504 may, optionally, include one or more of a microelectronic device structure (e.g., one or more of the microelectronic device structures 100, 200, 300, 400) and a microelectronic device previously described herein. While the memory device 502 and the electronic signal processor device 504 are depicted as two (2) separate devices in FIG. 5, in additional embodiments, a single (e.g., only one) memory/processor device having the functionalities of the memory device 502 and the electronic signal processor device 504 is included in the electronic system 500. In such embodiments, the memory/processor device may include one or more of a microelectronic device structure (e.g., one or more of the microelectronic device structures 100, 200, 300, 400) and a microelectronic device previously described herein. The electronic system 500 may further include one or more input devices 506 for inputting information into the electronic system 500 by a user, such as, for example, a mouse or other pointing device, a keyboard, a touchpad, a button, or a control panel. The electronic system 500 may further include one or more output devices 508 for outputting information (e.g., visual or audio output) to a user such as, for example, one or more of a monitor, a display, a printer, an audio output jack, and a speaker. In some embodiments, the input device 506 and the output device 508 may comprise a single touchscreen device that can be used both to input information to the electronic system 500 and to output visual information to a user. The input device 506 and the output device 508 may communicate electrically with one or more of the memory device 502 and the electronic signal processor device 504.

Thus, an electronic system in accordance with embodiments of the disclosure comprises an input device, an output device, a processor device operably connected to the input device and the output device, and a memory device operably connected to the processor device. The memory device comprises pillar structures vertically extending through blocks of a stack structure comprising conductive material vertically alternating with insulative material; conductive plug structures within horizontal areas of the blocks of the stack structure and coupled to different groups of the pillar structures than one another, each of the different groups of the pillar structures including multiple of the pillar structures positioned within different sub-blocks of one of the blocks than one another; horizontal sense transistors gated by the conductive plug structures; read source lines vertically overlying the horizontal sense transistors; digit lines vertical overlying the read source lines; first vertical read selector transistors vertically interposed between and in electrical communication with the read source lines and source regions of the horizontal sense transistors; second vertical read selector transistors vertically interposed between and in electrical communication with the digit lines structures and drain regions of the horizontal sense transistors; and vertical write selector transistors vertically interposed between and in electrical communication with the digit lines structures and the conductive plug structures.

The structures, devices, methods, and systems of the disclosure advantageously facilitate one or more of

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improved microelectronic device performance, reduced costs (e.g., manufacturing costs, material costs), increased miniaturization of components, and greater packaging density as compared to conventional structures, conventional devices, conventional methods, and conventional systems. The structure, devices, methods, and systems of the disclosure may also improve scalability, efficiency, and simplicity as compared to conventional structures, conventional devices, conventional methods, and conventional systems.

While the disclosure is susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and have been described in detail herein. However, the disclosure is not limited to the particular forms disclosed. Rather, the disclosure is to cover all modifications, equivalents, and alternatives falling within the scope of the following appended claims and their legal equivalent. For example, elements and features disclosed in relation to one embodiment may be combined with elements and features disclosed in relation to other embodiments of the disclosure.

What is claimed is:

1. A microelectronic device, comprising:

a stack structure comprising a vertically alternating sequence of conductive material and insulative material, the stack structure divided into blocks separated by dielectric slot structures, the blocks individually including sub-blocks horizontally extending in parallel with one another;

pillar structures comprising semiconductor material vertically extending through one of the blocks of the stack structure, each pillar structure of a group of the pillar structures horizontally positioned within a different one of the sub-blocks of the one of the blocks than each other pillar structure of the group of the pillar structures;

a conductive plug structure coupled to and horizontally extending across and between multiple of the pillar structures of the group of the pillar structures;

a sense transistor gated by the conductive plug structure; and

selector transistors coupling the sense transistor to a read source line structure and a digit line structure.

2. The microelectronic device of claim 1, wherein the stack structure comprises tiers each including the conductive material and the insulative material vertically adjacent the conductive material, the tiers grouped into tier sections comprising:

an access line tier section comprising a first group of the tiers;

stacked drain side select gate (SGD) tier section overlying the access line tier section and comprising a second group of the tiers; and

a sense node tier section overlying the SGD tier section and comprising a third group of the tiers.

3. The microelectronic device of claim 2, wherein the conductive plug structure and portions of each of the second group of the tiers and the third group of the tiers within a horizontal area of the one of the blocks of the stack structure occupied by the group of the pillar structures define a sense node of the one of the blocks.

4. The microelectronic device of claim 2, wherein:

the second group of the tiers of the SGD tier section comprises:

a program-inhibit SGD tier overlying the access line tier section;

multiple SGD tiers overlying the program-inhibit SGD tier; and

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a read-amplification SGD tier overlying a second SGD tier; and
 the third group of the tiers of the sense node tier section comprises multiple select gate programming (SGP) tiers overlying the read-amplification SGD tier.

5. The microelectronic device of claim 4, wherein: the multiple SGD tiers comprise:
 a first SGD bar tier overlying the program-inhibit SGD tier;
 a second SGD bar tier overlying the first SGD bar tier;
 a first SGD tier overlying the second SGD bar tier; and
 a second SGD tier overlying the first SGD tier; and
 the multiple SGP tiers comprise:
 a first SGP bar tier overlying the read-amplification SGD tier;
 a second SGP bar tier overlying the first SGP bar tier;
 a first SGP tier overlying the second SGP bar tier; and
 a second SGP tier overlying the first SGP tier.

6. The microelectronic device of claim 5, wherein each of the first SGD bar tier, the second SGD bar tier, the first SGD tier, and the second SGD tier comprises a different horizontal arrangement of higher V_t transistors and lower V_t transistors within vertical boundaries thereof than each other of the first SGD bar tier, the second SGD bar tier, the first SGD tier, and the second SGD tier.

7. The microelectronic device of claim 5, wherein each of the first SGP bar tier, the second SGP bar tier, the first SGP tier, and the second SGP tier comprises a different horizontal arrangement of higher V_t transistors and lower V_t transistors within vertical boundaries thereof than each other of the first SGP bar tier, the second SGP bar tier, the first SGP tier, and the second SGP tier.

8. The microelectronic device of claim 4, wherein intersections of the pillar structures and the conductive material of at least some of the tiers within of the second group of the tiers and the third group of the tiers define select transistors within the blocks of the stack structure, some of the select transistors having a relatively higher threshold voltage level than some other of the select transistors.

9. The microelectronic device of claim 4, wherein the third group of the tiers of the sense node tier section further comprises a gate-induced drain-leakage (GIDL) generation tier overlying the multiple SGP tiers.

10. The microelectronic device of claim 1, wherein the group of the pillar structures comprises four of the pillar structures.

11. The microelectronic device of claim 1, wherein:
 the sub-blocks of the blocks horizontally extend in parallel with the dielectric slot structures in a first direction; and
 a horizontal orientation the conductive plug structure is acutely angled relative to the first direction.

12. The microelectronic device of claim 1, wherein:
 conductive plug structure horizontally extends in parallel with the dielectric slot structures in a first direction; and
 the sub-blocks of the blocks horizontally extend in parallel in a second direction acutely angled relative to the first direction.

13. The microelectronic device of claim 1, wherein the sense transistor comprises a horizontal sense transistor comprising:
 the conductive plug structure;
 a channel structure vertically overlying and only partially horizontally extending across the conductive plug structure; and
 a gate dielectric structure vertically interposed between the conductive plug structure and the channel structure.

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14. The microelectronic device of claim 13, wherein the selector transistors comprise vertical selector transistors, each of the vertical selector transistors comprising:
 a semiconductive material comprising channel region vertically interposed between a source region and a drain region;
 a gate electrode horizontally neighboring the semiconductive material and within vertical boundaries of the channel region; and
 a gate dielectric material horizontally interposed between the semiconductive material and the gate electrode.

15. The microelectronic device of claim 1, the selector transistors comprise:
 a write selector transistor vertically interposed between and coupled to the digit line structure and the conductive plug structure;
 a first read selector transistor vertically interposed between and coupled to the read source line structure and a source side of a channel structure of the sense transistor; and
 a second read selector transistor vertically interposed between and coupled to the digit line structure and a drain side of the channel structure of the sense transistor.

16. The microelectronic device of claim 15, further comprising:
 a local strap structure coupled to each of the digit line structure, the write selector transistor, and the second read selector transistor; and
 a digit line contact vertically interposed between and coupled to the local strap structure and the digit line structure.

17. The microelectronic device of claim 1, wherein:
 the read source line structure vertically overlies the sense transistor;
 the digit line structure vertically overlies the read source line structure; and
 the selector transistors are vertically interposed between the read source line structure and the sense transistor.

18. The microelectronic device of claim 17, wherein the selector transistors comprise:
 a write selector transistor vertically interposed between and coupled to the digit line structure and the conductive plug structure;
 a read selector transistor vertically interposed between and coupled to one of:
 the read source line structure and a source side of a channel structure of the sense transistor; and
 the digit line structure and a drain side of the channel structure of the sense transistor.

19. The microelectronic device of claim 18, further comprising an additional read selector transistor having different V_t characteristics than the read selector transistor, the additional read selector transistor vertically interposed between and coupled to an other of:
 the read source line structure and the source side of the channel structure of the sense transistor; and
 the digit line structure and the drain side of the channel structure of the sense transistor.

20. The microelectronic device of claim 18, further comprising a conductive contact structure vertically interposed between and coupled to an other of:
 the read source line structure and the source side of the channel structure of the sense transistor; and
 the digit line structure and the drain side of the channel structure of the sense transistor.

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21. A method of forming a microelectronic device, comprising:

forming pillar structures to vertically extend through a stack structure comprising tiers each including sacrificial material vertically adjacent insulative material;

implanting P-type dopant into different horizontal portions of some of the tiers and into portions of the pillar structures at vertical positions of the some of the tiers; forming sense transistors vertically over the stack structure, the sense transistors coupled to different groups of the pillar structures than one another;

forming an additional stack structure over the sense transistors, the additional stack structure comprising additional tiers each including additional sacrificial material vertically adjacent additional insulative material;

forming additional pillar structures to vertically extend through the additional stack structure and to the sense transistors;

implanting additional P-type dopant into different horizontal portions of two or more of the additional tiers and into portions of the additional pillar structures at vertical elevations of the two or more of the additional tiers;

replacing the sacrificial material and the additional sacrificial material with conductive material; and forming conductive line structures over and in electrical communication with the additional pillar structures.

22. The method of claim 21, wherein forming pillar structures comprises forming the pillar structures to individually comprise:

cell film material on surfaces of the stack structure; semiconductor channel material inwardly horizontally adjacent the cell film material; and dielectric fill material inwardly horizontally adjacent the semiconductor channel material.

23. The method of claim 21, further comprising selecting the P-type dopant to comprise boron.

24. The method of claim 21, wherein implanting the P-type dopant comprises forming a different pattern of regions doped with the P-type dopant at a vertical position of at least one tier of the some of the tiers than at a vertical position of at least one other tier of the some of the tiers.

25. The method of claim 21, wherein forming sense transistors vertically over the stack structure comprises:

forming conductive plug material over the stack structure and the pillar structures;

forming gate dielectric material over the conductive plug material;

forming channel material over the gate dielectric material; forming back-side dielectric material over the channel material;

forming dielectric cap material over the back-side dielectric material; and

removing portions of the dielectric cap material, the back-side dielectric material, the channel material, the gate dielectric material, and the conductive plug material to respectively form dielectric cap structures, back-side dielectric structures, channel structures, gate dielectric structures, and conductive plug structures.

26. The method of claim 25, further comprising:

forming each of the conductive plug structures to be coupled to and horizontally extend across and between a different group of multiple of the pillar structures than each other of the conductive plug structures; and

forming the channel structures to be substantially confined within horizontal areas of the conductive plug

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structures and to have additional horizontal areas smaller than the horizontal areas of the conductive plug structures.

27. The method of claim 26, wherein forming additional pillar structures to vertically extend through the additional stack structure and to the sense transistors comprises:

forming read select pillar structures vertically extending through the additional stack structure, the dielectric cap structures, and the back-side dielectric material and into the channel structures; and

forming write select pillar structures vertically extending through the additional stack structure and into the conductive plug structures.

28. The method of claim 21, wherein replacing the sacrificial material and the additional sacrificial material with conductive material comprises:

forming slots vertically extending through the additional stack structure and the stack structure;

simultaneously removing the sacrificial material of the tiers of the stack structure and the additional sacrificial material of the additional tiers of the additional stack structure by way of the slots to form voids in the tiers of the stack structure and additional voids in the additional tiers of the additional stack structure; and

simultaneously filling the voids in the tiers of the stack structure and the additional voids in the additional tiers of the additional stack structure with the conductive material.

29. The method of claim 21, wherein forming conductive line structures over and in electrical communication with the additional pillar structures comprises:

forming read source line structures over and in electrical communication with some of the additional pillar structures coupled to channel structures of the sense transistors; and

forming digit line structures over and in electrical communication with:

some other of the additional pillar structures coupled to the channel structures of the sense transistors, and yet some other of the additional pillar structures coupled to gate electrodes of the sense transistors.

30. A memory device, comprising:

a stack structure divided into blocks separated by dielectric slot structures, each of the blocks comprising:

an access line section comprising tiers including access line structures;

a select gate section overlying the access line section and comprising additional tiers including drain side select gate (SGD) structures; and

a sense node section overlying the select gate section and comprising further tiers including select gate programming (SGP) structures;

pillar groups within horizontal areas of the blocks and individually comprising multiple pillar structures vertically extending completely through one of the blocks, each pillar structure of the multiple pillar structures horizontally positioned within a different sub-block of the one of the blocks than each other pillar structure of the multiple pillar structures;

horizontal sense transistors vertically overlying and coupled to the pillar groups;

vertical read selector transistors vertically overlying and coupled to the horizontal sense transistors;

vertical write selector transistors vertically overlying and coupled to the horizontal sense transistors, the vertical write selector transistors horizontally offset from the vertical read selector transistors; and

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conductive line structures vertically overlying and coupled to the vertical read selector transistors and the vertical write selector transistors.

31. The memory device of claim 30, wherein the SGD structures of each of the additional tiers of at least one of the blocks continuously horizontally extend from and between at least two of the dielectric slot structures horizontally neighboring the at least one of the blocks.

32. The memory device of claim 30, further comprising, horizontal areas of the blocks:

select gate electrodes vertically overlying the horizontal sense transistors;

read select pillar structures vertically extending through select gate electrodes and coupled to channel structures of the horizontal sense transistors, intersections of the read select pillar structures and the select gate electrodes defining the vertical read selector transistors; and write select pillar structures vertically extending through select gate electrodes and coupled to gate electrodes of the horizontal sense transistors, intersections of the write select pillar structures and the select gate electrodes defining the vertical write selector transistors.

33. The memory device of claim 32, wherein each of the horizontal sense transistors is individually coupled to two of the read select pillar structures and one of the write select pillar structures.

34. The memory device of claim 30, wherein the conductive line structures comprise read source line structures and digit line structures.

35. An electronic system, comprising:

an input device;

an output device;

a processor device operably connected to the input device and the output device; and

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a memory device operably connected to the processor device and comprising:

pillar structures vertically extending through blocks of a stack structure comprising conductive material vertically alternating with insulative material;

conductive plug structures within horizontal areas of the blocks of the stack structure and coupled to different groups of the pillar structures than one another, each of the different groups of the pillar structures including multiple of the pillar structures positioned within different sub-blocks of one of the blocks than one another;

horizontal sense transistors gated by the conductive plug structures;

read source lines vertically overlying the horizontal sense transistors;

digit lines structures vertically overlying the read source lines;

first vertical read selector transistors vertically interposed between and in electrical communication with the read source lines and source regions of the horizontal sense transistors;

second vertical read selector transistors vertically interposed between and in electrical communication with the digit lines structures and drain regions of the horizontal sense transistors; and

vertical write selector transistors vertically interposed between and in electrical communication with the digit lines structures and the conductive plug structures.

36. The electronic system of claim 35, wherein the memory device comprises a 3D NAND Flash memory device.

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